



**UNIVERSITÀ
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Thesis in **Deep Learning**

**Reliable RGB–Thermal Fusion for UAV-Based
Search and Rescue:
Reproducibility, Mechanistic Analysis, and
Architectural Transferability**

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Abstract

Unmanned Aerial Vehicles equipped with visible and thermal cameras can support wilderness Search and Rescue, but imperfect sensor registration, missing or uninformative modalities, limited data, and training variability complicate reliable detector comparison. This thesis studies RGB–thermal person detection on the WiSARD dataset, starting from a dual-backbone RT-DETR detector and an inherited Feature Alignment Module (FAM). It contributes a reliability-aware evaluation and extension of this starting point through paired multi-seed comparisons, internal diagnostics, a complete-video validation protocol, matched full-data confirmation, and transfer to other detector families.

Principal controlled comparisons use five paired seeds and fixed training budgets without early stopping. In the validation-led protocol, Stage A selects **best** on a held-out complete video and retains the final checkpoint as a diagnostic. A candidate satisfying the aggregate promotion rule is retrained from the pretrained initialization on all training data in Stage B and compared with a freshly trained matched control using final checkpoints only.

In the historical locked comparison, standard FAM improves Base RT-DETR in all five seeds, increasing mean mAP@50 from 0.3080 ± 0.0677 to 0.3780 ± 0.0440 . Its main behavioral effect is higher recall. A post-hoc analysis varies the detection confidence threshold and measures the resulting recall and average number of false positives per image. When comparing the highest recall attainable without exceeding an average of 0.5 false positives per image, FAM achieves a mean recall gain over additive fusion of 9.8–14.8 percentage points across the three evaluated populations, with positive differences in all five paired seeds within each population. Internal inspection confirms learned transformations at P3 and P4 without establishing physical RGB–IR registration, and reveals a compensated P5 collapse in one high-performing checkpoint. The validation-led campaign rejects the tested P2, global 800×800 , zero-offset initialization, post-FAM gating, scalar residual, mixed-consistency, and box-guided variants. Reliability-Conditioned Residual Alignment (RCRA) is the only candidate promoted from Stage A. In matched full-data Stage B, it obtains 0.3753 ± 0.0538 , against 0.3569 ± 0.0422 for the matched FAM control, but wins only three of five seeds and therefore does not replace FAM.

A post-selection audit of the matched Stage-B FAM checkpoints on the same 708 paired frames shows that fusion exceeds VIS in all five seeds by 0.0373 ± 0.0223 mAP@50. Masked IR performs poorly against VIS annotations, whereas native-IR evaluation confirms that the thermal branch remains functional; the two conditions use different coordinate contracts. On two previously unused internal WiSARD acquisitions, historical FAM exceeds historical Base in all ten acquisition–seed comparisons, with a macro-average gain of 0.1257. Controlled IR translations and rescaling nevertheless do

not establish a universally more favorable signed relative-drop curve for FAM.

Architectural transfer is also conditional. A separate five-seed, 200-epoch YOLOv10 protocol finds no consistent FAM benefit. RT-DETRv2 FAM has a positive mean Stage-A gain of 0.0220 but only three positive pairs, while YOLO26 FAM reduces the best-validation mean by 0.0234 and wins only one pair. These are within-family comparisons rather than a detector leaderboard. Standard FAM therefore remains the selected RT-DETR configuration, while the broader result is methodological: multimodal improvements require paired replication, explicit coordinate contracts, and predeclared model-selection rules.

Keywords: RGB–thermal fusion; Search and Rescue; UAV; RT-DETR; RT-DETRv2; YOLOv10; YOLO26; missing modalities; feature alignment; reproducibility.

Contents

Abstract	i
Acronyms	xii
1 Introduction	1
1.1 Motivation	1
1.2 Problem statement	1
1.3 Research questions	2
1.4 Scope and relationship with previous work	3
1.5 Contributions	3
1.6 Thesis organization	4
2 Background and Research Starting Point	5
2.1 UAV-based RGB–thermal detection for Search and Rescue	5
2.2 Object detection and evaluation	5
2.3 Object detector families	6
2.3.1 Transformer encoder–decoder and attention	6
2.3.2 DETR	7
2.3.3 Deformable DETR	8
2.3.4 DINO	9
2.3.5 RT-DETR	10
2.3.6 RT-DETRv2	11
2.3.7 YOLOv10	11
2.3.8 YOLO26	12
2.4 Multimodal fusion	12
2.5 Missing modalities and Modal Dropout	13
2.6 Feature-level alignment and fusion	13
2.6.1 Deformable convolution in the FAM	13
2.6.2 Inherited Feature Alignment Module	14
2.7 Dataset organization and evaluation conditions	16
2.7.1 WiSARD acquisitions and experimental roles	16
2.8 Inherited experimental starting point	18

3	Experimental Design and Methodology	19
3.1	Experimental design and evidence types	19
3.2	FAM implementation and regularization	21
3.2.1	Lazy registration bug	21
3.2.2	Eager, Frozen, and Spatial Dropout variants	21
3.2.3	Stochastic Spatial Jitter	21
3.3	Exploratory architectural extensions	22
3.3.1	Deformable DETR with channel fusion, FAM, and SSJ	22
3.3.2	Complete DINO implementation	24
3.4	Reproducibility investigation	25
3.4.1	Why single runs were insufficient	25
3.4.2	Pretrained single-class head	25
3.4.3	DCNv2 deterministic limitation	26
3.4.4	Modal Dropout and evaluation leakage	26
3.4.5	Validation and checkpoint policy	26
3.5	Historical locked RT-DETR protocol	27
3.6	Validation-led two-stage protocol	28
3.7	Alignment-extension hypotheses	29
3.7.1	Zero-offset initialization	29
3.7.2	Mixed native/paired consistency training	30
3.7.3	Supervised box-guided alignment	30
3.8	Detector-transfer comparisons under Stage A	31
3.9	Coordinate contracts for modality evaluation	31
3.10	Acquisition confirmation and geometric-stress protocols	32
3.10.1	Previously unused internal acquisitions	32
3.10.2	Controlled synthetic geometric stress	32
3.11	Statistical reporting	32
4	Experimental Evaluation	34
4.1	Historical locked RT-DETR multi-seed study	34
4.1.1	Raw and aggregate VIS+IR results	34
4.1.2	Paired contrasts and model selection	36
4.1.3	Historical native-sensor modality evaluation	36
4.2	Validation-led RT-DETR extension campaign	37
4.2.1	Stage-A checkpoint reference	37
4.2.2	Five-seed Stage-A candidate comparison	38
4.2.3	Zero-offset initialization	38
4.2.4	Consistency and box-guided alignment	39
4.2.5	Matched full-data Stage B	40
4.3	Post-selection paired modality characterization	40

4.3.1	Modal Dropout coordinate-system ablation	41
4.4	Mechanistic validation of the FAM	42
4.4.1	Diagnostic design	42
4.4.2	Offset behavior across levels and seeds	42
4.4.3	P5 collapse in seed 41	43
4.4.4	Factorial P3/P4/P5 ablation	45
4.4.5	Bounded-offset intervention	45
4.5	Detection error analysis	46
4.5.1	Protocol	46
4.5.2	Results across thresholds	46
4.5.3	Recall at matched false-positive budgets	48
4.5.4	Qualitative examples	50
4.6	Carnation scale-mismatch stress test	51
4.6.1	Frozen design	51
4.6.2	Results	51
4.7	Confirmation on previously unused WiSARD acquisitions	52
4.8	Controlled synthetic geometric stress	53
4.9	Computational cost	54
4.10	Architectural transferability to YOLOv10	54
4.10.1	Dual-backbone architecture	54
4.10.2	Exploratory history and checkpoint problem	57
4.10.3	Final YOLOv10 protocol	57
4.10.4	Standalone modality evaluation	57
4.10.5	Training-horizon interpretation	58
4.11	Five-seed Stage-A detector-transfer comparisons	59
4.11.1	RT-DETRv2 with FAM	59
4.11.2	YOLO26 dual-backbone Base and FAM	60
4.12	Cross-architecture summary	60
5	Discussion and Conclusions	61
5.1	Answers to the research questions	61
5.1.1	RQ1: experimental reliability	61
5.1.2	RQ2: RT-DETR effectiveness and modality use	62
5.1.3	RQ3: mechanism, levels, and improved alignment	62
5.1.4	RQ4: operational relevance	64
5.1.5	RQ5: architectural transferability	64
5.2	Overall interpretation	65
5.3	Threats to validity	66
5.3.1	Internal validity	66
5.3.2	Construct validity	66

5.3.3	Statistical conclusion validity	67
5.3.4	External validity	67
5.4	Future work	68
5.4.1	Independent confirmation of the model-selection protocol	68
5.4.2	Improving the alignment architecture	68
5.4.3	Broader evaluation and deployment	69
5.5	Conclusions	69
A	Supplementary Results and Artifact Provenance	71
A.1	Secondary RT-DETR metrics	71
A.2	Low-threshold qualitative montage	71
A.3	Versioned compact artifacts	71
A.4	Code availability	75

List of Figures

2.1	The original Transformer encoder–decoder architecture. The encoder (left) alternates multi-head self-attention and feed-forward sublayers. The autoregressive decoder (right) adds masked self-attention and cross-attention over the encoder output. Residual connections and normalization surround every sublayer, while positional encodings retain sequence order. Source: Vaswani et al. [11].	7
2.2	DETR architecture. A CNN backbone produces image features that are combined with positional encodings, contextualized by a Transformer encoder, and decoded from learned object queries into class and bounding-box predictions. Reproduced from Fig. 2 of Carion et al. [4].	8
2.3	Deformable DETR uses multi-scale deformable self-attention in the encoder and deformable cross-attention in the decoder. Each attention module samples a sparse set of learned locations instead of attending densely over every feature-map cell. Reproduced from Fig. 1 of Zhu et al. [20].	9
2.4	RT-DETR architecture. Its efficient hybrid encoder combines attention-based intra-scale interaction with convolutional cross-scale fusion, while IoU-aware query selection initializes the Transformer decoder. Source: Zhao et al. [18].	10
2.5	YOLOv10 consistent dual assignments. A shared backbone and PAN support one-to-many and one-to-one detection heads during training; only the one-to-one branch is retained for NMS-free inference. Source: Wang et al. [12].	12
2.6	Dual-backbone RT-DETR and the inherited Feature Alignment Module. At each pyramid level, RGB and IR features jointly predict 18 offset components and nine modulation masks, DCNv2 then transforms the IR feature before additive fusion. In the final data path, training-only Modal Dropout zero-pads the missing input channels while preserving the four-channel interface.	15
3.1	Overview of the experimental design. Stage-A configurations share the five-seed comparison and checkpoint-selection rules. Only a candidate satisfying both aggregate promotion conditions enters full-data Stage B. The locked comparison and post-selection diagnostics answer separate questions and are not pooled with development-validation estimates. . .	20

3.2	RGB–IR frontend of the dual-backbone Deformable DETR adaptation. FAM is applied independently to the three native feature levels when enabled, otherwise the raw IR features bypass it. Each RGB–IR pair is then fused along the channel dimension. The fourth encoder level is generated from the fused C5 feature and therefore does not receive a separate FAM.	23
3.3	Standard single-stage Deformable DETR backend used after the multi-modal frontend. The four flattened feature levels enter the deformable-attention encoder. Learned query content, positional embeddings, and reference points then condition the decoder, whose final outputs feed the classification and bounding-box heads.	24
4.1	Final RT-DETR VIS+IR mAP@50 from the <code>latest</code> checkpoints. Panel A connects Base and FAM runs initialized with the same training seed; FAM wins all five pairs. Panel B reports, for each seed, the difference between a variant and standard FAM, with black bars denoting paired means. Each seed-specific run/checkpoint is an experimental unit; points are not repeated measurements of one checkpoint.	35
4.2	Mechanistic diagnostic of the collapsed P5 branch in FAM seed 41 for the prespecified MtErie sample 35. Panels (a)–(c) are pseudo-colour PCA projections obtained with a basis shared across the standardized feature tensors; panels (d)–(e) are qualitative 50% blends and are not tensors consumed by the detector. Panel (f) summarizes absolute DCNv2 offset components in feature-cell coordinates and the modulation-mask statistics. The transformed IR feature is spatially constant, while the mean absolute offset component corresponds to 5,227 input pixels. These quantities diagnose a learned feature-space transformation and must not be interpreted as a camera-registration field.	44
4.3	Sensitivity of precision, recall, false positives per image, and non-empty frames with a false negative to the five predefined confidence thresholds. Error bars are sample standard deviations across checkpoints.	47
4.4	Recall at matched false-positive budgets for historical Base (additive fusion) and FAM RT-DETR checkpoints in VIS+IR mode. Curves show five-seed means and shaded bands show sample standard deviations at common FPPI budgets, with one-to-one confidence-ordered matching at IoU 0.50. The bands describe training variability, not confidence intervals over frames or unseen acquisitions.	49
4.5	Predefined qualitative examples at confidence 0.25. Each row contains one small-target frame and one empty frame from a MtErie sequence. The same frames and display rules are used for Base RT-DETR and FAM. These examples illustrate, but do not replace, the aggregate analysis.	50
4.6	Paired Carnation results for the three modality conditions. The stress test shows relative advantage under mismatch by FAM, not correction of the severe mismatch.	52

4.7	Signed relative mAP@50 drop under IR-only translations and rescaling, averaged first across directions within checkpoint and then over five seeds. Non-monotone negative drops show why these transformations cannot be interpreted as calibrated displacement from a known optimum.	53
4.8	Implemented YOLOv10-s transfer architecture. The controlled comparison changes the P3/P4/P5 fusion operation while retaining the dual backbones, feature gating, FPN/PAN neck, detection head, and final protocol. The FAM operation is the same as the one expanded in Figure 2.6 and is therefore not duplicated internally. During unimodal inference the excluded backbone is not executed and FAM is bypassed. Consequently, unimodal Base–FAM differences are indirect effects of training rather than direct alignment effects.	56
4.9	Final YOLOv10 paired modality evaluation. The favorable but very small IR-only delta does not compensate for the negative fusion result.	58
A.1	The six predefined qualitative frames at confidence threshold 0.01. Titles count all detections, while at most twenty high-score predictions are drawn for readability. The density of low-confidence false positives illustrates why 0.01 is a metric-collection threshold rather than an operational setting.	74

List of Tables

2.1	Conceptual role of the detector families in this thesis. The table summarizes architectural distinctions rather than the study-specific fusion implementations.	12
2.2	WiSARD paired acquisitions used in the study. Frame counts are reported after matching the VIS and IR streams.	16
2.3	Effective composition of the historical full-data split after preprocessing and pairing filters. Median box area is expressed as a percentage of the evaluation image area.	17
3.1	Exploratory Deformable DETR results, reported as median [minimum–maximum] mAP@50 over five runs.	24
3.2	DINO base result under the previous protocol, reported as median [minimum–maximum] mAP@50 over five runs.	25
3.3	Historical locked RT-DETR protocol used by all six configurations in the original multi-seed comparison.	27
3.4	Common five-seed Stage-A/Stage-B decision protocol. Training budgets remain detector-specific.	28
4.1	Final RT-DETR VIS+IR mAP@50 by paired training seed.	34
4.2	Aggregate final RT-DETR VIS+IR mAP@50. Standard deviations are sample standard deviations across five checkpoints.	34
4.3	Paired VIS+IR mAP@50 contrasts. A win means that the first configuration exceeds its reference for the same training seed.	36
4.4	Historical native-sensor RT-DETR mAP@50 by input condition, mean $\pm$ sample standard deviation across five checkpoints. Columns use their original evaluation populations and should not be ranked against one another as a same-frame modality ablation.	37
4.5	Stage-A FAM checkpoint audit. Validation uses the complete FHL 0401/0402 video. MtErie is evaluated only after all five runs are complete.	37
4.6	Stage-A best-validation mAP@50 across five seeds. A promotion requires mean delta at least +0.01 and at least four paired wins.	38
4.7	Final-epoch validation diagnostics for zero-offset initialization, consistency, and box-guided alignment. Values are epoch-10 latest mAP@50 over five seeds; these scores do not replace the best-checkpoint promotion criterion.	39
4.8	Matched full-data Stage-B VIS+IR mAP@50.	40

4.9	Paired modality characterization of the selected current-code FAM checkpoints. Every column uses the same frames and VIS ground truth. . . .	41
4.10	Effect of changing the IR-only Modal Dropout coordinate convention in the seed-40 Stage-A experiment, evaluated on the 896-pair validation sequence.	41
4.11	Mean absolute offset components in input-image pixels. Values are aggregated over thirty frozen samples per checkpoint.	43
4.12	Post-hoc factorial level ablation. Delta is the paired mean difference from the full trained FAM.	45
4.13	Automatic VIS+IR mAP@50 for the five bounded-offset runs and the corresponding frozen standard-FAM reference.	46
4.14	Paired RT-DETR error behavior across confidence thresholds. Values are mean $\pm$ sample standard deviation over five checkpoints.	47
4.15	Recall at the three predefined FPPI budgets. Base, FAM, and paired FAM–Base differences are mean $\pm$ sample standard deviation across five seeds. Wins count strictly positive paired differences.	49
4.16	Carnation mAP@50 across five frozen checkpoints.	51
4.17	Frozen acquisition-level confirmation. Values are mean mAP@50 over five paired checkpoints; intervals and standard deviations refer to the five within-seed differences.	52
4.18	RT-DETR detector-only computational benchmark. The operation count is a proxy and the latency uncertainty is the sample standard deviation across three trial means.	54
4.19	Final automatic YOLOv10 VIS+IR mAP@50 at <code>last.pt</code>	57
4.20	Final standalone YOLOv10 mAP@50 by modality, mean $\pm$ SD over five checkpoints.	58
4.21	Detector-transfer Stage-A validation mAP@50, mean $\pm$ sample standard deviation across five seeds. Comparisons are within each detector: budgets and optimization recipes differ across families.	59
4.22	Paired FAM minus Base contrasts on Stage-A validation. Promotion uses only the best-checkpoint row and requires both a mean gain of at least 0.01 and at least four wins.	59
A.1	Final VIS+IR secondary metrics, mean $\pm$ sample standard deviation over five training checkpoints. COCO mAP averages IoU thresholds from 0.50 to 0.95.	71

Acronyms

AIFI	Attention-based Intra-scale Feature Interaction
AP	Average Precision
CDN	Contrastive DeNoising
CI	Confidence Interval
COCO	Common Objects in Context
CUDA	Compute Unified Device Architecture
DCNv2	Deformable Convolutional Networks v2
DETR	DEtection TRansformer
FAM	Feature Alignment Module
FN	False Negative
FP	False Positive
FPN	Feature Pyramid Network
FPPI	False Positives Per Image
GFLOPs	Giga Floating-Point Operations
GT	Ground Truth
IR	Infrared
IoU	Intersection over Union
LFT	Look-Forward-Twice
mAP	mean Average Precision
mAR	mean Average Recall
MQS	Mixed Query Selection
NMS	Non-Maximum Suppression
PAN	Path Aggregation Network

PCA	Principal Component Analysis
RCRA	Reliability-Conditioned Residual Alignment
RGB	Red, Green, and Blue visible spectrum
SAR	Search and Rescue
SD	Standard Deviation
SSJ	Stochastic Spatial Jitter
UAV	Unmanned Aerial Vehicle

Chapter 1

Introduction

1.1 Motivation

Wilderness Search and Rescue (SAR) requires locating missing or injured people over large and visually complex areas while time, personnel, visibility, and accessibility are constrained. Unmanned Aerial Vehicles (UAVs) can reduce the area that must be inspected from the ground and provide a viewpoint that is otherwise difficult or dangerous to obtain. However, turning aerial images into a reliable detection signal is challenging since a person may occupy only a small fraction of the frame, vegetation and terrain create clutter, and appearance changes substantially with illumination, altitude, season, and weather.

Visible and thermal infrared cameras provide complementary information. The visible stream contains texture, colour, and scene context, whereas the thermal stream responds to emitted radiation and can retain a useful human signature when visible contrast is weak. The WiSARD dataset was introduced specifically to support the study of visual–thermal perception for wilderness SAR across varied acquisition conditions [3]. Complementarity, however, does not imply that using both streams is automatically beneficial. The two cameras are physically distinct, their fields of view and effective scales may differ, and their projections are not perfectly registered. Moreover, either stream may be absent or uninformative in practice. A multimodal detector must therefore learn not only how to combine two signals, but also when and how much to rely on each one.

These deployment concerns interact with an experimental one. The datasets and training budgets available for UAV-based RGB–thermal detection are modest in comparison with general-purpose detection corpora. Small changes in initialization, batch order, stochastic modality masking, or nondeterministic GPU kernels can then lead to visibly different final scores. If architectures are compared through a single execution, the resulting ranking may describe a particular training trajectory rather than a repeatable architectural effect.

1.2 Problem statement

This thesis investigates person detection from paired RGB and thermal images on the WiSARD dataset. The main detector family is RT-DETR, adapted to a four-channel

input through two independent backbones and feature-level fusion. The baseline sums corresponding RGB and infrared feature maps at the P3, P4, and P5 pyramid levels. The studied alternative transforms the infrared feature map with a Feature Alignment Module (FAM) before the same additive fusion is applied.

The FAM uses RGB and IR features jointly to predict deformable-convolution offsets and modulation masks. This creates a spatially adaptive transformation of the infrared representation, but the module receives no ground-truth supervision indicating the physical geometric correspondence between the RGB and IR images. Therefore a higher detection score is insufficient to claim that physical sensor registration has been learned. Establishing what the module does requires inspecting its internal variables and intervening on its levels, in addition to measuring the detector output.

The study addresses four connected forms of robustness:

1. **modality robustness**, when RGB or IR is unavailable;
2. **geometric robustness**, when paired streams are imperfectly aligned or differ in scale;
3. **experimental robustness**, when training is repeated under different seeds and nondeterministic execution;
4. **architectural robustness**, when the same fusion idea is moved to another detector family.

1.3 Research questions

The thesis is organized around the following research questions.

RQ1 — Experimental reliability. How strongly do run-to-run variability, validation design, and checkpoint selection affect the ranking of RGB–thermal detectors, and which protocol can support a fair architecture-development campaign?

RQ2 — RT-DETR effectiveness and modality use. Does FAM provide a repeatable advantage over the Base RT-DETR configuration, which uses direct additive fusion, and does the selected detector obtain a measurable benefit from the thermal stream when all modality conditions are evaluated on the same paired frames and ground truth?

RQ3 — Mechanism and improved alignment. What geometric-functional transformation does FAM learn across P3, P4, and P5, which failure modes does it exhibit, and can a reliability-conditioned local modulation of the FAM correction improve it more consistently than simpler alignment, resolution, or gating interventions?

RQ4 — Operational relevance. How do the detector’s error profile, computational cost, and behavior under a severe sensor-scale mismatch delimit its usefulness for UAV Search and Rescue?

RQ5 — Architectural transferability. Does the FAM advantage observed in RT-DETR transfer to other detector families under the implemented Deformable DETR, YOLOv10, RT-DETRv2, and YOLO26 configurations?

1.4 Scope and relationship with previous work

This thesis builds on preliminary RGB–thermal detection project work conducted before this thesis. That work examined DETR, Modal Dropout, the Base RT-DETR configuration with direct additive fusion, fixed tiling, and CMX. Subsequent preliminary work introduced FAM on RT-DETR, developed a Hybrid CMX variant, and adapted Deformable DETR to channel fusion with FAM. These components, preserved in the public project repository identified in Appendix A, define the starting point of the present work.

The FAM is therefore *not* presented as a new architecture invented in this thesis. It is described in full so that the document remains self-contained, but its provenance is stated explicitly. The present contribution is the reliability-aware validation of that module: its performance across paired runs, its internal behavior, its failure modes, the effect of targeted corrections, and the extent to which the result transfers outside the original RT-DETR setting. The thesis additionally develops Reliability-Conditioned Residual Alignment (RCRA), an extension developed in this thesis that locally selects how much of the FAM correction to retain.

Model comparisons are reported within their respective data splits, training budgets and checkpoint-selection protocols. Analysis of FAM’s internal behavior complements these comparisons, but does not replace the performance criteria used for Stage-A/B model selection.

1.5 Contributions

The contributions developed in the present work are:

1. a reliability-aware experimental framework for RGB–IR detection, combining paired training seeds, transferred heads, training-only Modal Dropout, isolated processes, complete-video validation, and a two-stage checkpoint policy;
2. a controlled multi-seed evaluation of six RT-DETR configurations, establishing the consistent internal advantage of standard FAM and its behavior when either modality is unavailable;
3. a mechanistic study of FAM across pyramid levels, combining internal diagnostics, factorial interventions, identification of a compensated P5 collapse, and a five-seed bounded-offset test;
4. a five-seed validation-led comparison of tiny-object, resolution, offset-initialization, reliability, residual-alignment, mixed-consistency, and supervised box-guided hypotheses under the common Stage-A/B protocol, including configuration-matched full-data confirmation of RCRA and FAM;

5. a coordinate-controlled modality audit on the same 708 paired frames, distinguishing paired missing-RGB robustness from native-IR detection;
6. an explicit delimitation of the result through paired error analysis and recall at matched false-positive budgets, the frozen Carnation scale-mismatch stress test, a detector-only compute benchmark, and a controlled negative transfer result on dual-backbone YOLOv10;
7. a predeclared and versioned confirmation on two previously unused internal WiS-ARD acquisitions and a controlled synthetic geometric stress test;
8. paired five-seed Stage-A transfer comparisons of Base and FAM fusion in RT-DETRv2 and YOLO26, with validation-selected checkpoints as primary endpoints and final-epoch checkpoints as diagnostics.

Exploratory DINO and Deformable DETR studies are retained as development evidence, but are not presented as central contributions. Negative results are integral to the final contribution: they delimit which mechanistic changes improve internal behavior without improving detection and which RT-DETR findings fail to transfer under the studied conditions.

1.6 Thesis organization

Chapter 2 introduces the dataset, detector families, multimodal fusion, missing-modality training, and the inherited architectural starting point. Chapter 3 defines the experimental methods, distinguishes the historical locked comparison from validation-led model development, and specifies the common five-seed, two-stage protocol. Chapter 4 presents the main RT-DETR results, the validation-led architecture campaign and modality audit, internal FAM diagnostics, bounded-offset intervention, error analysis, Carnation stress test, confirmation on previously unused acquisitions, controlled geometric stress, computational cost, and five-seed detector-transfer studies. Finally, Chapter 5 answers the research questions, states the validity boundaries, and outlines future work.

Chapter 2

Background and Research Starting Point

2.1 UAV-based RGB–thermal detection for Search and Rescue

Person detection in aerial wilderness images differs from conventional street-level pedestrian detection. Subjects are frequently small, partially occluded, and observed from changing altitudes and viewing angles. Natural backgrounds provide repeated textures, warm rocks may resemble thermal human signatures, and targets may exhibit weak visible or thermal contrast. The WiSARD dataset addresses this setting with labeled visual and thermal images collected from UAV flights over diverse terrains and acquisition conditions [3].

RGB–thermal sensing seeks to reduce dependence on a single observation channel. Multispectral pedestrian benchmarks have long shown the potential of combining visible and thermal cues [6], while the broader multimodal-learning literature emphasizes both complementarity and the difficulty of learning balanced representations [2, 14]. In this thesis, robustness is evaluated explicitly in three input conditions: full fusion (VIS+IR), visible-only (VIS), and infrared-only (IR).

2.2 Object detection and evaluation

An object detector maps an image to a set of class scores and bounding boxes. For a predicted box B_p and a ground-truth box B_g , Intersection over Union is

$$\text{IoU}(B_p, B_g) = \frac{|B_p \cap B_g|}{|B_p \cup B_g|}. \quad (2.1)$$

Average Precision (AP) integrates precision as a function of recall after ranking detections by confidence. The primary metric in the final experiments is mAP@50, for which a match requires IoU at least 0.50. COCO mAP averages AP over IoU thresholds from 0.50 to 0.95 and provides a stricter localization summary [8]. mAP@75 and mean Average Recall (mAR) are retained as secondary metrics.

The confidence value used to collect detections is not itself the IoU match criterion.

A low confidence threshold preserves candidates for AP computation, but may be unusable as an operational alarm threshold. This distinction is important in Section 4.5, where threshold 0.01 produces high recall and tens of false positives per image.

2.3 Object detector families

2.3.1 Transformer encoder–decoder and attention

The Transformer was originally introduced as an encoder–decoder architecture built around attention rather than recurrent processing [11]. For a set of query, key, and value vectors, scaled dot-product attention is

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V, \quad (2.2)$$

where d_k is the key dimension. The dot products measure which values are relevant to each query, while the softmax produces normalized aggregation weights. Multi-head attention repeats this operation in several learned subspaces and combines their outputs, allowing different heads to represent different relationships.

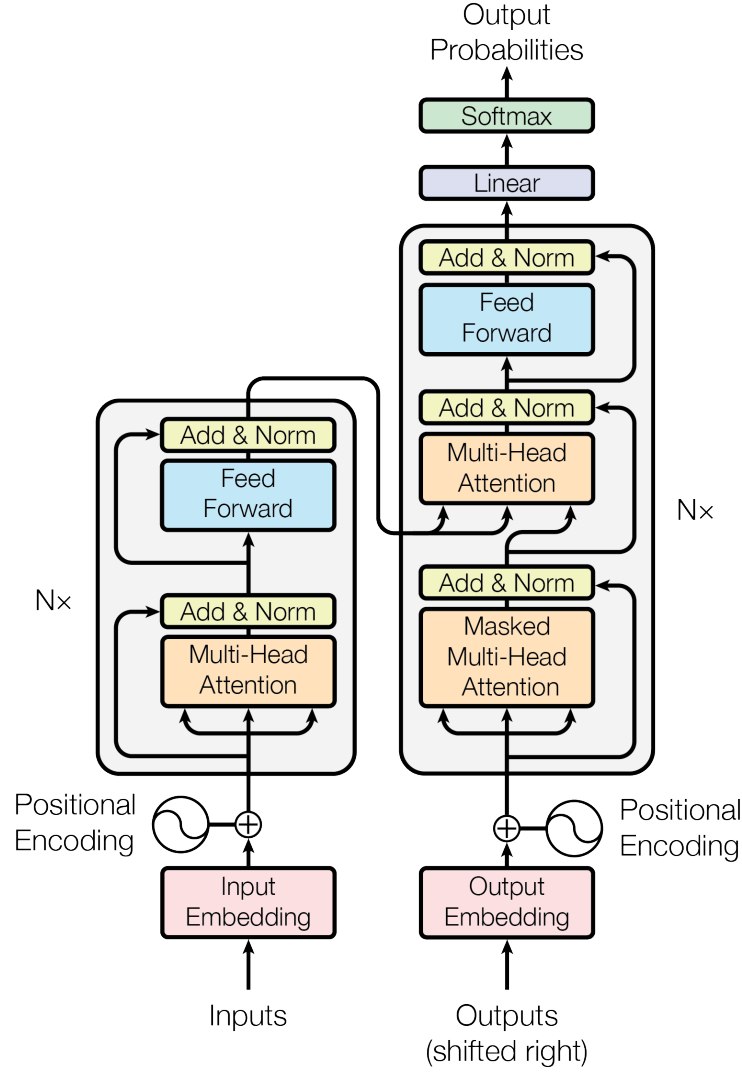


Figure 2.1: The original Transformer encoder–decoder architecture. The encoder (left) alternates multi-head self-attention and feed-forward sublayers. The autoregressive decoder (right) adds masked self-attention and cross-attention sublayers over the encoder output. Residual connections and normalization surround every sublayer, while positional encodings retain sequence order. Source: Vaswani et al. [11].

Figure 2.1 shows the original sequence-to-sequence model. Its encoder builds contextual representations of the complete input sequence, while the autoregressive decoder combines its previous output tokens with the encoder representation through cross-attention. The original model therefore uses a causal mask in decoder self-attention, while vision-detection variants modify this setup according to their prediction task.

2.3.2 DETR

DETR applies this architecture to direct set prediction [4]. After the backbone and a channel projection, the original model flattens a single feature map into a sequence and passes it through the Transformer encoder. A fixed number of learned object queries enters the decoder, and each query produces either one object prediction or the special no-object class. The output size is therefore fixed, while the number of real objects can

vary between images.

Unlike the original sequence-to-sequence Transformer, DETR replaces input-word embeddings with image features and shifted output tokens with object queries, while class and box heads replace the vocabulary softmax. Image tokens are augmented with positional encodings to retain their two-dimensional locations. The object queries are decoded in parallel, so DETR does not use an autoregressive causal mask. Decoder self-attention models relations among candidate objects, whereas cross-attention retrieves relevant evidence from the encoded image features.

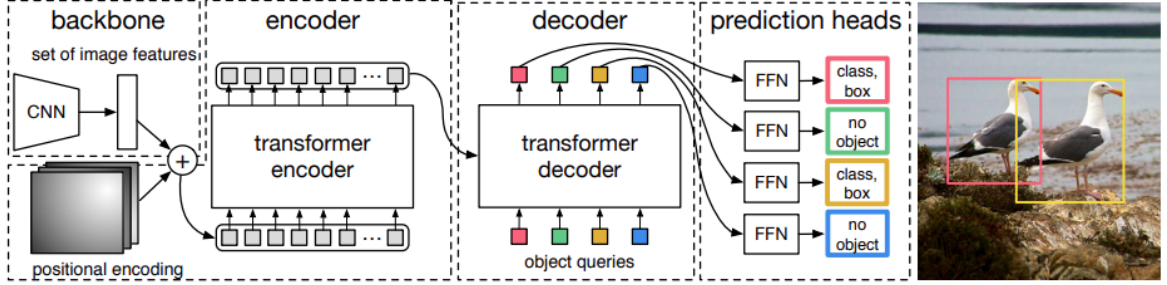


Figure 2.2: DETR architecture. A CNN backbone produces image features that are combined with positional encodings, contextualized by a Transformer encoder, and decoded from learned object queries into class and bounding-box predictions. Reproduced from Fig. 2 of Carion et al. [4].

Training must decide which predicted query corresponds to which ground-truth object. DETR solves this with one-to-one bipartite matching: the Hungarian algorithm selects the assignment with minimum classification and localization cost, after which matched pairs receive classification and box-regression losses and unmatched queries are trained as no-object. Because every target is assigned to only one query, duplicate predictions are discouraged by the training objective itself. The original inference pipeline consequently does not require handcrafted anchors or non-maximum suppression.

This simplification has costs. Dense encoder self-attention compares all spatial tokens, so its memory and computation grow quadratically with feature resolution. The use of a relatively coarse, single-scale feature map also limits the detail available for small objects. Moreover, one-to-one matching is initially unstable because object queries do not yet specialize, leading to the slow convergence reported for the original DETR. These limitations motivate the later variants used or explored in this study.

In the work that preceded this thesis DETR was used to establish the value of stochastic missing-modality training. DETR is not rerun in the final campaign, its role is to motivate the training regime inherited by later architectures.

2.3.3 Deformable DETR

Deformable DETR replaces dense attention over all spatial locations with multi-scale deformable attention around learned reference points [20]. Instead of comparing every query with every feature location, each attention head predicts a small number of sampling offsets around a reference point at each feature level. Values at fractional

coordinates are obtained by bilinear interpolation and combined through learned attention weights. The sampling cost therefore depends on the small number of selected points rather than on all spatial cells.

The model consumes a feature pyramid rather than only the deepest backbone map. High-resolution levels retain spatial detail for small targets, while lower-resolution levels provide wider context. Reference points also give each query an explicit spatial anchor, and iterative box refinement can update this reference as it passes through decoder layers. Together, sparse sampling, multi-scale features, and spatial references improve convergence and are particularly relevant to aerial images, where a person may occupy very few pixels.

The configuration used in this thesis provides four feature levels to the detector. The three native ResNet-50 outputs, $C3$, $C4$, and $C5$, are fused across RGB and IR; a fourth, coarser level is derived from the fused $C5$ feature by stride-2 projection. FAM operates only at the three native levels before the fourth is formed.

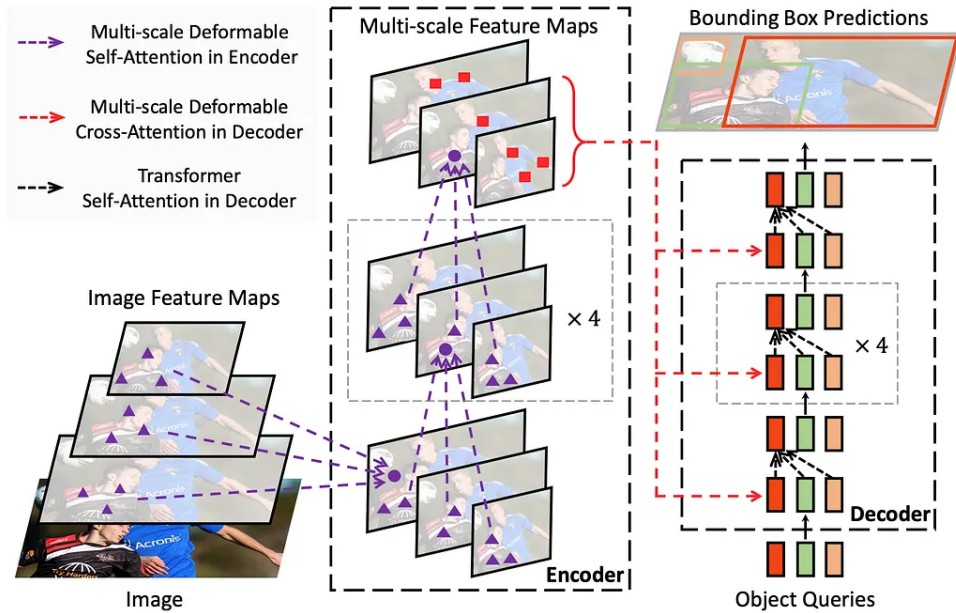


Figure 2.3: Deformable DETR uses multi-scale deformable self-attention in the encoder and deformable cross-attention in the decoder. Each attention module samples a sparse set of learned locations instead of attending densely over every feature-map cell. Reproduced from Fig. 1 of Zhu et al. [20].

Deformable attention should not be confused with the deformable convolution used by the FAM. The former is an attention rule that samples multi-scale features for each query, while the latter transforms a dense IR feature map before fusion. They share the idea of learned sampling offsets but act at different places in the detector and have different parameters and outputs.

2.3.4 DINO

DINO builds on the multi-scale deformable DETR line and targets two remaining problems: unstable matching early in training and the quality of decoder query initial-

ization [15]. Contrastive DeNoising (CDN) adds noisy copies of known labels and boxes during training and asks the decoder to reconstruct them. Positive and harder negative denoising queries teach the model how small perturbations should affect classification and localization, providing a more direct learning signal than Hungarian matching alone at the start of training.

Mixed Query Selection (MQS) initializes the positional part of decoder queries from selected encoder features while retaining learned content queries. This separates the questions of *where* a candidate object may be and *what* it represents. Look-Forward-Twice (LFT) strengthens iterative box refinement by allowing the parameters involved in one refinement step to receive supervision from the following decoder layer as well. These mechanisms are complementary: CDN stabilizes training queries, MQS improves their spatial initialization, and LFT improves the learning signal across refinement stages. A complete implementation was developed and is documented separately in Section 3.3.2. It retains the same four-level multi-scale representation as the Deformable DETR adaptation.

2.3.5 RT-DETR

RT-DETR retains end-to-end set prediction while redesigning the multi-scale encoder for lower latency [18]. Its efficient hybrid encoder separates two operations that a standard multi-scale Transformer would perform together. Attention-based intra-scale feature interaction is applied only to the deepest, lowest-resolution feature level, where global semantic interaction is comparatively inexpensive. A convolutional cross-scale feature fusion module then propagates information through the feature pyramid. This division reduces the cost of applying self-attention densely at several resolutions.

The decoder is initialized through uncertainty-minimal query selection. The encoder scores candidate features and selects high-quality spatial locations for the initial queries, seeking candidates whose classification and localization estimates are reliable. Decoder layers then refine these queries into the final detection set. Since the decoder is the repeated part of the detection head, its number of layers can also be adjusted to trade accuracy for latency without retraining the encoder.

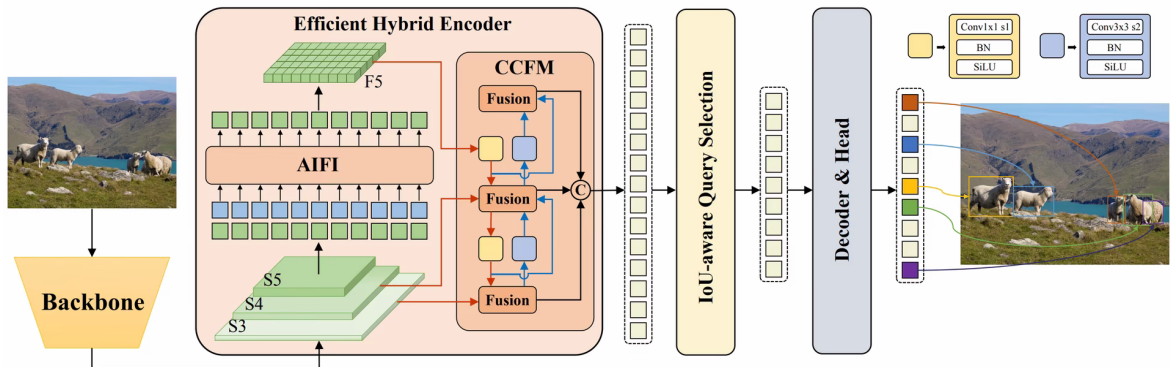


Figure 2.4: RT-DETR architecture. Its efficient hybrid encoder combines attention-based intra-scale interaction with convolutional cross-scale fusion, while IoU-aware query selection initializes the Transformer decoder. Source: Zhao et al. [18].

The implementation used here starts from the pretrained RT-DETR R50vd check-

point and exposes three backbone outputs, denoted P3, P4, and P5 with strides 8, 16, and 32. The dual-backbone adaptation preserves the downstream hybrid encoder and decoder: the experimental variable is how corresponding RGB and IR features are transformed and fused before entering them.

RT-DETR is designed for real-time object detection. In this thesis, computational efficiency is characterized through detector-only forward-pass latency measured on a specified GPU. Preprocessing and postprocessing are excluded, so these measurements compare the computational cost of the detector configurations rather than establish end-to-end real-time deployment performance.

2.3.6 RT-DETRv2

RT-DETRv2 retains the RT-DETR encoder–decoder framework while revising the decoder’s deformable-attention sampling and the associated training strategy [9]. It allows the number of sampling points to vary across feature scales, provides an optional discrete sampling operator, and introduces dynamic augmentation and scale-adaptive hyperparameters. The five-seed comparison uses the official COCO-trained detector checkpoint `PekingU/rtdetr_v2_r50vd`, rather than an ImageNet-only backbone checkpoint. Base and FAM share the same dataset split, training configuration, and checkpoint-selection rule defined in Section 3.6. The comparison evaluates the fusion intervention within this detector.

2.3.7 YOLOv10

YOLOv10 is an end-to-end member of the YOLO family that combines consistent dual assignments with an efficiency-oriented convolutional architecture [12]. As in other one-stage YOLO detectors, a backbone extracts a feature pyramid, a neck combines fine spatial detail with coarser semantic information, and a dense head predicts classes and boxes at multiple locations, which is structurally different from decoding a small fixed set of object queries.

Traditional dense detectors can produce several boxes for the same object and therefore rely on non-maximum suppression. YOLOv10 instead trains two assignment branches consistently: a one-to-many branch supplies abundant positive matches for optimization, while a one-to-one branch learns the non-redundant predictions used for end-to-end inference. The matching rules are coordinated so that the richer supervision of the first branch benefits the second rather than teaching incompatible rankings. At inference time, the one-to-one output removes the need for NMS.

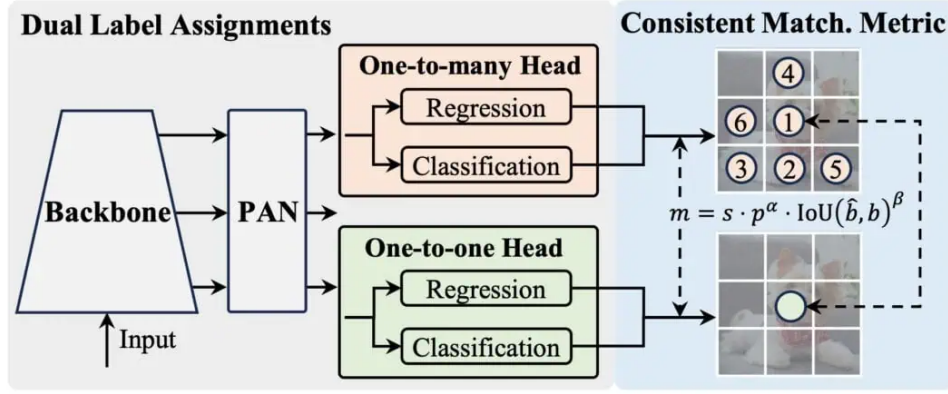


Figure 2.5: YOLOv10 consistent dual assignments. A shared backbone and PAN support one-to-many and one-to-one detection heads during training; only the one-to-one branch is retained for NMS-free inference. Source: Wang et al. [12].

The thesis adapts YOLOv10-s to two modality-specific backbones and fuses their P3, P4, and P5 outputs before the shared FPN/PAN neck. It therefore provides a useful transfer target: both detector families receive corresponding multi-scale RGB and IR features, but RT-DETR processes them with a hybrid encoder and query decoder, whereas YOLOv10 processes them through a convolutional neck and dense detection head.

2.3.8 YOLO26

YOLO26 is a later Ultralytics detector family with a lighter distribution-focal-loss-free regression head, optional native end-to-end inference, and a revised optimization and label-assignment recipe [7]. The pretrained YOLO26s model is adapted to use separate RGB and IR backbones, whose features are fused at pyramid levels P3, P4, and P5. Base and FAM are compared over five paired seeds under the common Stage-A/Stage-B decision procedure, with a fixed fifty-epoch Stage-A budget. The measured contrast concerns this dual-backbone adaptation, not the stock single-stream detector.

Table 2.1: Conceptual role of the detector families in this thesis. The table summarizes architectural distinctions rather than the study-specific fusion implementations.

Family	Image representation	Prediction mechanism	Role in this thesis
DETR	single-scale tokens	learned object queries	inherited starting point
Deformable DETR	sparse multi-scale samples	reference-point queries	exploratory extension
DINO	deformable multi-scale features	denoised and mixed queries	exploratory negative result
RT-DETR	efficient hybrid pyramid	uncertainty-selected queries	primary detector
RT-DETRv2	revised deformable decoder	uncertainty-selected queries	five-seed Stage-A transfer
YOLOv10	convolutional pyramid and neck	consistent dense assignments	five-seed fixed-budget transfer
YOLO26	convolutional pyramid and lighter head	dense/end-to-end heads	five-seed Stage-A transfer

2.4 Multimodal fusion

Fusion methods can be grouped by where the modalities interact. Early fusion combines inputs or low-level representations, late fusion combines independent predictions, and

intermediate fusion combines learned features. This work uses intermediate fusion because each sensor receives a dedicated backbone while the detector still operates on a joint multi-scale representation.

In the RT-DETR, RT-DETRv2, and YOLO comparisons, *Base* denotes the dual-backbone RGB-IR configuration without FAM, not the stock single-stream detector. Both Base and FAM use additive fusion but the FAM configuration first transforms the IR features, while the RT-DETR Base configuration uses direct element-wise addition:

$$F_{\text{fused}}^{(i)} = F_{\text{RGB}}^{(i)} + F_{\text{IR}}^{(i)}, \quad i \in \{P3, P4, P5\}. \quad (2.3)$$

Addition introduces no trainable fusion parameters and preserves feature-map shape, but assumes that corresponding cells are meaningfully compatible. A sensor displacement or scale difference violates this assumption. More complex interaction does not automatically solve the problem: multimodal networks can overfit or optimize their modalities at different rates [14].

Previous work also examined CMX, a cross-modal rectification and fusion framework originally proposed for RGB-X semantic segmentation [16]. The specific RT-DETR integration was not competitive on WiSARD. This finding is retained as an exploratory result of that implementation, not as a general rejection of attention-based fusion.

2.5 Missing modalities and Modal Dropout

Modal Dropout was proposed to preserve modality-specific representations by randomly dropping input channels during multimodal training [10]. This study applies the same principle at the modality level. For a paired sample $(x_{\text{RGB}}, x_{\text{IR}})$, training draws one of three conditions:

$$\tilde{x} = \begin{cases} (x_{\text{RGB}}, 0), & \text{with probability } p_{\text{VIS}}, \\ (0, x_{\text{IR}}), & \text{with probability } p_{\text{IR}}, \\ (x_{\text{RGB}}, x_{\text{IR}}), & \text{with probability } p_{\text{fusion}}. \end{cases} \quad (2.4)$$

The inherited schedule uses $p_{\text{VIS}} = 0.2$, $p_{\text{IR}} = 0.2$, and $p_{\text{fusion}} = 0.6$. It is a motivated exploratory choice from the preceding DETR study, not a statistically demonstrated optimum. The equation specifies which sensor is present, but not its coordinate contract. In the RT-DETR data loader used here, an IR-only draw uses the native IR canvas and native IR annotations. This distinction becomes important in Section 3.9. A software audit conducted for this thesis also found that modality dropout was being applied during validation and test. The pipeline was corrected so that the sampling in Equation (2.4) is active only during training.

2.6 Feature-level alignment and fusion

2.6.1 Deformable convolution in the FAM

This subsection introduces DCNv2, the operation used by the Feature Alignment Module (FAM) described next to transform IR feature maps before fusion. It is distinct

from the multi-scale deformable attention used inside Deformable DETR: DCNv2 is a convolutional operation on a dense feature map, whereas deformable attention samples encoder or decoder features for a query.

Standard convolution samples a fixed grid $\mathcal{R}$ around each spatial position p . Modulated deformable convolution augments every kernel point p_k with an input-dependent offset Δp_k and a modulation value m_k [19]:

$$y(p) = \sum_{k \in \mathcal{R}} w_k x(p + p_k + \Delta p_k) \sigma(m_k). \quad (2.5)$$

Sampling at non-integer positions uses bilinear interpolation. The operation therefore combines a geometric sampling change with learned filtering and channel mixing. Its output cannot be interpreted as a pure warp unless those additional effects are controlled.

2.6.2 Inherited Feature Alignment Module

For every pyramid level, the FAM concatenates RGB and IR features and predicts 18 offset channels and 9 mask logits for a 3×3 DCNv2 kernel:

$$[\Delta, m] = \text{Conv}_{3 \times 3}([F_{\text{RGB}}, F_{\text{IR}}]), \quad (2.6)$$

$$\tilde{F}_{\text{IR}} = \text{DCNv2}(F_{\text{IR}}, \Delta, \sigma(m)), \quad (2.7)$$

$$F_{\text{fused}} = F_{\text{RGB}} + \tilde{F}_{\text{IR}}. \quad (2.8)$$

The standalone FAM constructor initializes the offset/mask predictor to zero, which gives null offsets and masks of 0.5. In the standard complete RT-DETR configuration, however, the subsequent Hugging Face `post_init()` step replaces the predictor weights with seed-dependent random values. RT-DETRv2 preserves this effective initialization. Null offsets and constant masks are therefore not guaranteed at the start of training in these models. The standard FAM does *not* start as a mathematical identity. Even with null offsets, the DCNv2 filter weights and modulation masks do not in general make the filtered output equal to the input.

The initialization ablation in Section 3.7.1 explicitly zeros only the offset-predictor rows after complete pretrained model construction. It preserves the standard mask and DCNv2 initialization and leaves the offsets trainable, separating the initial sampling displacement from an identity-initialized operator or permanently fixed sampling locations.

Because there is no ground-truth displacement field, the thesis uses the term *geometric-functional transformation*. Nonzero offsets show that the geometric path is active, while they do not prove recovery of the physical mapping between the cameras. This distinction motivates the internal diagnostics in Section 4.4.

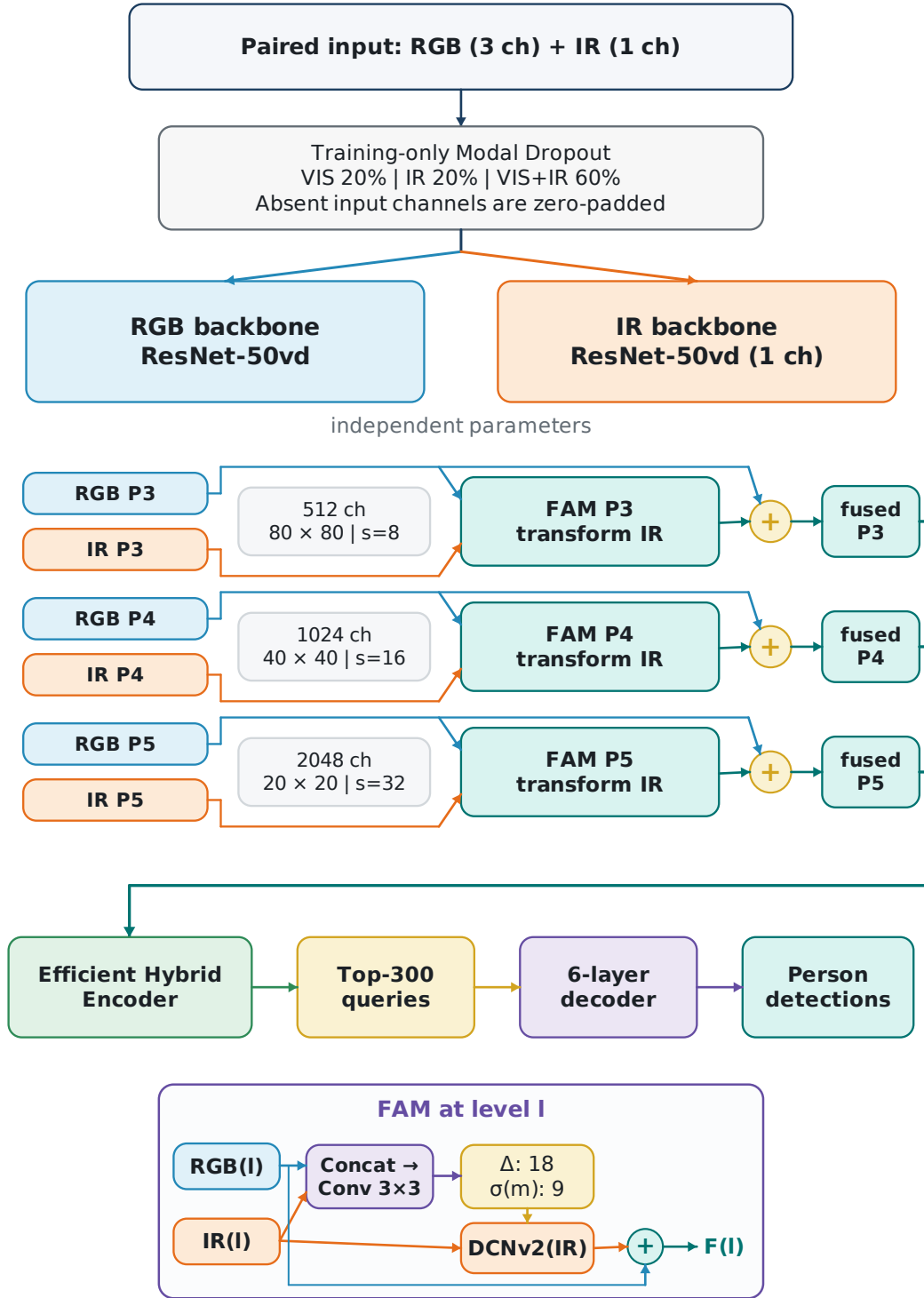


Figure 2.6: Dual-backbone RT-DETR and the inherited Feature Alignment Module. At each pyramid level, RGB and IR features jointly predict 18 offset components and nine modulation masks, DCNv2 then transforms the IR feature before additive fusion. In the final data path, training-only Modal Dropout zero-pads the missing input channels while preserving the four-channel interface.

2.7 Dataset organization and evaluation conditions

The primary fusion comparisons use synchronized WiSARD RGB-IR pairs. The effective split audit is summarized in Table 2.3. Earlier experiments also used unpaired visible or thermal folders to vary the training inventory. The final training inventory is restricted to paired samples. Modality evaluations follow two distinct protocols: paired missing-input comparisons retain the same frames and VIS annotations, whereas native-sensor evaluations use sensor-specific images and annotations.

2.7.1 WiSARD acquisitions and experimental roles

WiSARD is organized into acquisition sequences identified by location and by the corresponding VIS and IR stream numbers. These names identify different recorded sequences. The subsets used in this thesis and their experimental roles are summarized in Table 2.2.

Table 2.2: WiSARD paired acquisitions used in the study. Frame counts are reported after matching the VIS and IR streams.

Acquisition	Paired frames	Experimental role
FHL 0401/0402	896	Stage-A validation; restored to the training inventory in Stage B.
FHL 0405/0406	943	Training in both Stage A and Stage B.
Baker pair 1	2,180	Training in both Stage A and Stage B.
FHL 0134/0135	273	Pre-existing validation sequence used by the historical runs; retained as evidence of distribution shift rather than for current checkpoint selection.
MtErie, three pairs	708	Previously consulted internal development benchmark used for the main paired comparisons.
Carnation 0023/0024	739	Separately frozen stress test for severe cross-modal scale mismatch.
Carnation 0025/0026	1,313	Previously unused internal acquisition-level confirmation set.
FHL 0407/0408	1,035	Previously unused internal acquisition-level confirmation set.

The full paired training inventory contains 4,019 frames: 896 from FHL 0401/0402, 943 from FHL 0405/0406, and 2,180 from Baker pair 1. Stage A holds out the complete FHL 0401/0402 sequence, leaving 3,123 frames for optimization. Stage B restores that sequence and trains on all 4,019 pairs. FHL 0134/0135 is the much smaller validation sequence inherited from the earlier protocol and is not part of either training inventory.

The pre-existing 273-frame validation set contains one FHL session, 184 empty frames, and 148 boxes. Its median target area is approximately twelve times smaller than in training. It is informative as evidence of a strong distribution shift, but it is

Table 2.3: Effective composition of the historical full-data split after preprocessing and pairing filters. Median box area is expressed as a percentage of the evaluation image area.

Split	Sessions	Frames	Empty frames	Boxes/frame	Median box area
Train	3	4,019	17.82%	2.465	0.1640%
Validation	1	273	67.40%	0.542	0.0133%
Internal test	3	708	2.68%	2.500	0.0764%

not a representative selector for performance on the three MtErie sequences. This is distinct from the 896-frame complete-video validation introduced for Stage A.

MtErie denotes the three synchronized WiSARD VIS–IR sequences used for the main 708-frame benchmark. Their VIS annotations define the common reference coordinate system and ground truth for the fusion and paired missing-input comparisons. MtErie was consulted during architectural development and is therefore an *internal development benchmark*. Final paired comparisons remain useful, particularly after the protocol was frozen, but may be optimistic with respect to a new acquisition. Carnation 0023/0024 denotes a separate paired WiSARD acquisition used once as a predefined 739-frame acquisition-level stress case: its visible camera is substantially more zoomed-in than its thermal camera, creating a severe cross-modal scale mismatch. It is not a new validation set or an estimate of general domain generalization.

Carnation 0025/0026 and FHL 0407/0408 are distinct, larger paired acquisitions first evaluated in the later frozen confirmation and subsequently reused for post-hoc recall–FPPI characterization. Exact identifier intersection produces 1,313 and 1,035 pairs, respectively. They were previously unused and not inspected for experimental selection, but remain internal WiSARD data and not external holdouts.

In the paired modality evaluations, the three input conditions retain a common dataset and annotation set:

VIS+IR both streams from each synchronized pair are supplied;

VIS the same paired frame and VIS annotations are retained, but its thermal input is masked, so no IR signal is available to the model;

IR the same paired frame and VIS annotations are retained, but its visible input is masked, so no RGB signal is available to the model.

Under this protocol, VIS and IR are controlled missing-input evaluations rather than datasets in which the opposite sensor file is absent. RT-DETR represents the missing stream as zero-filled input while retaining its four-channel interface; the YOLOv10 evaluator uses an explicit modality mask and bypasses the excluded backbone and FAM path. The conditions therefore share images and annotations, but their internal execution is not architecturally identical.

Native-sensor evaluations instead retain the relevant sensor’s own image coordinates and annotations and may also use a different frame inventory, as distinguished in Section 3.9. They are reported separately from these paired missing-input comparisons.

2.8 Inherited experimental starting point

Previous work established DETR with Modal Dropout, the RT-DETR Base configuration, a fixed 2×2 tiling experiment related to slicing-aided small-object detection [1], and the initial CMX comparison. It subsequently introduced FAM on RT-DETR, Hybrid CMX, and Deformable DETR with nonlinear channel fusion and FAM. The resulting historical scores were based primarily on the earlier checkpoint policy and, for several RT-DETR variants, on individual executions.

Those results generated the central hypothesis that transforming IR features before fusion could be preferable to direct addition. They did not yet answer whether the advantage was consistent across runs, whether the learned offsets were geometrically meaningful, whether all feature levels behaved correctly, or whether the result transferred to another detector. These questions define the point at which the present thesis begins.

Chapter 3

Experimental Design and Methodology

3.1 Experimental design and evidence types

The study distinguishes architectural comparisons, implementation diagnostics, and post-selection robustness evaluations. Principal controlled comparisons use five paired training seeds and aggregate reporting. The historical locked RT-DETR comparison evaluates six configurations under full-data, ten-epoch training with final checkpoints. It establishes the reference contrast between Base additive fusion and standard FAM.

The validation-led comparisons follow a two-stage decision procedure. Stage A trains each configuration on 3,123 paired frames and selects its **best** checkpoint on the same 896-frame whole-video validation split. All five seeds complete their fixed budget: ten epochs for RT-DETR and RT-DETRv2, fifty for YOLO26. The final checkpoint is retained as a mandatory diagnostic. This design covers the spatial-resolution, offset-initialization, reliability, residual-alignment, mixed-consistency, box-guided, and detector-transfer comparisons without making a performance decision from an individual run.

Stage B is conditional on the aggregate Stage-A promotion rule. It restores all 4,019 training frames and trains each promoted candidate and its matched control, using only final checkpoints for confirmation on MtErie. RCRA is the only candidate that reaches this stage. The common decision procedure does not make different detector-specific optimization recipes identical, nor does it make validation mAP directly comparable with full-data MtErie mAP. Results are compared within the declared split, budget, checkpoint rule, and control.

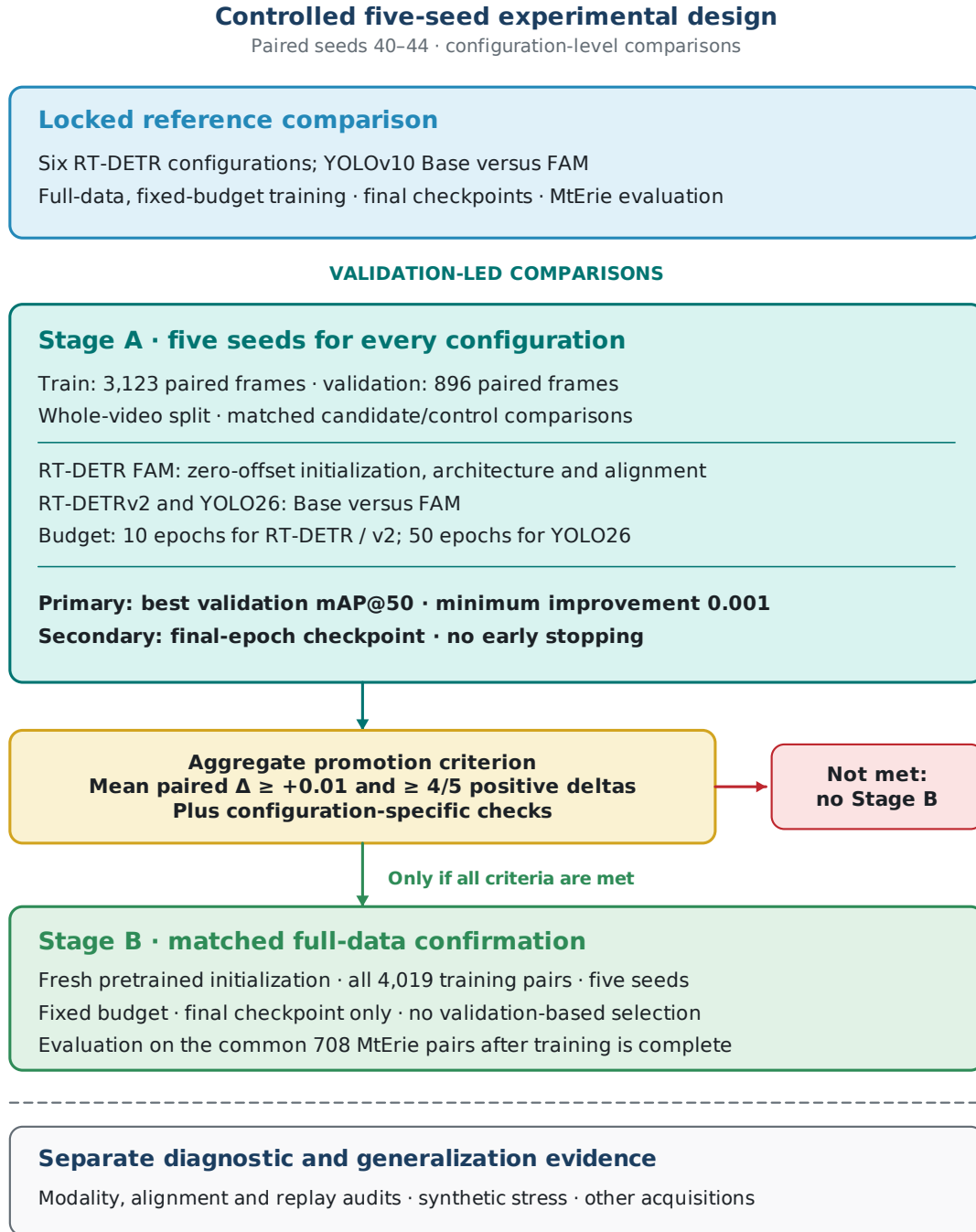


Figure 3.1: Overview of the experimental design. Stage-A configurations share the five-seed comparison and checkpoint-selection rules. Only a candidate satisfying both aggregate promotion conditions enters full-data Stage B. The locked comparison and post-selection diagnostics answer separate questions and are not pooled with development-validation estimates.

3.2 FAM implementation and regularization

3.2.1 Lazy registration bug

The first RT-DETR FAM implementation instantiated its alignment layers lazily during the first forward pass. The optimizer was created earlier and therefore did not contain those parameters. The offset predictor remained at zero, while the deformable-convolution weights remained fixed at their random initialization. This model did not implement an identity transformation and did not train a functional FAM, even though its fusion score was high in one run.

The extremely low IR-only score of that checkpoint is consistent with a weak or unusable thermal path, but the final metrics do not prove literal cancellation of the IR backbone. The run is useful as a debugging case, not as a competitive model.

3.2.2 Eager, Frozen, and Spatial Dropout variants

The implementation was corrected by registering all FAM layers in the model constructor before optimizer creation. This *Eager FAM* allowed the alignment parameters to train from the first optimization step. A Frozen variant then tested a fixed, eagerly registered transform: gradients could propagate through it to the IR backbone, but the FAM parameters themselves did not change. A further variant applied channel-wise spatial dropout to the transformed IR features immediately before addition.

These interventions did not isolate a single causal explanation for the historical Lazy score. In particular, the Lazy bug generated a fixed random filter with null offsets, not stochastic geometric jitter. Their role was to motivate a cleaner hypothesis about perturbing sampling geometry during training.

3.2.3 Stochastic Spatial Jitter

Stochastic Spatial Jitter adds zero-mean Gaussian noise to the predicted offsets only during training:

$$\Delta_{\text{train}} = \Delta_{\text{FAM}} + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2). \quad (3.1)$$

Unlike feature dropout, SSJ does not explicitly remove channels. It perturbs the positions sampled by DCNv2 around the correction currently predicted by the FAM. At inference, ε is absent. The intended regularization is therefore local uncertainty around a learned transformation, rather than a return to the original sensor displacement.

The relative effect of SSJ is assessed through the five-seed locked comparison in Section 4.1, which selects standard FAM without SSJ. Implementation diagnostics were used to identify potential failure modes and motivate the tested interventions. Performance conclusions, however, are based on the five-seed comparisons rather than on individual diagnostic runs.

3.3 Exploratory architectural extensions

3.3.1 Deformable DETR with channel fusion, FAM, and SSJ

The inherited Deformable DETR adaptation uses independent ResNet-50 RGB and IR backbones. The IR input stem is initialized by averaging the three RGB input channels. At the three native backbone levels, corresponding features are concatenated along the channel dimension and passed through a nonlinear fusion block:

$$F_{\text{fused}}^{(i)} = \text{ReLU}\left(\text{GN}_{32}\left(\text{Conv}_{1\times 1}([F_{\text{RGB}}^{(i)}, F_{\text{IR}}^{(i)}])\right)\right). \quad (3.2)$$

The model retains the standard four-scale Deformable DETR configuration. The three native ResNet levels (C3,C4,C5) are fused explicitly; a fourth level is then derived from the fused (C5) feature through a stride-2 projection. This is the usual multi-scale construction of Deformable DETR, rather than an additional backbone output. The pretrained configuration retains six deformable-attention encoder layers and six decoder layers.

This preserves the spatial dimensions required by the pretrained attention reference points. It replaced an unsuccessful spatial-concatenation attempt that doubled feature-map height, forced a reduction in pyramid levels because of memory use, and produced near-zero detection performance.

Preliminary work preceding this thesis had already integrated FAM before the channel-fusion block. The present study extends that configuration by adding SSJ under the same earlier experimental regime. Five runs are available for each configuration, but they precede the final RT-DETR head initialization, checkpoint rule, and validation policy. Their median and range are reported in Table 3.1 as exploratory transfer evidence.

The architecture is divided across Figures 3.2 and 3.3 to keep both the multimodal processing and the standard detector components legible. The first diagram follows the RGB–IR streams through the optional FAM, channel fusion, four-level projection, and multi-scale flattening. The second continues from that token sequence through the standard single-stage Deformable DETR encoder, learned queries, decoder, and prediction heads. SSJ is not shown because it is used only during training and does not modify the inference graph.

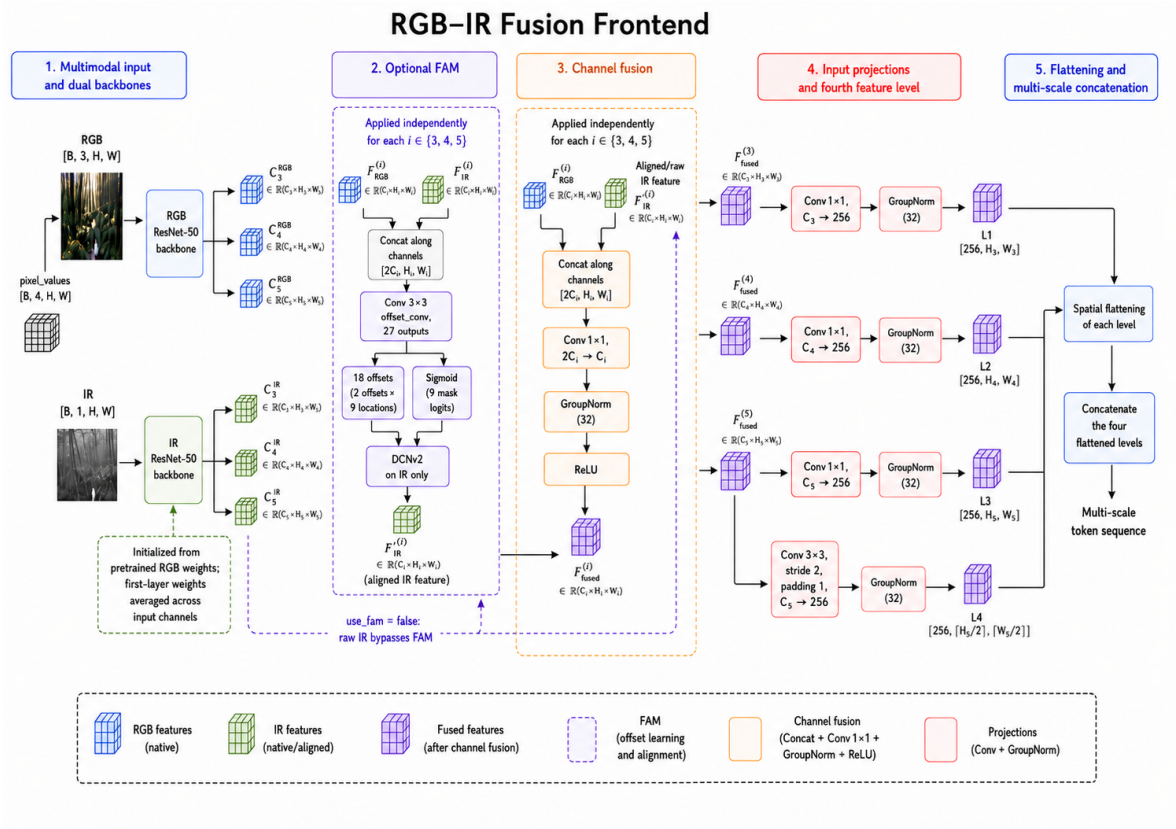


Figure 3.2: RGB-IR frontend of the dual-backbone Deformable DETR adaptation. FAM is applied independently to the three native feature levels when enabled, otherwise the raw IR features bypass it. Each RGB-IR pair is then fused along the channel dimension. The fourth encoder level is generated from the fused C5 feature and therefore does not receive a separate FAM.

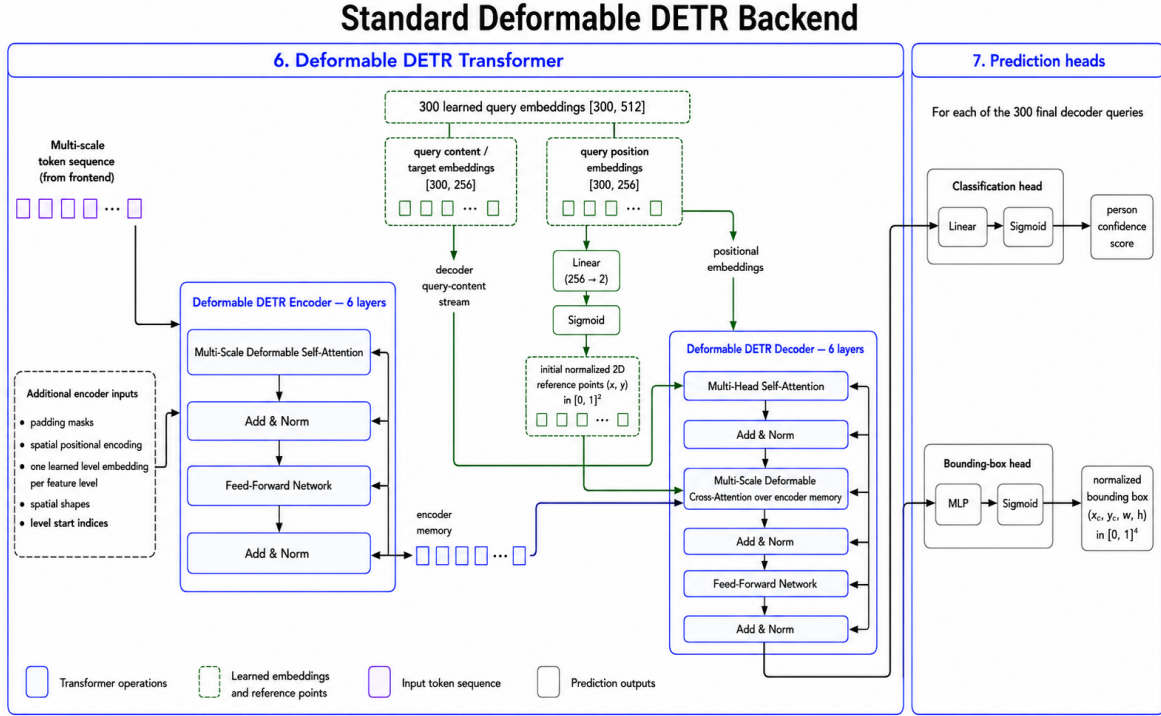


Figure 3.3: Standard single-stage Deformable DETR backend used after the multi-modal frontend. The four flattened feature levels enter the deformable-attention encoder. Learned query content, positional embeddings, and reference points then condition the decoder, whose final outputs feed the classification and bounding-box heads.

Table 3.1: Exploratory Deformable DETR results, reported as median [minimum–maximum] mAP@50 over five runs.

Configuration	VIS	IR	VIS+IR
Deformable DETR	0.157 [0.131–0.197]	0.115 [0.086–0.142]	0.249 [0.199–0.303]
+ FAM	0.184 [0.171–0.253]	0.131 [0.097–0.165]	0.294 [0.213–0.314]
+ FAM + SSJ	0.142 [0.115–0.206]	0.088 [0.073–0.109]	0.263 [0.170–0.350]

FAM has the highest observed median, but the ranges overlap substantially. SSJ does not reproduce the gain suggested by the preliminary RT-DETR run and has the widest VIS+IR range. A definitive cross-architecture ranking would require retraining these models under a protocol aligned with the final RT-DETR study.

3.3.2 Complete DINO implementation

The complete DINO implementation retains the dual-backbone and channel-fusion base of Deformable DETR: four feature levels, six encoder layers, and six decoder layers. It activates all three defining mechanisms. MQS uses the top encoder proposals as spatial anchors while drawing query content from an independent learned embedding. LFT keeps the refined reference passed to the next attention layer detached while preserving a graph-connected copy for the following prediction loss. CDN inserts noisy positive

and negative target queries during training and applies the required decoder attention mask.

The complete implementation was trained five times and produced the values in Table 3.2. Since even its maximum fusion result remained below the median of the historical Deformable DETR base, complete FAM and SSJ variants were not launched without a new diagnostic hypothesis.

Table 3.2: DINO base result under the previous protocol, reported as median [minimum–maximum] mAP@50 over five runs.

Configuration	VIS	IR	VIS+IR
Complete DINO base	0.0565 [0.0232–0.1091]	0.0881 [0.0671–0.1473]	0.1088 [0.0669–0.1635]

This is a negative result for the implemented configuration and training budget, while it does not show that DINO is intrinsically unsuitable for WiSARD.

3.4 Reproducibility investigation

3.4.1 Why single runs were insufficient

Two native-CUDA RT-DETR training runs with the same nominal seed and pretrained person head produced final test mAP@50 values of 0.4171 and 0.2934. Their first forward loss was identical, but their weight hashes diverged after the first backward pass. A difference of 0.1236 is large enough to reverse many of the historical architectural comparisons.

The investigation therefore recorded runtime fingerprints, initialization hashes, sample indices, modality-dropout decisions, input hashes, losses, and post-update weight hashes. Native execution diverged at the first optimizer step even when inputs and the initial loss matched. A deterministic interpolation substitute for the deformable-attention sampling path produced identical traces, metrics, and final checkpoint hashes in two complete RT-DETR runs. This establishes that strict reproduction is possible for that substitute path, while native RT-DETR CUDA is not fully deterministic.

3.4.2 Pretrained single-class head

The original COCO checkpoint has 80 classes, whereas the task has one class, *person*. Randomly initialized classification and denoising heads made the seed affect a substantial part of the initial model. The final setup transfers the COCO person row into all compatible RT-DETR classification heads and the denoising embedding. In the historical RT-DETR configuration examined in that audit, the transferred initialization made the initial model hash independent of the experiment seed. This is not a whole-model hash guarantee for every extension or detector: newly introduced FAM parameters can remain seed-dependent. Head transfer reduces, but does not eliminate, amplification of native CUDA perturbations.

In 20-step probes, the mean range of the loss across identical-seed native runs decreased from 4.0763 with random heads to 0.4393 with the transferred person head. The maximum range decreased from 20.8075 to 1.6424. The change is therefore a mitigation and a more controlled transfer-learning initialization, not a determinism guarantee.

3.4.3 DCNv2 deterministic limitation

The strict FAM probe stops at the first backward pass because the CUDA backward of `torchvision.ops.DeformConv2d` has no deterministic implementation in the studied environment. A numerically equivalent PyTorch fallback was too slow at the 512, 1024, and 2048 channel feature levels for full training. FAM must consequently be evaluated statistically under native CUDA rather than through a strictly deterministic full campaign.

3.4.4 Modal Dropout and evaluation leakage

The data pipeline originally sampled Modal Dropout during validation and test as well as training. This made evaluation inputs stochastic and inconsistent with the intended VIS, IR, and VIS+IR conditions. The loader was corrected and regression-tested so that dropout is active only in training while all final evaluations disable it.

3.4.5 Validation and checkpoint policy

The official FHL validation split contains 273 frames, of which 184 are empty, and only 148 boxes. Its objects are substantially smaller than those in the training sessions and MtErie. Near-zero scores on this split are evidence of a strong domain and scale mismatch, rather than a validator defect. It could not provide a stable checkpoint ranking for the detector studied here.

A first train-derived pilot instead reserved the temporal tail of every training video after a 30-frame embargo. Its seed-40 validation score was high, but the split retained near-adjacent content from the same continuous clips and its person distribution was dominated by Baker. The validation-selected epoch was also worse than epoch 10 on the already-consulted MtErie benchmark. The pilot was therefore retired before running seeds 41–44.

The definitive Stage-A split holds out the complete paired FHL 0401/0402 video. Training uses 3,123 frames from FHL 0405/0406 and Baker pair 1, while validation uses all 896 paired FHL 0401/0402 frames. This removes direct same-video temporal adjacency. It remains an internal split because the two FHL videos belong to the same acquisition campaign.

Every run completes its declared budget: validation does not implement early stopping. It updates `best` when validation mAP@50 exceeds the previously selected value by more than 0.001, while the final checkpoint stores epoch 10 for RT-DETR/RT-DETRv2 or epoch 50 for YOLO26. Consequently, `best` denotes the checkpoint selected using this tolerance and may differ from the epoch with the numerically highest validation score when the improvement is smaller than 0.001.

3.5 Historical locked RT-DETR protocol

The first locked RT-DETR comparison uses six independently trained configurations:

1. Base (direct additive fusion);
2. standard FAM with DCNv2;
3. FAM with IR feature dropout probability 0.4;
4. FAM with SSJ standard deviation 0.5;
5. an identity-initialized DCNv2 FAM;
6. a direct bilinear Grid Sample warp.

IR feature dropout is active only during training. Immediately after FAM has transformed the IR feature map and before additive fusion, `Dropout2d` independently zeros complete IR feature channels with probability 0.4 at each pyramid level; the remaining channels are rescaled in the usual way. It does not remove complete frames or disable the IR sensor, and it is inactive during all evaluations.

The identity DCNv2 ablation retains the standard FAM structure but initializes the central kernel as a channel-wise identity with weight two, exactly compensating the initial mask value 0.5. The Grid Sample variant predicts one two-dimensional displacement per spatial cell and applies a bilinear warp without a learned post-sampling convolution. It is exactly the identity at zero offset through a correction that subtracts numerical sampling of the base grid.

Table 3.3: Historical locked RT-DETR protocol used by all six configurations in the original multi-seed comparison.

Component	Fixed choice
Training seeds	Paired seeds 40–44; five runs per configuration
Execution	Native CUDA; one isolated Python process per seed
Training horizon	10 fixed epochs
Checkpoint	Final <code>latest</code> ; no test-based or validation-based selection
Initialization	Pretrained RT-DETR R50vd and transferred COCO person heads
Transformer depth	One AIFI encoder layer and six decoder layers
Modal Dropout	VIS/IR/fusion probabilities 0.2/0.2/0.6, training only
Validation	Not used during training or for checkpoint selection
Primary benchmark	MtErie internal development benchmark
Primary metric	mAP@50
Evaluation	VIS, IR, and VIS+IR with a fixed evaluation seed

The same seed index is paired across configurations to reduce the noise in architectural contrasts, while it does not make native executions deterministic. This campaign

remains valid evidence about the six configurations under its frozen full-data, epoch-10 policy. It is not used as the direct comparator for new validation-selected candidates trained at a later code revision.

3.6 Validation-led two-stage protocol

All validation-led alignment and detector-transfer comparisons separate architectural development from full-data confirmation. The two stages are summarized in Table 3.4.

Table 3.4: Common five-seed Stage-A/Stage-B decision protocol. Training budgets remain detector-specific.

Component	Stage A: development	Stage B: confirmation
Optimization data	3,123 paired frames: FHL 0405/0406 and Baker pair 1	all 4,019 paired training frames
Validation	complete FHL 0401/0402 video, 896 pairs	none; all frames restored to optimization
Horizon	10 epochs RT-DETR/RT-DETRv2; 50 YOLO26; no early stopping	same family-specific budget; no early stopping
Primary checkpoint	validation-selected best ; latest diagnostic	final latest / last.pt only
Role of MtErie	disabled during training and candidate selection	evaluated only after all matched runs exist
Comparison	alignment candidate versus FAM, or FAM versus Base within a detector	promoted candidate versus freshly trained matched control

All Stage-A comparisons use paired seeds 40–44 and validation after every epoch. The default input is 640×640 , except for the declared 800×800 ablation. Physical batch size is four unless a memory-matched control is specified. Within a comparison, the control and candidate share their base optimizer and training-only Modal Dropout schedule, with any additional parameter groups or objectives explicitly declared. Advancement requires a mean paired validation gain of at least 0.01 and positive differences in at least four of five seeds. These engineering thresholds are frozen before each five-seed decision. All five seeds are retained, including those for which the candidate underperforms the control. Stage-A comparisons and promotion decisions use the validation-selected **best** checkpoints. Final-epoch checkpoints are reported only as secondary diagnostics and are not used to revise the decision criterion after the results are observed.

Stage B restores the held-out video and retrain both the selected candidate and its control from the pretrained weights, without resuming Stage-A best checkpoints. For the promoted RT-DETR RCRA comparison, this uses the full-data, ten-epoch **latest** recipe. Confirmation requires both a mean paired MtErie gain of at least 0.01 and at least four wins in five seeds. The historical FAM mean remains a useful reference for the earlier implementation, but it cannot replace the matched Stage-B FAM control.

Although the FAM architecture is unchanged, Stage B retrain FAM using the same current training and evaluation pipeline.

3.7 Alignment-extension hypotheses

The Stage-A campaign tests several separable hypotheses rather than introducing all changes simultaneously. A P2 variant adds a stride-4 feature level to increase spatial detail for tiny targets. A resolution control instead keeps P3–P5 unchanged and increases the input from 640×640 to 800×800 . Two post-FAM reliability gates test whether RGB and thermal contributions should be modulated after FAM alignment and before addition, first at the shared detector learning rate and then with a dedicated ten-times-larger gate rate.

The principal new candidate is Reliability-Conditioned Residual Alignment (RCRA). Related work already combines feature alignment and reliability-aware fusion for weakly aligned multispectral detection [17, 5], while uncertainty-conditioned modulation has also been studied for unaligned RGB–thermal fusion [13].

At pyramid level l , let $I_{\text{raw}}^{(l)}$ be the IR-backbone feature, $I_{\text{aligned}}^{(l)}$ the FAM output, and $R^{(l)} = I_{\text{aligned}}^{(l)} - I_{\text{raw}}^{(l)}$ the alignment residual. RCRA predicts a spatial coefficient $\alpha^{(l)} \in (0, 2)$:

$$\begin{aligned}\alpha^{(l)} &= 2 \sigma(g^{(l)}(d^{(l)})), \\ I_{\text{selected}}^{(l)} &= I_{\text{aligned}}^{(l)} + (\alpha^{(l)} - 1) R^{(l)}, \\ F_{\text{fused}}^{(l)} &= F_{\text{RGB}}^{(l)} + I_{\text{selected}}^{(l)}.\end{aligned}\tag{3.3}$$

Thus $\alpha = 0$ recovers raw IR, $\alpha = 1$ retains the ordinary FAM output, and $\alpha > 1$ amplifies its correction. Each level uses local means, RMS magnitudes, RGB–IR agreement, alignment-residual magnitude, agreement gain, and modality-presence indicators as descriptors. They enter a 3×3 convolution with 16 hidden channels and a 1×1 output. The three predictors add 5,283 parameters. Their final convolution is zero-initialized, making RCRA exactly equivalent to standard FAM at initialization, and their dedicated learning rate is 2×10^{-4} while all existing parameters retain 2×10^{-5} .

A frozen attribution control replaces the descriptors and spatial predictors with one input-independent coefficient per pyramid level, for only three new parameters. It uses the same residual equation, neutral initialization, and dedicated learning rate. This tests whether any RCRA gain can be reproduced by simple global per-level calibration rather than conditional local selection.

3.7.1 Zero-offset initialization

This ablation tests whether standard FAM benefits from starting with null sampling offsets. The complete pretrained RT-DETR v1 model is first built, including Hugging Face initialization and transfer of the COCO person head. Only then are the weights and biases of predictor rows 0:18 set to zero at P3, P4, and P5. Mask-predictor rows 18:27, DCNv2 filters, backbones, and detection heads retain their control initialization. The masks are therefore not forced to 0.5, while all offset parameters remain trainable.

Unlike the Identity DCNv2 ablation, this intervention changes only the initial offsets and does not make the full FAM operator an identity.

The five candidate runs use the common Stage-A split, seeds 40–44, ten complete epochs, AdamW at 2×10^{-5} , batch size four, and native-coordinate Modal Dropout at 20/20/60. The five standard-FAM Stage-A controls are reused. Before each run, the reconstructed standard-model hash must match the paired control’s recorded initialization; the intervention must preserve all unaffected parameters, parameter identities, and random-number-generator state.

Each serialized **best** checkpoint is strictly reloaded in a separate process and evaluated on the same 896 validation pairs. The allowed absolute live-replay mAP@50 difference is 0.0002; the live checkpoint-selection score remains primary. Only after all five training runs and replays are complete is the common mean-gain and four-win promotion rule applied. Epoch-10 **latest** scores remain diagnostic. This comparison does not use the historical full-data test mean of 0.3780 as its control.

3.7.2 Mixed native/paired consistency training

Mixed consistency preserves the native-coordinate Modal Dropout supervision and adds an online teacher–student objective. The detector first processes the paired VIS+IR input in evaluation mode and without gradient tracking, producing class probabilities and bounding boxes that serve as pseudo-targets. A student forward pass of the same model then receives either masked RGB or masked IR in the VIS coordinate system. The consistency objective combines classification, box, and generalized-IoU terms with weights 2, 5, and 2. Teacher predictions use confidence threshold 0.2 and at most twenty queries. The objective is inactive during the first epoch and reaches full weight through a two-epoch warm-up. From its activation onward, training requires an additional teacher forward pass and student forward/backward pass.

This is a five-seed comparison against standard FAM under the common Stage-A fusion criterion. Had the candidate passed this necessary criterion, promotion would also have required a mean paired masked-IR/VIS-ground-truth gain of at least 0.03 and a mean native-IR change of at least -0.03 . Because the primary fusion criterion failed, these modality-specific evaluations were not performed for the new five-seed campaign, and their outcomes are not inferred from the fusion scores. Additional student-view random draws can also alter subsequent Modal Dropout realizations: paired seeds do not guarantee identical tensors at every update, and the experiment evaluates the complete training intervention rather than an isolated loss on fixed inputs.

3.7.3 Supervised box-guided alignment

Box-Guided Common-Offset P3 predicts an additional two-dimensional displacement shared by the deformable kernel points at P3. Matched VIS/IR box centres supply weak training supervision; P4 and P5 retain standard FAM. Existing detector parameters retain learning rate 2×10^{-5} , while the guide uses 10^{-4} . The auxiliary Smooth L1 objective has weight 0.2, a two-epoch warm-up, and a transition scale of 0.25 feature cells.

Its detection effect is evaluated against the same five standard-FAM controls. Geometric target fit and inference-time contribution are distinct questions: an audit of

the learned field against zero and constant-vector controls, and a validation intervention that zeros only the guide output, would be needed for a mechanism claim. These audits are not part of the completed five-seed fusion results and are not assumed from detection improvement or parameter updates.

3.8 Detector-transfer comparisons under Stage A

RT-DETRv2 uses the official COCO-trained detector checkpoint `PekingU/rtdetr_v2_r50vd` with independent RGB and IR backbones. Its Base and FAM branches share the Stage-A split, pretrained initialization contract, AdamW learning rate 2×10^{-5} , batch four, ten-epoch budget, and native training-only Modal Dropout. The controlled contrast isolates the effect of FAM within RT-DETRv2; it neither reproduces the complete COCO training procedure nor constitutes a controlled detector-only comparison with RT-DETR v1.

YOLO26s likewise compares two RGB-IR backbones with P3/P4/P5 Base additive fusion against the same model with FAM applied before addition. Both branches complete fifty epochs for every seed. The shared recipe uses AdamW with initial learning rate 10^{-3} , nominal batch size 16 with physical batch four, three warm-up epochs, and `warmup_bias_lr=0.0`. The checkpoint selector applies the same minimum improvement of 0.001. A validation replay checks the serialized best against its live selection metric with absolute tolerance 0.003; the replay is an integrity check, not an alternative primary score.

For both detector families, promotion requires a mean paired FAM-Base gain of at least 0.01 and four positive differences among the five seeds. There is no separate single-run performance gate. Sharing this decision procedure does not imply identical optimization budgets or evaluators across families and their absolute validation scores are not a controlled cross-architecture ranking.

3.9 Coordinate contracts for modality evaluation

Two distinct modality evaluations are required. A *native-sensor* evaluation presents VIS or IR on its own native canvas with the corresponding native annotations. It describes standalone deployment on each sensor, but VIS, IR, and fusion may then contain different frame populations, coordinate systems, and ground-truth boxes. Their AP values cannot by themselves establish same-frame sensor complementarity.

The post-selection *paired intervention* instead fixes the population to the 708 common MtErie pairs and keeps the same 1,770 VIS boxes in every condition. VIS-only zeros the adapted IR channel, while paired masked-IR zeros RGB after IR has been adapted to the VIS canvas. The existing `adapt_ir2rgb` operation resizes and pads IR to the visible canvas but does not implement calibrated geometric registration. Since FAM predicts its offsets jointly from RGB and IR and transforms IR toward RGB, paired masked-IR also removes the reference used by this asymmetric alignment path.

Historical RT-DETR Modal Dropout uses the native coordinate contract for an IR-only training draw: native IR image, native IR canvas, and native IR annotations. A separately frozen seed-40 ablation changes only those draws to the paired

VIS-coordinate convention. The experiment tests the effect of this coordinate mismatch; it is not a new five-seed performance campaign. Its result cannot be presented as interchangeable with native IR evaluation.

3.10 Acquisition confirmation and geometric-stress protocols

Acquisition confirmation and geometric stress evaluate frozen checkpoints without further training or model selection. Their populations and interventions have different evidential roles and are not pooled with the five-seed MtErie or Stage-A validation estimates.

3.10.1 Previously unused internal acquisitions

Two paired WiSARD acquisitions absent from all preserved training, validation, and test campaigns were reserved before inference: Carnation 0025/0026 and FHL 0407/0408. Their common paired populations contain 1,313 and 1,035 frames, respectively. The campaign evaluates five historical RT-DETR Base checkpoints, five FAM checkpoints, and the five matched Stage-B FAM/RCRA pairs. Across the two acquisitions and declared modality contrasts this yields 50 checkpoint–acquisition evaluations. These data are an internal confirmation set: they reduce direct reuse of MtErie but remain WiSARD acquisitions, not an external or blind population-level holdout.

3.10.2 Controlled synthetic geometric stress

The same frozen checkpoints are evaluated after transforming only the IR stream. The intervention uses translations of 8, 16, and 32 input pixels in four directions and isotropic scales 0.9 and 1.1, always relative to each checkpoint’s unmodified identity condition. Translation directions are first averaged within each checkpoint so that the trained checkpoint, rather than a direction or frame, remains the experimental unit. The primary response is the signed relative drop in mAP@50: positive values indicate degradation, whereas negative values indicate that the perturbation improved performance. These improvements are retained rather than clipped to zero. No observed transformation is adopted as preprocessing or used to reselect a model.

3.11 Statistical reporting

The post-hoc recall–FPPI analysis in Section 4.5.3 reuses the historical Base and FAM epoch-10 `latest` checkpoints, seeds 40–44, on MtErie, Carnation 0025/0026, and FHL 0407/0408. It summarizes recall at common false-positive budgets within each evaluation set, without changing any training, checkpoint-selection, or confirmation rule. These descriptive curves do not select confidence thresholds for deployment.

Five-seed configuration summaries report means and sample standard deviations; architectural contrasts report within-seed mean differences, 95% Student *t*-intervals, and paired wins. The compact artifacts retain per-seed values, medians, and ranges for

the Stage-A comparisons. These supporting records are not individual-run performance claims. With five seeds, the interval on a paired mean is $\bar{d} \pm 2.776445 s_d / \sqrt{5}$. Paired t -tests and exact Wilcoxon signed-rank tests in the historical comparison are exploratory summaries, not requirements applied to every Stage-A table.

With only five pairs, intervals are imprecise, p-values are discrete or sensitive to assumptions, and no correction is applied for the many possible comparisons. Conclusions therefore prioritize effect size, sign consistency, and agreement across metrics over a binary significance threshold. In the validation-led campaign, predeclared mean-gain and paired-win rules govern promotion and confirmation. Confidence intervals communicate uncertainty but do not retroactively change those engineering decisions.

The experimental unit for architectural comparison is the independently trained run/checkpoint. Frames, bounding boxes, samples used in a diagnostic, feature-map cells, and the nine DCNv2 kernel points are nested observations, not independent replicates. Diagnostics first aggregate within a checkpoint and only then compare the five checkpoints.

Chapter 4

Experimental Evaluation

4.1 Historical locked RT-DETR multi-seed study

4.1.1 Raw and aggregate VIS+IR results

All thirty training runs completed the fixed ten-epoch protocol and produced a final `latest` checkpoint. Table 4.1 reports the primary VIS+IR mAP@50 result for every experimental unit.

Table 4.1: Final RT-DETR VIS+IR mAP@50 by paired training seed.

Seed	Base	FAM	FAM + IR Dropout	FAM + SSJ	Identity DCNv2	Grid Sample
40	0.2566	0.3783	0.4087	0.3663	0.4336	0.4050
41	0.4029	0.4335	0.3234	0.3734	0.2113	0.3483
42	0.2958	0.3129	0.4452	0.3528	0.2926	0.4083
43	0.3476	0.3964	0.3986	0.4030	0.2860	0.3326
44	0.2372	0.3690	0.3598	0.3792	0.4162	0.3564

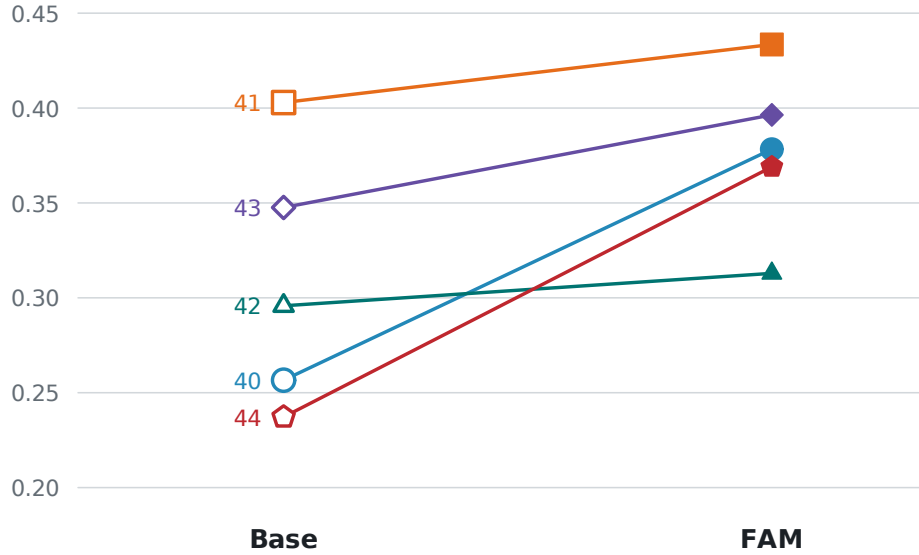
The aggregate distributions in Table 4.2 show why a single run is insufficient. Identity DCNv2 ranges from 0.2113 to 0.4336, and the highest value of several individual seeds belongs to a configuration that is not best on average or consistently paired with the reference.

Table 4.2: Aggregate final RT-DETR VIS+IR mAP@50. Standard deviations are sample standard deviations across five checkpoints.

Configuration	Mean	Median	SD	Min-max	95% CI of mean
Base	0.3080	0.2958	0.0677	0.2372–0.4029	0.2239–0.3921
FAM	0.3780	0.3783	0.0440	0.3129–0.4335	0.3234–0.4326
FAM + IR Dropout	0.3871	0.3986	0.0469	0.3234–0.4452	0.3289–0.4453
FAM + SSJ	0.3749	0.3734	0.0185	0.3528–0.4030	0.3519–0.3980
Identity DCNv2	0.3280	0.2926	0.0943	0.2113–0.4336	0.2109–0.4451
Grid Sample	0.3701	0.3564	0.0345	0.3326–0.4083	0.3273–0.4129

FAM increases mean mAP@50 over Base RT-DETR by 0.0700 and wins in all five paired seeds. Its mean COCO mAP also increases from 0.1059 ± 0.0243 to 0.1351 ± 0.0096 , mAP@75 from 0.0474 ± 0.0128 to 0.0692 ± 0.0043 , and mAR@100 from 0.2474 ± 0.0302 to 0.2798 ± 0.0141 . The direction is therefore not limited to the primary IoU threshold.

A VIS+IR mAP@50: FAM versus Base



B Delta mAP@50 relative to standard FAM

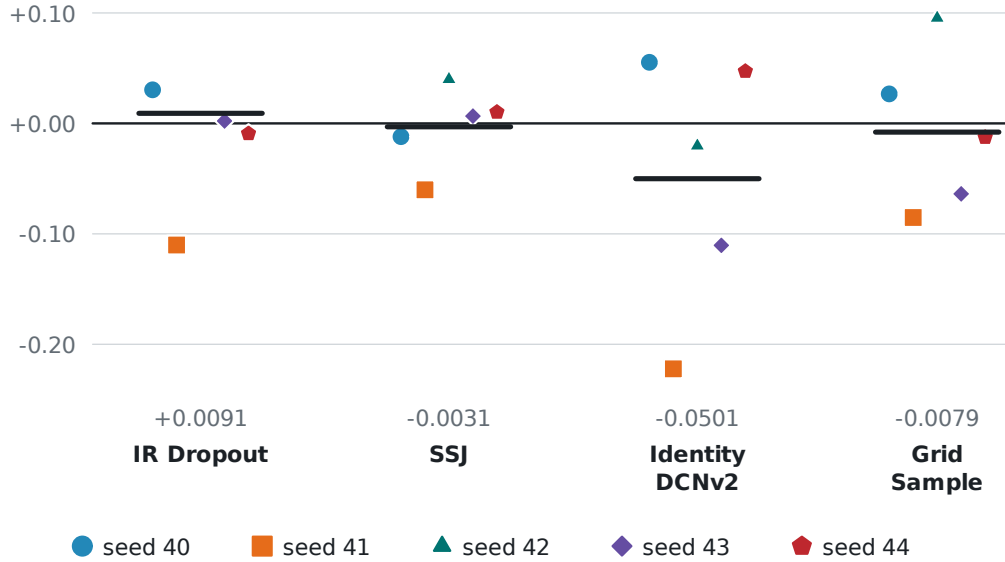


Figure 4.1: Final RT-DETR VIS+IR mAP@50 from the `latest` checkpoints. Panel A connects Base and FAM runs initialized with the same training seed; FAM wins all five pairs. Panel B reports, for each seed, the difference between a variant and standard FAM, with black bars denoting paired means. Each seed-specific run/checkpoint is an experimental unit; points are not repeated measurements of one checkpoint.

4.1.2 Paired contrasts and model selection

Table 4.3 compares FAM with Base RT-DETR and compares each FAM regularization or ablation with the standard FAM. With five pairs, the tests and intervals are exploratory and are not used as a binary decision rule.

Table 4.3: Paired VIS+IR mAP@50 contrasts. A win means that the first configuration exceeds its reference for the same training seed.

Contrast	Mean delta	SD delta	95% CI	Wins	Paired t -test p / Wilcoxon p
FAM – Base	+0.0700	0.0531	[+0.0040,+0.1359]	5/5	0.0421 / 0.0625
FAM + IR Dropout – FAM	+0.0091	0.0869	[-0.0988,+0.1171]	3/5	0.8260 / 0.8125
FAM + SSJ – FAM	-0.0031	0.0369	[-0.0488,+0.0427]	3/5	0.8617 / 1.0000
Identity DCNv2 – FAM	-0.0501	0.1170	[-0.1953,+0.0952]	2/5	0.3928 / 0.6250
Grid Sample – FAM	-0.0079	0.0724	[-0.0979,+0.0820]	2/5	0.8192 / 1.0000

IR Dropout has the highest raw mean, but its mean improvement over FAM is only 0.0091, changes sign across seeds, and has a wide interval. It is not selected as a reliable improvement. SSJ has the smallest descriptive standard deviation in the six-configuration campaign but does not increase mean performance. Five runs are insufficient to infer that it reduces population variance.

Identity initialization does not solve final instability. Grid Sample is close to standard FAM, but its parameterization is not equivalent to DCNv2: it predicts one displacement per cell and performs no post-sampling filtering or channel mixing. Its result supports the relevance of a geometric component, but does not causally assign all FAM improvement to physical alignment.

The main model for subsequent analysis is therefore standard FAM with DCNv2, without SSJ and without IR feature dropout. Standard FAM is the only final configuration that exceeds Base in all five paired seeds. The variants are compared directly with FAM because those contrasts isolate the effect of the additional intervention: IR feature dropout does not provide a consistent increment over FAM despite its higher raw mean.

4.1.3 Historical native-sensor modality evaluation

Each checkpoint was also evaluated with Modal Dropout disabled using the historical native-sensor protocol. VIS and IR use their respective native images and annotations, whereas VIS+IR uses paired visible-coordinate samples. The aggregate mAP@50 results are reported in Table 4.4. Comparisons between configurations within one column remain meaningful, but the three columns do not describe the same frame population or ground truth.

FAM improves Base RT-DETR within the historical VIS population in 5/5 seeds, with a mean delta of 0.0899, and within the native IR population in 4/5, with a mean delta of 0.0462. These are valid cross-model contrasts because each column holds its evaluator fixed. Native VIS, native IR, and paired fusion use different inventories and annotations, so their numerical ordering does not support a direct cross-modality ranking or a claim of same-frame complementarity.

In RT-DETR, a missing input is zero-filled while both backbones and the FAM path

Table 4.4: Historical native-sensor RT-DETR mAP@50 by input condition, mean $\pm$ sample standard deviation across five checkpoints. Columns use their original evaluation populations and should not be ranked against one another as a same-frame modality ablation.

Configuration	VIS	IR	VIS+IR
Base	0.1543 $\pm$ 0.0331	0.1567 $\pm$ 0.0464	0.3081 $\pm$ 0.0677
FAM	0.2442 $\pm$ 0.0307	0.2028 $\pm$ 0.0635	0.3780 $\pm$ 0.0439
FAM + IR Dropout	0.2368 $\pm$ 0.0428	0.2079 $\pm$ 0.0267	0.3870 $\pm$ 0.0466
FAM + SSJ	0.2312 $\pm$ 0.0163	0.1794 $\pm$ 0.0416	0.3751 $\pm$ 0.0185
Identity DCNv2	0.2150 $\pm$ 0.0561	0.1768 $\pm$ 0.0686	0.3281 $\pm$ 0.0942
Grid Sample	0.2614 $\pm$ 0.0267	0.1895 $\pm$ 0.0211	0.3701 $\pm$ 0.0345

remain in the model. A unimodal result therefore measures robustness of the complete learned system under a specified coordinate contract. The later post-selection audit in Section 4.3 provides the required same-population comparison.

4.2 Validation-led RT-DETR extension campaign

4.2.1 Stage-A checkpoint reference

The standard-FAM reference uses the complete-video split and checkpoint rule in Section 3.6 for seeds 40–44. All runs completed ten epochs. The selected epochs were 1, 4, 6, 1, and 1, showing that the decline after early epochs was repeatable rather than a logging anomaly.

Table 4.5: Stage-A FAM checkpoint audit. Validation uses the complete FHL 0401/0402 video. MtErie is evaluated only after all five runs are complete.

Seed	Best epoch	Val best	Val latest	MtErie best	MtErie latest
40	1	0.1521	0.0397	0.3490	0.2132
41	4	0.1424	0.1030	0.3354	0.2887
42	6	0.1655	0.0946	0.3077	0.2740
43	1	0.1939	0.0790	0.3994	0.2876
44	1	0.1689	0.1037	0.4036	0.2811

Across seeds, validation-selected **best** obtains 0.3590 ± 0.0416 MtErie mAP@50, whereas epoch-10 **latest** obtains 0.2689 ± 0.0317 . The paired gain is 0.0901 ± 0.0466 , positive in 5/5 seeds, with a 95% t -interval of $[0.0323, 0.1479]$. This supports the selector for Stage-A development. It does not show that the new split improves absolute performance over historical FAM: the Stage-A model optimizes only 3,123 frames, while the historical 0.3780 ± 0.0440 result optimizes all 4,019 and uses epoch 10.

4.2.2 Five-seed Stage-A candidate comparison

Table 4.6 summarizes the ablations performed under the common complete-video validation protocol. Standard FAM is the paired reference for every delta, including zero-offset initialization, mixed consistency, and box-guided alignment. All configurations complete ten epochs for every seed and selection is based on the validation best.

Table 4.6: Stage-A best-validation mAP@50 across five seeds. A promotion requires mean delta at least +0.01 and at least four paired wins.

Candidate	Mean $\pm$ SD	Mean delta	Wins	95% CI of delta	Decision
FAM reference	0.1646 $\pm$ 0.0196	–	–	–	reference
FAM, zero-offset init.	0.1521 $\pm$ 0.0167	-0.0125	1/5	[-0.0444,+0.0195]	reject
FAM + P2	0.0991 $\pm$ 0.0207	-0.0655	0/5	[-0.1032,-0.0278]	reject
FAM at 800 $\times$ 800	0.1447 $\pm$ 0.0151	-0.0199	1/5	[-0.0484,+0.0087]	reject
Post-FAM reliability gate	0.1662 $\pm$ 0.0087	+0.0016	3/5	[-0.0242,+0.0274]	reject
Gate with 10 $\times$ LR	0.1546 $\pm$ 0.0200	-0.0099	3/5	[-0.0512,+0.0314]	reject
Scalar residual control	0.1603 $\pm$ 0.0127	-0.0043	2/5	[-0.0284,+0.0199]	reject
Mixed consistency	0.1094 $\pm$ 0.0328	-0.0551	0/5	[-0.0980,-0.0122]	reject
Box-Guided P3	0.1723 $\pm$ 0.0233	+0.0077	2/5	[-0.0327,+0.0481]	reject
RCRA	0.1825 $\pm$ 0.0257	+0.0179	4/5	[-0.0350,+0.0709]	promote

P2 loses to FAM in every seed. A matched seed-40 FAM run using P2’s smaller micro-batch and gradient accumulation obtains 0.1593, compared with 0.1521 for direct-batch FAM and 0.1014 for P2, ruling out the changed micro-batch as the main explanation under the declared control. Increasing global input resolution also fails its rule. The shared-LR reliability gate remains within approximately 0.002 of neutral weight, while the higher gate learning rate activates the mechanism without improving detection reliably.

RCRA is the only candidate to pass both frozen Stage-A requirements. Its paired gain is positive in four seeds, although the confidence interval includes both signs. Its mechanism audit shows mean fusion α values of 0.9469, 0.9079, and 0.9887 at P3–P5: the model mainly suppresses part of the FAM residual at P3 and P4 while leaving P5 close to ordinary FAM. Removing either modality changes the mean coefficient by at least 0.01 in all five seeds. The gate is therefore active and condition-dependent.

The three-parameter scalar control also moves away from identity, with mean coefficients 0.9576, 0.9788, and 0.9950, but fails to improve FAM. This rules out an inactive control and supports carrying the conditional RCRA candidate to Stage B. It does not establish statistical superiority over the scalar model, which RCRA beats in only 3/5 direct pairs.

4.2.3 Zero-offset initialization

Initializing only FAM’s offset predictors to zero yields best-validation mAP@50 of 0.1521 $\pm$ 0.0167, against 0.1646 $\pm$ 0.0196 for the paired standard-FAM controls. The mean paired difference is -0.0125 ± 0.0257 , with a 95% interval of [-0.0444, +0.0195] and only one positive pair. Neither promotion condition is met. The interval includes

zero, so the observed lower mean does not establish a universal disadvantage of zero initialization. No Stage-B training or MtErie evaluation is performed for this candidate.

At epoch 10, the direction of the mean contrast reverses: zero-offset initialization obtains 0.0958 ± 0.0245 , compared with 0.0840 ± 0.0267 for standard FAM. The paired difference is $+0.0118 \pm 0.0181$, with three positive pairs and a 95% interval of $[-0.0106, +0.0342]$. These final-epoch results are included in Table 4.7. Every candidate finishes below its own validation-selected best. The smaller mean best-to-final decline is descriptive evidence and does not establish improved training stability or generalization; it cannot replace the primary checkpoint criterion.

All five serialized-best replays exactly reproduce the live selection score. The saved audits confirm that only the intended offset rows were reset and that unaffected parameters and random-number-generator state were preserved. Inspection of all thirty offset predictors across five seeds, two checkpoint types, and three pyramid levels finds finite parameters, with all eighteen weight rows nonzero and mutually distinct and all eighteen biases nonzero.

4.2.4 Consistency and box-guided alignment

Mixed consistency reduces best-validation mAP@50 in all five pairs. Its mean is 0.1094 ± 0.0328 , compared with 0.1646 ± 0.0196 for standard FAM. The mean paired difference is -0.0551 , with a 95% interval entirely below zero. For four of the five seeds, the validation-selected checkpoint comes from the first epoch, before the consistency loss is activated. Only seed 41 selects its best checkpoint after consistency training has begun.

Box-Guided P3 obtains 0.1723 ± 0.0233 , a mean gain of 0.0077. The gain is positive in only two pairs, its median is -0.0037 , and the interval includes zero. It meets neither the required mean improvement nor the four-win condition. The five standard-FAM reference runs are shared with the other Stage-A alignment comparisons rather than retrained for these two candidates.

Table 4.7: Final-epoch validation diagnostics for zero-offset initialization, consistency, and box-guided alignment. Values are epoch-10 `latest` mAP@50 over five seeds; these scores do not replace the best-checkpoint promotion criterion.

Configuration	Mean $\pm$ SD	Mean paired delta	Wins	95% CI of delta
FAM reference	0.0840 ± 0.0267	—	—	—
FAM, zero-offset init.	0.0958 ± 0.0245	+0.0118	3/5	$[-0.0106, +0.0342]$
Mixed consistency	0.0291 ± 0.0154	-0.0549	0/5	$[-0.0881, -0.0217]$
Box-Guided P3	0.0845 ± 0.0263	+0.0005	3/5	$[-0.0235, +0.0246]$

The final-epoch diagnostic preserves the mixed-consistency regression and shows essentially no mean box-guided advantage. Both candidates fail the necessary fusion criterion and do not enter Stage B. Native-IR and paired masked-IR evaluations for mixed consistency, and geometric and active-versus-zero guide audits for box-guided alignment, were not carried out across these five-seed configurations. No aggregate result is claimed for those unmeasured conditions. In particular, parameter updates

or an auxiliary training loss are not evidence that the guide improves registration or contributes directly to detection at inference.

4.2.5 Matched full-data Stage B

Stage B restores all 4,019 training frames and trains RCRA and a matched current-code FAM control for ten epochs, without validation or automatic MtErie testing. The primary evaluation uses the same 708 paired MtErie frames and epoch-10 **latest** checkpoints.

Table 4.8: Matched full-data Stage-B VIS+IR mAP@50.

Seed	Matched FAM	RCRA	RCRA – FAM
40	0.4291	0.3969	-0.0323
41	0.3498	0.3737	+0.0239
42	0.3233	0.4185	+0.0952
43	0.3308	0.2835	-0.0474
44	0.3513	0.4038	+0.0525
Mean $\pm$ SD	0.3569 $\pm$ 0.0422	0.3753 $\pm$ 0.0538	+0.0184 $\pm$ 0.0591

RCRA passes the required mean-gain condition but wins only 3/5 seeds, rather than the required 4/5. Its paired 95% interval is [-0.0550, 0.0918]. The frozen decision therefore retains matched FAM as the selected model and reference. This is not evidence that RCRA is uniformly worse: its mean and median effects are positive, but the improvement is too unstable to support replacement.

Secondary results point in the same uncertain direction. COCO mAP changes from 0.1244 to 0.1336 and mAP@75 from 0.0581 to 0.0701; RCRA wins 3/5 and 4/5 pairs, respectively. Mean training time increases from approximately 2 h 22 min to 3 h 28 min per run. These secondary observations do not override the frozen mAP@50 rule. The historical FAM mean of 0.3780 remains close in absolute scale, but the current matched value of 0.3569 is the valid Stage-B comparator.

4.3 Post-selection paired modality characterization

After model selection was closed, the retained five FAM checkpoints were evaluated in three conditions on exactly the same 708 MtErie pairs and the same 1,770 VIS boxes. The VIS condition zeros IR, while paired masked-IR zeros RGB after adapting IR to the VIS canvas.

Fusion exceeds VIS in 5/5 seeds. The paired difference has a 95% interval of [0.0095, 0.0650], supporting a thermal contribution beyond VIS on this fixed internal population. The masked-IR value does not represent general native-IR performance. It removes the RGB reference used by the asymmetric FAM while still requesting VIS-coordinate boxes, and the IR adaptation is only a resize-and-pad operation.

A post-hoc diagnostic therefore evaluates the same 708 IR counterparts on their native canvas with 1,824 native IR boxes. Native IR obtains 0.5618 ± 0.0498 mAP@50

Table 4.9: Paired modality characterization of the selected current-code FAM checkpoints. Every column uses the same frames and VIS ground truth.

Seed	VIS+IR	VIS	Masked IR / VIS GT	Fusion – VIS
40	0.4291	0.3764	0.0213	+0.0527
41	0.3498	0.3054	0.0333	+0.0444
42	0.3233	0.3174	0.0170	+0.0059
43	0.3308	0.3077	0.0093	+0.0232
44	0.3513	0.2911	0.0265	+0.0602
Mean $\pm$ SD	0.3569 ± 0.0422	0.3196 ± 0.0331	0.0215 ± 0.0091	$+0.0373 \pm 0.0223$

with a 95% interval of [0.5001, 0.6236], compared with 0.0215 ± 0.0091 under paired masked-IR/VIS ground truth. This rules out a collapsed IR branch, but it does not show that native IR is better than fusion: the coordinate systems and annotations differ. It also cannot replace the historical native-IR benchmark, which uses a broader test population.

4.3.1 Modal Dropout coordinate-system ablation

The paired-modality audit revealed that the missing-RGB condition used at evaluation time did not reproduce the IR-only examples used by the original Modal Dropout. During the original training, an IR-only draw contains the native IR image and its native IR annotations. In the paired evaluation, instead, IR is adapted to the VIS canvas, the RGB channels are set to zero, and detections are compared with VIS annotations. The very low masked-IR result therefore measures a condition that the detector was not trained to reproduce exactly.

To test whether this discrepancy explains the result, one additional Stage-A training was performed with seed 40. Only the IR-only branch of Modal Dropout was changed: it first constructs the same paired sample used for fusion, then sets the RGB channels to zero while retaining the adapted IR channel and the VIS annotations. The fusion and RGB-only branches, detector architecture, optimizer, data split, and ten-epoch budget remain unchanged. This is a single-seed diagnostic ablation, not a new final model comparison.

Table 4.10: Effect of changing the IR-only Modal Dropout coordinate convention in the seed-40 Stage-A experiment, evaluated on the 896-pair validation sequence.

Condition	Original dropout	VIS-coordinate variant	Delta
Paired VIS+IR / VIS GT	0.152148	0.156899	+0.004752
Paired VIS / VIS GT	0.151107	0.143206	-0.007901
Paired masked-IR / VIS GT	0.002519	0.002836	+0.000317
Native IR / IR GT	0.263709	0.027067	-0.236642

Fusion changes only from 0.152148 to 0.156899 mAP@50, while masked-IR with VIS annotations changes from 0.002519 to 0.002836. The modification therefore does

not solve localization in VIS coordinates when RGB is missing. At the same time, native-IR performance falls from 0.263709 to 0.027067, consistently with the removal of IR-only supervision in the native IR coordinate system. The IR-only sampling trace remains close to the configured 20%, ruling out the possibility that this outcome is caused by an inactive training branch.

The ablation consequently fails the predeclared masked-IR improvement and native-IR preservation requirements. It is rejected after the single screening seed and is not extended to seeds 41–44, MtErie, or Stage B. The result supports retaining the original native-IR Modal Dropout in the final FAM model: matching the paired evaluation convention removes useful native-IR supervision without making the detector robust to a missing RGB reference.

4.4 Mechanistic validation of the FAM

4.4.1 Diagnostic design

The main detection result establishes a predictive advantage but does not identify its mechanism. A frozen diagnostic campaign therefore examined standard FAM, FAM+SSJ, and Grid Sample for seeds 40–44. For every checkpoint, ten predefined samples were drawn from MtErie, ten from the FHL training session, and ten from the Baker training session. Each sample was inspected at P3, P4, and P5:

$$3 \text{ configurations} \times 5 \text{ seeds} \times 30 \text{ samples} \times 3 \text{ levels} = 1,350 \quad (4.1)$$

sample-level observations. They are aggregated within each checkpoint before the five checkpoints are compared.

Forward hooks capture RGB and IR features before FAM, the transformed IR feature, and the predicted offsets and masks. Offset components are converted to input-image pixels by multiplying them by the level stride. Feature maps are projected to three shared PCA components for visualization. These pseudo-colour maps are abstract feature projections and not registered RGB and thermal images.

4.4.2 Offset behavior across levels and seeds

Table 4.11 reports the mean absolute offset component after averaging the thirty samples within each checkpoint. Standard FAM has nonzero transformations at P3 and P4 in every seed. Excluding the pathological P5 entry, offset scale varies both by seed and level.

At P3, SSJ increases the mean offset magnitude in all five paired seeds; the mean difference is $(+1.230 \pm 1.001)$ input pixels. It also reduces the ratio of output-to-input spatial variation in all five checkpoints, producing a smoother transformed feature. P4 offset changes are less consistent, and P5 cannot be paired directly with standard FAM without accounting for the seed-41 collapse. The diagnostic conclusion agrees with the detection result: SSJ changes the learned module in a repeatable way at P3, but that change does not improve final detection accuracy.

Grid Sample has smaller and less dispersed P3/P4 offsets. Unlike DCNv2, it performs only a bilinear warp, without learned post-sampling filtering or channel mixing.

Table 4.11: Mean absolute offset components in input-image pixels. Values are aggregated over thirty frozen samples per checkpoint.

Configuration	Seed	P3	P4	P5
FAM	40	2.69	3.23	13.69
FAM	41	3.34	7.47	5029.83
FAM	42	3.48	6.32	34.20
FAM	43	3.04	3.74	14.44
FAM	44	5.99	4.80	26.63
FAM + SSJ	40	4.42	9.11	34.45
FAM + SSJ	41	3.45	10.71	23.42
FAM + SSJ	42	5.41	4.73	20.95
FAM + SSJ	43	5.23	6.84	18.02
FAM + SSJ	44	6.17	4.18	36.85
Grid Sample	40	1.65	2.60	25.25
Grid Sample	41	1.70	3.59	7.66
Grid Sample	42	2.08	2.89	11.65
Grid Sample	43	2.29	4.60	15.86
Grid Sample	44	2.41	2.49	10.17

Its output can therefore be interpreted more directly as a spatially transformed IR feature, whereas the DCNv2 output is a broader feature-space transformation. Neither operation establishes physical camera alignment.

4.4.3 P5 collapse in seed 41

FAM seed 41 exhibits a complete and repeated P5 degeneration across all thirty samples and all three sessions. Its mean absolute offset is approximately 5,030 input pixels, orders of magnitude larger than the normal checkpoints. The transformed IR output has zero spatial standard deviation and equals the learned DCNv2 bias at every location.

The scale anomaly is already present before FAM: mean P5 global standard deviations are 17.30 for RGB and 20.18 for IR, compared with 0.17–0.69 and 0.056–0.101 in the other seeds. The offset predictor itself has ordinary weight norms. Its enormous offsets are nearly identical across samples (correlation above 0.999) and sample outside the feature map, leaving only the deformable-convolution bias.

The detector nevertheless obtains 0.4335 mAP@50, the highest of the five FAM checkpoints. High output accuracy therefore does not guarantee that every internal alignment level behaves sensibly. The main FAM advantage is not created by the anomaly: after excluding seed 41, FAM still beats Base RT-DETR in 4/4 paired seeds with mean delta (+0.0799).

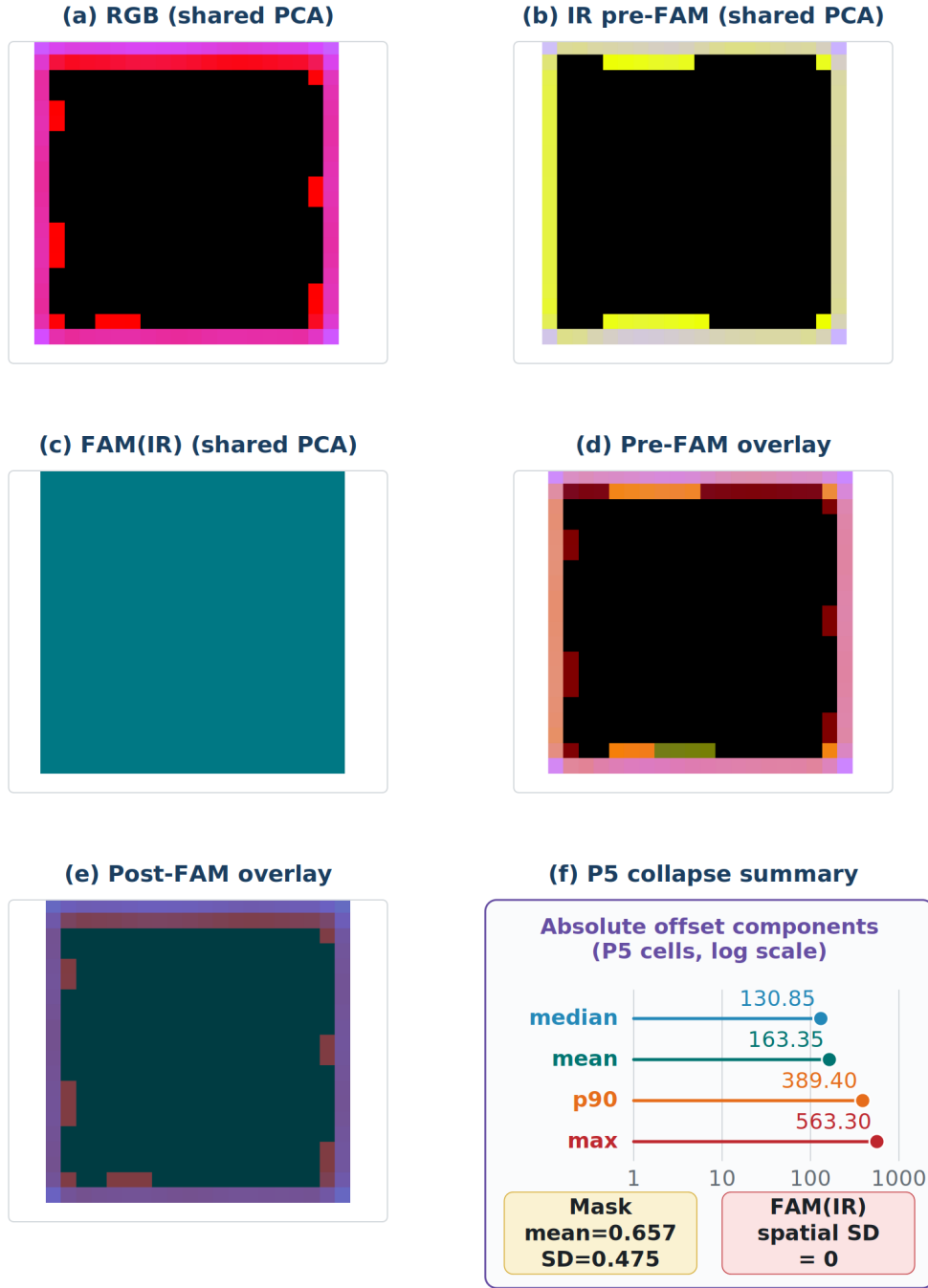


Figure 4.2: Mechanistic diagnostic of the collapsed P5 branch in FAM seed 41 for the prespecified MtErie sample 35. Panels (a)–(c) are pseudo-colour PCA projections obtained with a basis shared across the standardized feature tensors; panels (d)–(e) are qualitative 50% blends and are not tensors consumed by the detector. Panel (f) summarizes absolute DCNv2 offset components in feature-cell coordinates and the modulation-mask statistics. The transformed IR feature is spatially constant, while the mean absolute offset component corresponds to 5,227 input pixels. These quantities diagnose a learned feature-space transformation and must not be interpreted as a camera-registration field.

4.4.4 Factorial P3/P4/P5 ablation

All eight subsets of active FAM levels were evaluated post-hoc in each of the five trained FAM checkpoints. Bypassing a level replaces its transformed IR feature with the pre-FAM IR feature. The intervention measures sensitivity of the trained system; it does not reproduce training a network with that level absent.

Table 4.12: Post-hoc factorial level ablation. Delta is the paired mean difference from the full trained FAM.

Active FAM levels	Mean mAP@50	SD	Delta from full
None	0.2259	0.0654	-0.1521
P3	0.3223	0.0660	-0.0558
P4	0.2396	0.0983	-0.1384
P5	0.2791	0.0377	-0.0989
P3 + P4	0.3221	0.0701	-0.0559
P3 + P5	0.3648	0.0503	-0.0132
P4 + P5	0.2984	0.0254	-0.0796
P3 + P4 + P5	0.3780	0.0440	0.0000

P3 has the largest and most stable factorial main effect. P5 has a positive mean effect but high variability, dominated by seed 41. Removing P5 from that checkpoint reduces mAP@50 from 0.4335 to 0.2399. The decoder has adapted to, or learned to compensate for, the collapsed representation. Static removal after training is therefore not a repair, and the result does not predict the score of a newly trained P3+P4-only model.

4.4.5 Bounded-offset intervention

The proximal mechanism of the constant output is unbounded sampling far outside the feature map. A single corrective hypothesis was therefore frozen before new training. For every raw offset component, the bounded variant uses

$$\Delta_{\text{eff}} = 4 \tanh\left(\frac{\Delta_{\text{raw}}}{4}\right), \quad (4.2)$$

where the limit is expressed in cells of the current feature level. The derivative at zero is one, and no gate, normalization, residual connection, or backbone modification is introduced. The limit of four cells was chosen from the non-pathological checkpoint distributions before observing the five new training results.

The bounded model wins in only 2/5 seeds and has greater dispersion. Its standalone modality evaluation obtains 0.2252 ± 0.0544 VIS, 0.1943 ± 0.0272 IR, and 0.3496 ± 0.0685 VIS+IR, all below the standard-FAM means.

The internal intervention nevertheless succeeds. Every observed effective offset remains below four cells, including a maximum of 3.9915 at P5 seed 41, and none of the 150 inspected P5 checkpoint-sample cases has a spatially constant output. Raw offsets would still reach 13.69 cells, showing that the bound is active. The experiment

Table 4.13: Automatic VIS+IR mAP@50 for the five bounded-offset runs and the corresponding frozen standard-FAM reference.

Seed	Standard FAM	Bounded FAM	Bounded – standard
40	0.3783	0.4208	+0.0426
41	0.4335	0.3319	-0.1015
42	0.3129	0.4214	+0.1085
43	0.3964	0.2735	-0.1229
44	0.3690	0.2998	-0.0692
Mean $\pm$ SD	0.3780 $\pm$ 0.0440	0.3495 $\pm$ 0.0686	-0.0285

therefore separates two outcomes: the targeted failure mode is removed, but detection does not improve. Standard FAM remains the selected model.

4.5 Detection error analysis

These analyses use the historical locked Base and FAM checkpoints from Section 4.1. They characterize the established FAM gain over Base RT-DETR and do not compare RCRA with the matched current-code Stage-B FAM control.

4.5.1 Protocol

The error analysis compares the five Base RT-DETR and five standard-FAM final checkpoints on all 708 MtErie frames and 1,770 objects. Predictions are matched one-to-one to annotations by descending IoU, with a minimum IoU of 0.50. The primary evaluation retains all predictions with confidence at least 0.01, as in the final mAP evaluation. For the predefined sensitivity analysis, this same prediction set is additionally thresholded at 0.05, 0.10, 0.25, and 0.50 when reporting precision, recall, and false positives. These thresholds are not used for model selection or to select a deployment threshold.

Of the 1,770 objects, 1,451 are COCO-small, 301 medium, and 18 large at the evaluation resolution. Large-object results are therefore particularly uncertain.

4.5.2 Results across thresholds

Recall is the most consistent difference. FAM is better in 5/5 seeds at thresholds 0.10, 0.25, and 0.50. At 0.25, recall increases from 0.2340 to 0.3330, a mean paired gain of 0.0990, and the fraction of non-empty frames with at least one missed target decreases by 0.0824.

The gain is not limited to the 18 large objects. At threshold 0.25, mean recall increases by 0.0815 for small and 0.1781 for medium targets. False positives do not improve uniformly: FAM produces slightly fewer on average at low thresholds and slightly more at 0.25 and 0.50, with signs that vary across seeds.

Table 4.14: Paired RT-DETR error behavior across confidence thresholds. Values are mean $\pm$ sample standard deviation over five checkpoints.

Threshold	Model	Precision	Recall	FP/image	Non-empty frames with FN
0.01	Base	0.0418 ± 0.0175	0.7677 ± 0.0490	51.006 ± 21.591	0.4508 ± 0.0777
0.01	FAM	0.0443 ± 0.0132	0.8044 ± 0.0328	47.623 ± 17.873	0.4009 ± 0.0524
0.05	Base	0.2713 ± 0.0909	0.5932 ± 0.1134	4.844 ± 3.199	0.6476 ± 0.1155
0.05	FAM	0.3010 ± 0.0940	0.6539 ± 0.0972	4.336 ± 2.191	0.5835 ± 0.0775
0.10	Base	0.4721 ± 0.0923	0.4582 ± 0.1535	1.513 ± 1.115	0.7466 ± 0.1223
0.10	FAM	0.5194 ± 0.1148	0.5507 ± 0.1333	1.447 ± 0.792	0.6554 ± 0.0725
0.25	Base	0.7290 ± 0.0866	0.2340 ± 0.1468	0.249 ± 0.218	0.8682 ± 0.0879
0.25	FAM	0.7715 ± 0.0742	0.3330 ± 0.1506	0.276 ± 0.168	0.7858 ± 0.0525
0.50	Base	0.9140 ± 0.0536	0.1088 ± 0.0705	0.029 ± 0.033	0.9257 ± 0.0588
0.50	FAM	0.9350 ± 0.0504	0.1627 ± 0.0663	0.036 ± 0.037	0.8790 ± 0.0359

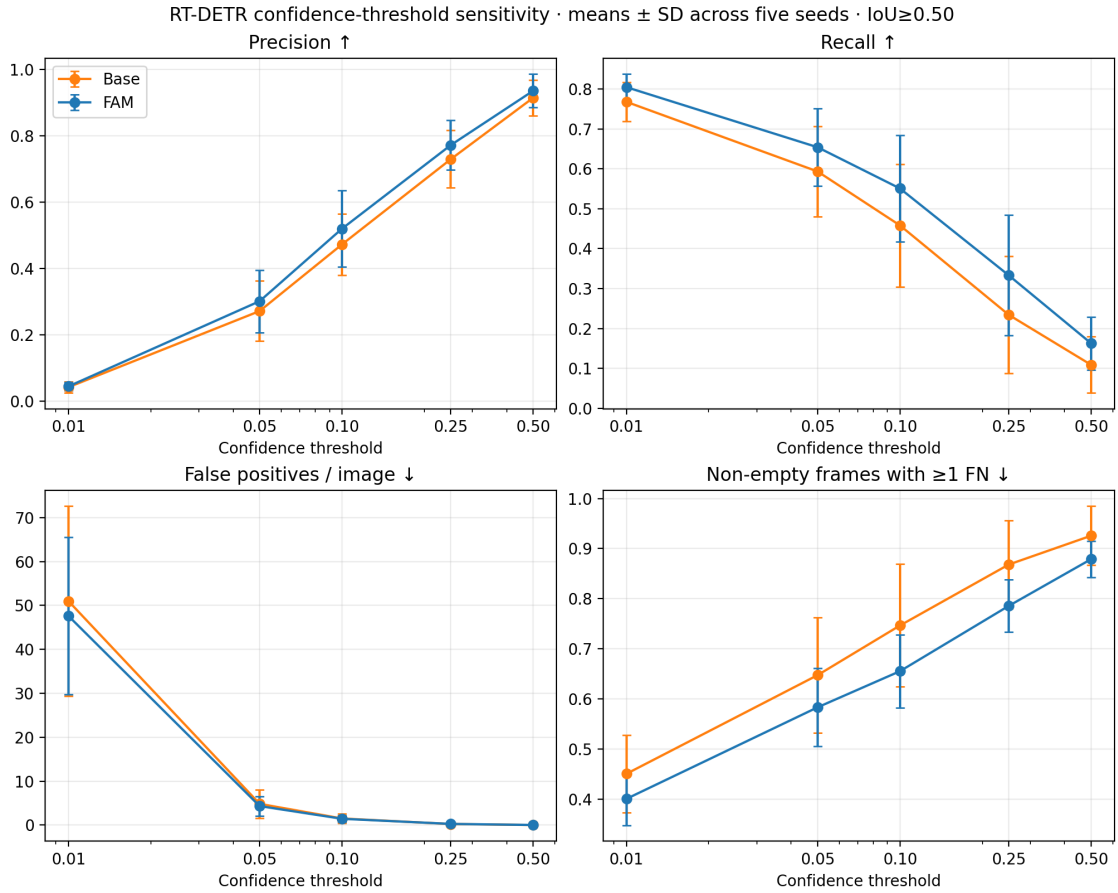


Figure 4.3: Sensitivity of precision, recall, false positives per image, and non-empty frames with a false negative to the five predefined confidence thresholds. Error bars are sample standard deviations across checkpoints.

At threshold 0.01, both configurations generate approximately 48–51 false positives per image. This threshold is appropriate for collecting a broad set of candidates for AP computation but is not an operational alarm threshold.

4.5.3 Recall at matched false-positive budgets

A common confidence threshold does not necessarily produce the same number of false positives in two detectors. A subsequent post-hoc analysis therefore asks whether the historical FAM recall advantage persists at a matched budget of false positives per image (FPPI). It reuses the five epoch-10 `latest` checkpoints per configuration, seeds 40–44, in VIS+IR (fusion) mode without further training or model selection. The three evaluation populations are the 708 paired MtErie frames and the 1,313 Carnation 0025/0026 and 1,035 FHL 0407/0408 pairs from Section 4.7. Carnation 0025/0026 is distinct from the 739-frame scale-mismatch acquisition in Section 4.6. Each contrast preserves its frozen pairing and uses identical VIS annotations for Base and FAM. The additional populations contain 5,238 and 2,022 VIS ground-truth boxes, respectively.

One inference pass collects all 300 native queries per image with collection threshold zero and no added non-maximum suppression. Confidence is then swept offline over every distinct prediction score, retaining predictions with $s \geq \tau$, where s is the confidence score of a predicted box and τ is the confidence threshold. Within each image, predictions are processed in descending confidence order and matched to the highest-IoU unassigned annotation at IoU at least 0.50. Duplicates are false positives. This differs from the descending-IoU matching used in Table 4.14: lower-confidence predictions cannot reassign matches already made at a higher threshold. The earlier analysis is retained with its original matching rule.

For each checkpoint, $\text{recall} = \text{TP}/N_{\text{GT}}$ and $\text{FPPI} = \text{FP}/N_{\text{images}}$. All frames, including those without annotated persons, contribute to N_{images} ; no frames are excluded based on their false-positive counts. At each FPPI budget, the reported recall is the highest empirically attainable value without exceeding that budget, without interpolation or splitting tied scores. For each model, the five seed-specific curves are evaluated at the same 301 logarithmically spaced FPPI budgets between 0.01 and 10. The mean and sample standard deviation of these recall values are then computed across the five seeds at each budget. The budgets 0.1, 0.5, and 1 FPPI were specified before this dense sweep and are all reported in Table 4.15. The confidence sweep is unconstrained by these budgets: recall and FPPI are first computed at every distinct prediction score. The three budgets are then used only to summarize the resulting curves at common false-positive limits. These are descriptive reference points, not operationally validated SAR limits.

FAM has higher mean recall at all nine population–budget combinations and at every displayed grid point in Figure 4.4. At 0.5 FPPI, the mean paired gains are 11.02 percentage points on MtErie, 9.81 on Carnation, and 14.79 on FHL, with positive differences in all five seeds within each population. The only negative paired difference at the predefined budgets is Carnation seed 43 at 0.1 FPPI: recall decreases from 0.2367 to 0.2310 (−0.57 percentage points). Individual-seed curves also cross elsewhere; the mean advantage does not establish universal dominance.

The recall gain therefore persists at a fixed false-positive budget rather than arising solely from accepting more detections at the same confidence. This is descriptive evi-

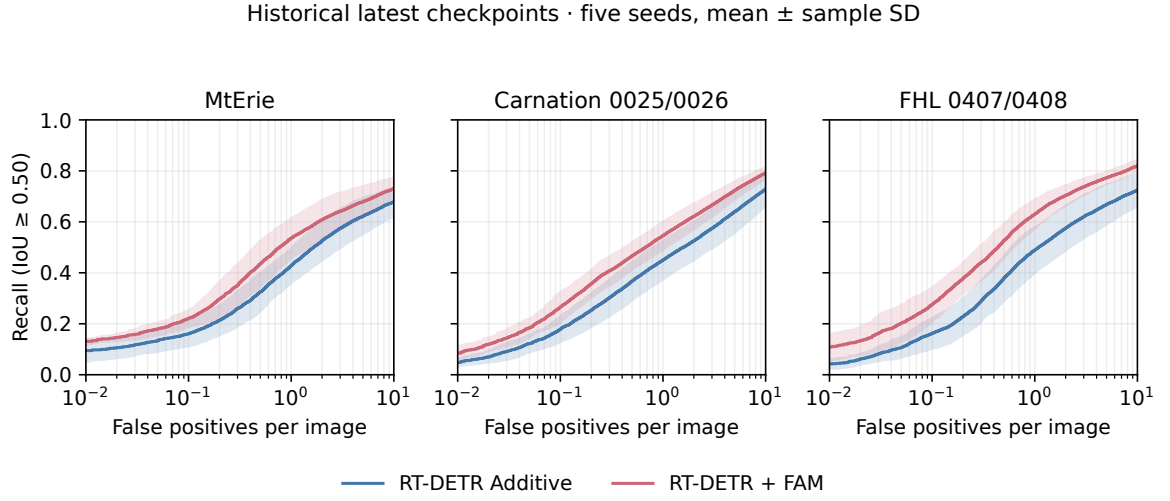


Figure 4.4: Recall at matched false-positive budgets for historical Base (additive fusion) and FAM RT-DETR checkpoints in VIS+IR mode. Curves show five-seed means and shaded bands show sample standard deviations at common FPPI budgets, with one-to-one confidence-ordered matching at IoU 0.50. The bands describe training variability, not confidence intervals over frames or unseen acquisitions.

Table 4.15: Recall at the three predefined FPPI budgets. Base, FAM, and paired FAM–Base differences are mean $\pm$ sample standard deviation across five seeds. Wins count strictly positive paired differences.

Population	FPPI $\leq$	Base	FAM	Paired delta	Wins
MtErie	0.1	0.1610 ± 0.0493	0.2217 ± 0.0338	$+0.0607 \pm 0.0393$	5/5
	0.5	0.3266 ± 0.0739	0.4367 ± 0.0885	$+0.1102 \pm 0.0615$	5/5
	1.0	0.4296 ± 0.0662	0.5368 ± 0.0778	$+0.1072 \pm 0.0600$	5/5
Carnation 0025/0026	0.1	0.1797 ± 0.0496	0.2670 ± 0.0579	$+0.0873 \pm 0.0550$	4/5
	0.5	0.3688 ± 0.0745	0.4669 ± 0.0531	$+0.0981 \pm 0.0318$	5/5
	1.0	0.4507 ± 0.0757	0.5467 ± 0.0542	$+0.0960 \pm 0.0388$	5/5
FHL 0407/0408	0.1	0.1632 ± 0.0604	0.2777 ± 0.0666	$+0.1145 \pm 0.0946$	5/5
	0.5	0.3804 ± 0.0884	0.5283 ± 0.0664	$+0.1479 \pm 0.1255$	5/5
	1.0	0.4903 ± 0.0895	0.6310 ± 0.0567	$+0.1407 \pm 0.1112$	5/5

dence on already-consulted populations, not a blind confirmation or a new architecture-selection result. The confidence thresholds corresponding to each FPPI budget vary across checkpoints and evaluation sets and are not selected for deployment. FPPI counts false boxes per frame, not video alarm events: temporal correlation and alarm aggregation must be considered before inferring operator workload.

4.5.4 Qualitative examples

The qualitative manifest was built from ground-truth annotations before inference. For each of the three MtErie sequences, it selects one frame with a small target and one empty frame. Seed 43 is used because its FAM–Base mAP@50 difference equals the median of the five paired differences; it is neither the best nor worst pair.

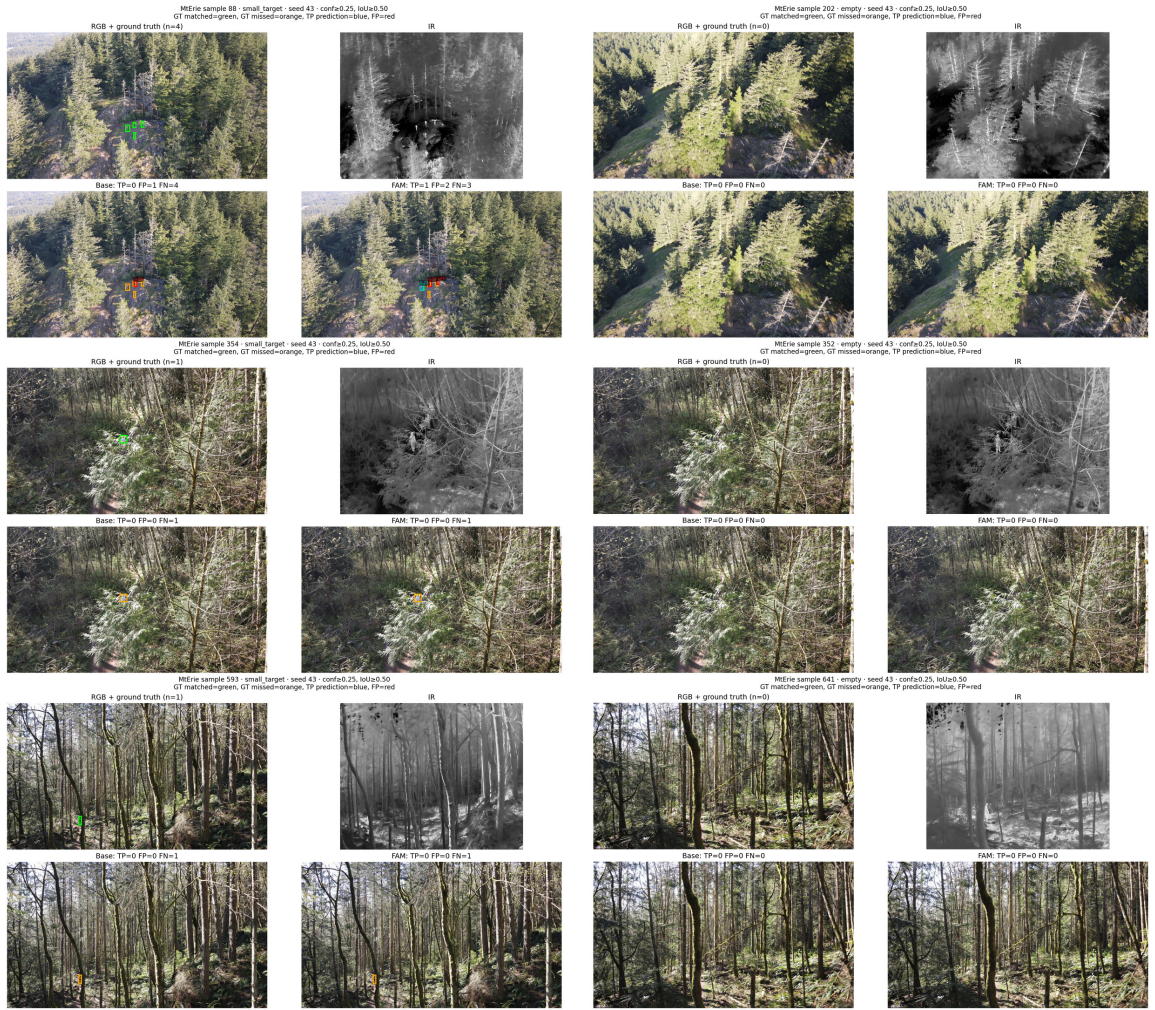


Figure 4.5: Predefined qualitative examples at confidence 0.25. Each row contains one small-target frame and one empty frame from a MtErie sequence. The same frames and display rules are used for Base RT-DETR and FAM. These examples illustrate, but do not replace, the aggregate analysis.

The corresponding threshold-0.01 montage is retained in the appendix because the large number of low-confidence boxes makes it difficult to read. In the three predefined small-target frames at 0.25, FAM recovers one target missed by Base RT-DETR in one

frame; both models miss the targets in the other two. No general conclusion is drawn from these six examples.

4.6 Carnation scale-mismatch stress test

4.6.1 Frozen design

Carnation contains a paired visual and thermal acquisition with a substantially stronger scale mismatch than the main splits. Its protocol was frozen before inference and admits only the five final Base RT-DETR and five final FAM checkpoints. The intersection of frame identifiers contains 739 examples. Each checkpoint is evaluated in VIS+IR, VIS, and IR, yielding thirty evaluations without training, tuning, checkpoint selection, or threshold selection.

VIS+IR and VIS are evaluated in the VIS coordinate system against the VIS annotations, so those two input conditions are directly comparable. The IR-only condition instead uses the native IR images and IR annotations. It supports a within-condition comparison between Base RT-DETR and FAM, but its absolute score is not directly comparable with the VIS+IR or VIS rows.

Carnation is a deliberately difficult single acquisition. It is not a representative sample of SAR domains, a new validation set, or an estimate of general domain generalization.

4.6.2 Results

Table 4.16: Carnation mAP@50 across five frozen checkpoints.

Input	Base RT-DETR		FAM	Mean paired delta	FAM wins
VIS+IR	0.1445 ± 0.0709	0.2408 ± 0.0380		+0.0962	4/5
VIS	0.1555 ± 0.0669	0.3196 ± 0.0518		+0.1641	5/5
Native IR / IR GT	0.1288 ± 0.0776	0.0979 ± 0.0652		-0.0308	1/5

FAM outperforms Base RT-DETR under this severe scale mismatch, with a mean gain of 0.0962 and four paired wins. This is not sufficient to make fusion the best input condition. For the FAM checkpoints, VIS-only exceeds VIS+IR in all five seeds by a mean of 0.0789. Adding the strongly mismatched IR signal is therefore harmful relative to the available visible stream even after FAM transformation.

The Carnation result defines a boundary of the internal paired-modality finding: fusion can outperform VIS on the matched MtErie population, but not under this extreme cross-sensor scale mismatch.

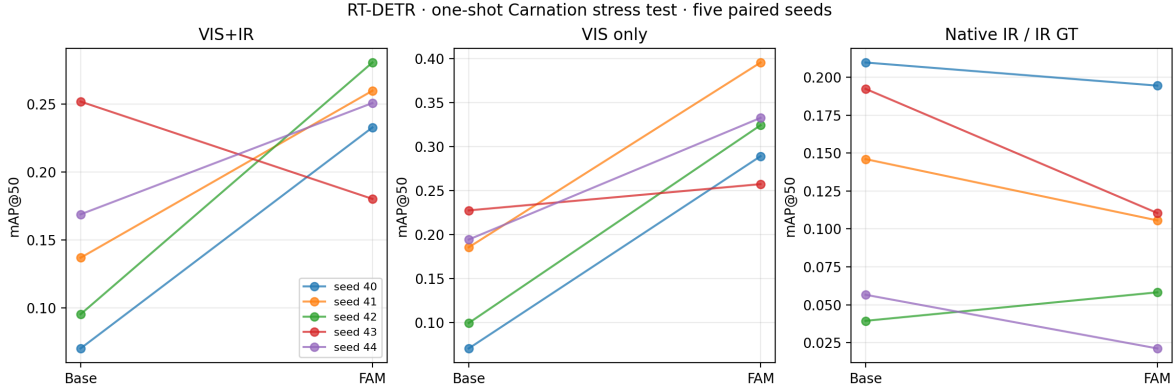


Figure 4.6: Paired Carnation results for the three modality conditions. The stress test shows relative advantage under mismatch by FAM, not correction of the severe mismatch.

4.7 Confirmation on previously unused WiSARD acquisitions

The frozen confirmation campaign evaluates 20 existing checkpoints on Carnation 0025/0026 and FHL 0407/0408. Neither acquisition appeared in the preserved model-development configurations or outputs. Pairing is by exact numeric identifier, yielding 1,313 Carnation pairs and 1,035 FHL pairs. These are previously unused internal acquisition sets, not an external holdout.

Table 4.17: Frozen acquisition-level confirmation. Values are mean mAP@50 over five paired checkpoints; intervals and standard deviations refer to the five within-seed differences.

Contrast	Acquisition	Reference	Candidate	Paired delta	Wins
Historical FAM – Base	Carnation	0.3945	0.4905	(+0.0960 ± 0.0595)	5/5
Historical FAM – Base	FHL	0.2953	0.4507	(+0.1554 ± 0.0909)	5/5
Stage-B FAM fusion – VIS	Carnation	0.4316	0.4336	(+0.0019 ± 0.0402)	3/5
Stage-B FAM fusion – VIS	FHL	0.4064	0.3808	(−0.0256 ± 0.0170)	1/5
Stage-B RCRA – FAM	Carnation	0.4336	0.4151	(−0.0184 ± 0.1149)	2/5
Stage-B RCRA – FAM	FHL	0.3808	0.4174	(+0.0366 ± 0.1201)	3/5

The central historical contrast replicates: FAM exceeds the baseline in all ten acquisition–seed comparisons. Giving both acquisitions equal weight produces a descriptive macro-average paired gain of (0.1257 ± 0.0521) , with a 95% interval of $[0.0610, 0.1904]$.

The additional contrasts prevent a stronger interpretation. On Stage-B FAM, fusion is essentially tied with VIS on Carnation and is 0.0256 lower on FHL. RCRA alternates large positive and negative seed effects and retains a macro-average interval spanning zero. FAM can therefore be a better fusion operator than the baseline without guaranteeing that IR helps every checkpoint on every acquisition.

4.8 Controlled synthetic geometric stress

The synthetic stress applies IR-only translations and isotropic scaling to the same two acquisitions. In total, 560 new checkpoint–transformation inferences are combined with 40 frozen identity results. Figure 4.7 reports the mean signed relative drop. Positive values are losses, while negative values indicate that the perturbation accidentally improved mAP@50.

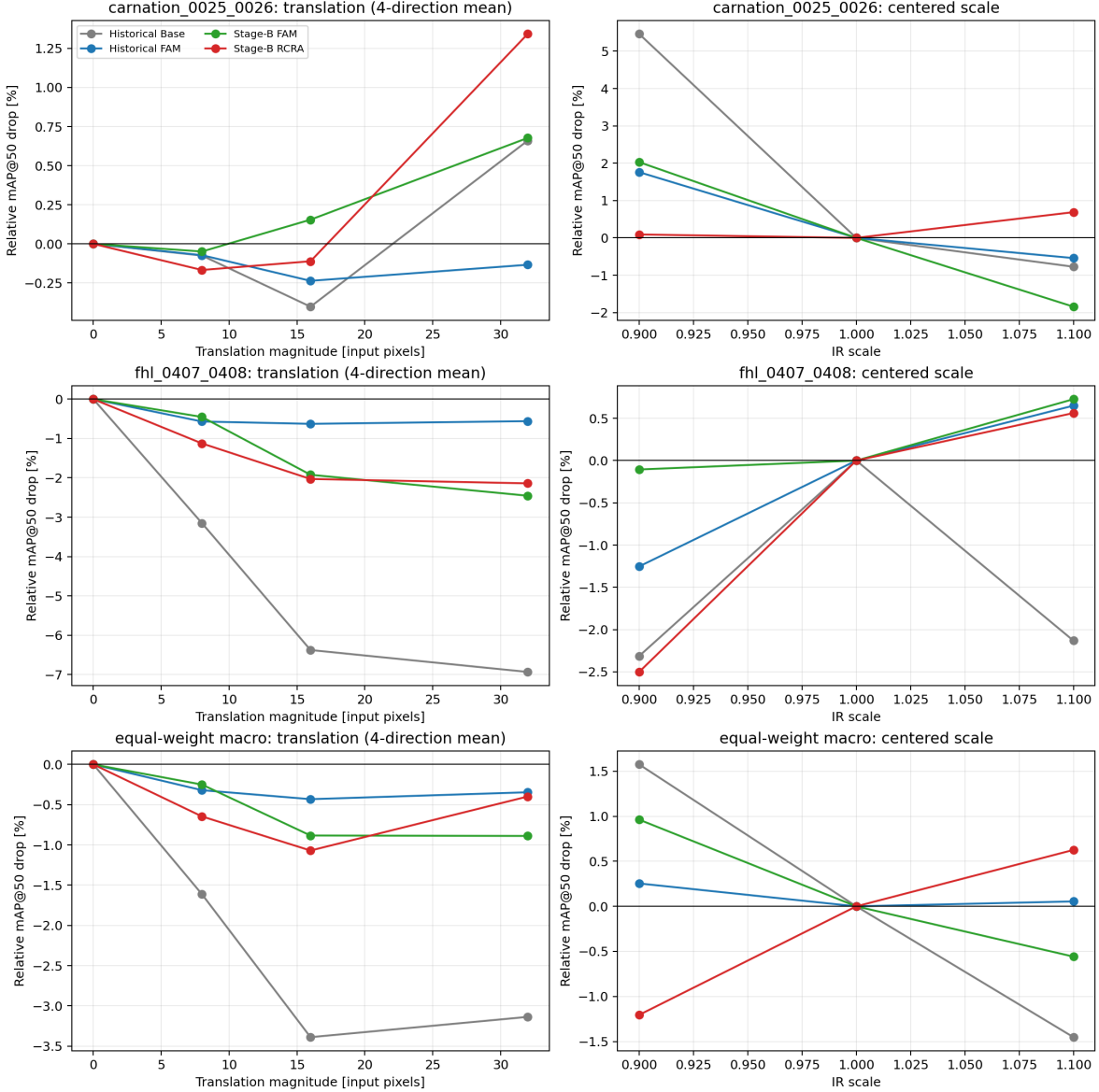


Figure 4.7: Signed relative mAP@50 drop under IR-only translations and rescaling, averaged first across directions within checkpoint and then over five seeds. Non-monotone negative drops show why these transformations cannot be interpreted as calibrated displacement from a known optimum.

The response is acquisition-dependent and non-monotone. On Carnation, FAM often limits the effect of perturbation, including a smaller mean loss under scale 0.9 than Base. On FHL, however, translations improve the baseline more than FAM in all five seeds. This yields signed robustness contrasts favoring Base even though FAM retains higher absolute accuracy. A synthetic shift may be compensating an existing

sensor mismatch or changing features favorably, rather than measuring distance from correct registration.

The predeclared strong hypothesis—that FAM is always more tolerant to geometric perturbation—is therefore not supported. Likewise RCRA has no stable advantage over FAM across acquisition, magnitude, and seed. This does not undo the absolute FAM gain in Section 4.7, but it limits its mechanism: the gain cannot generally be attributed to a superior physical-registration curve.

4.9 Computational cost

The compute comparison uses the seed-43 Base RT-DETR and FAM RT-DETR checkpoints, chosen in advance as the same typical pair used for qualitative illustrations. Each configuration is measured in three isolated processes on an NVIDIA GeForce RTX 4070 Laptop GPU with PyTorch 2.4.0, CUDA 12.1, batch size one, FP32 input of 640×640 , thirty warm-up iterations, and one hundred timed forward passes per trial. TF32, autocast, preprocessing, and postprocessing are excluded.

Table 4.18: RT-DETR detector-only computational benchmark. The operation count is a proxy and the latency uncertainty is the sample standard deviation across three trial means.

Model	Parameters	Proxy GFLOPs	Latency	Throughput	CUDA peak
Base	66.20 M	208.37	66.91 ± 0.30 ms	14.95 image/s	436.26 MiB
FAM	117.49 M	304.54	78.87 ± 0.37 ms	12.68 image/s	712.18 MiB

Relative to Base RT-DETR, FAM adds 77.5% parameters, 46.2% proxy computation, 17.9% detector latency, and 275.92 MiB of total peak CUDA memory. The proxy adds an analytical dense-convolution equivalent for DCNv2 because the profiler does not count that operator fully; it still omits parts of bilinear sampling, mask application, and sigmoid evaluation. Measured latency is therefore the more operational quantity, while remaining specific to this GPU and detector scope.

The FAM improvement is not computationally free. Deployment decisions must weigh its recall and mAP@50 gains against memory, throughput, energy, and the unmeasured cost of the full UAV pipeline.

4.10 Architectural transferability to YOLOv10

4.10.1 Dual-backbone architecture

The YOLOv10-s transfer model receives a four-channel paired tensor and splits it into a three-channel RGB stream and one-channel IR stream. Two independent backbones produce P3, P4, and P5 features at strides 8, 16, and 32. The Base configuration sums each pair; the FAM configuration applies the same standard DCNv2 module to IR before addition. The resulting features enter the shared YOLOv10 FPN/PAN neck and detection head.

Missing modalities are implemented through feature gating. During training, one branch or both are selected with probabilities 0.2/0.2/0.6. In standalone VIS or IR evaluation, the excluded backbone is not executed and FAM is bypassed because a paired reference is unavailable. Unimodal differences therefore reflect how training with FAM shaped the remaining weights, not a direct alignment operation at inference.

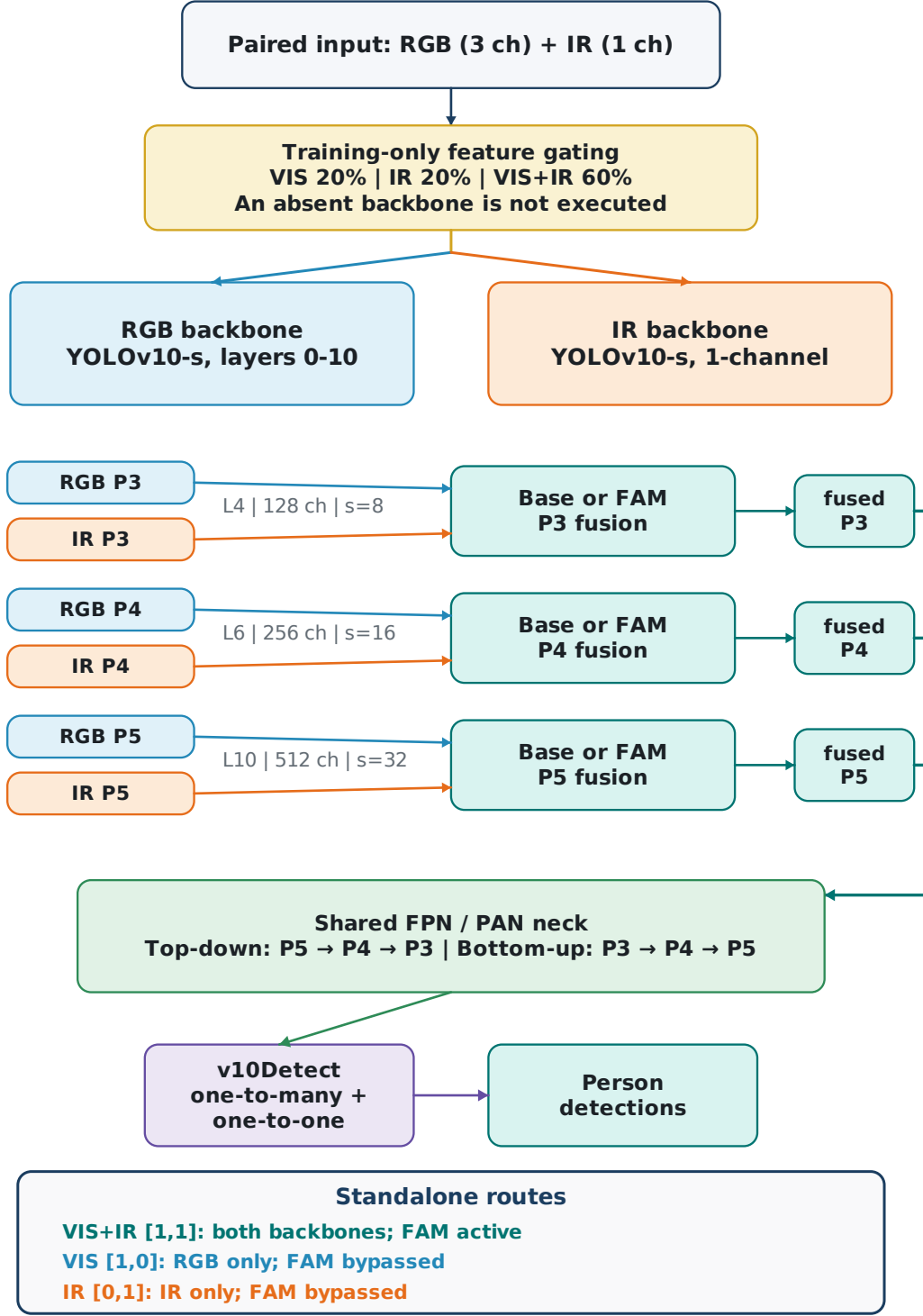


Figure 4.8: Implemented YOLOv10-s transfer architecture. The controlled comparison changes the P3/P4/P5 fusion operation while retaining the dual backbones, feature gating, FPN/PAN neck, detection head, and final protocol. The FAM operation is the same as the one expanded in Figure 2.6 and is therefore not duplicated internally. During unimodal inference the excluded backbone is not executed and FAM is bypassed. Consequently, unimodal Base–FAM differences are indirect effects of training rather than direct alignment effects.

4.10.2 Exploratory history and checkpoint problem

Initial YOLO experiments used single runs, `best.pt`, and variable early stopping selected through the small FHL validation set. They explored fusion-only training, input-level modality masking, feature gating, FAM, and SSJ. These runs helped identify feature-level gating as the appropriate missing-modality contract but do not provide a final transfer claim.

The historical seed-42 Base and FAM checkpoints were selected at epochs 16 and 89 and achieved higher MtErie values than the final 200-epoch checkpoints. Because FHL is small and strongly shifted, its curve does not define a stable general rule. Selecting 100 epochs, reusing `best.pt`, or choosing a new epoch after observing MtErie would be retrospective test tuning.

4.10.3 Final YOLOv10 protocol

The final comparison fixes two configurations: dual-backbone Base and dual-backbone with FAM, both with feature gating 20/20/60. Each uses paired seeds 40–44, 200 fixed epochs, disabled early stopping, no validation-based selection, and the final `last.pt` checkpoint. The 200-epoch horizon was retained because it was declared in the historical YAML schedule before the final results were observed.

All ten runs completed 200 epochs. Their VIS+IR results are shown in Table 4.19.

Table 4.19: Final automatic YOLOv10 VIS+IR mAP@50 at `last.pt`.

Seed	Base	FAM	FAM – Base
40	0.2304	0.1883	-0.0421
41	0.2740	0.2749	+0.0009
42	0.2494	0.1947	-0.0546
43	0.2159	0.2603	+0.0444
44	0.2728	0.1802	-0.0926
Mean $\pm$ SD	0.2485 $\pm$ 0.0256	0.2197 $\pm$ 0.0443	-0.0288 $\pm$ 0.0528

FAM wins in 2/5 seeds. The paired 95% t -interval is $[-0.0944, +0.0368]$, with exploratory paired t -test $p = 0.2894$ and exact Wilcoxon $p = 0.4375$. The final protocol does not support a YOLOv10 fusion benefit.

4.10.4 Standalone modality evaluation

The same standalone evaluator was used for all three input conditions on the ten final checkpoints. Its results are in Table 4.20.

Table 4.20: Final standalone YOLOv10 mAP@50 by modality, mean $\pm$ SD over five checkpoints.

Input	Base	FAM	Mean delta	FAM wins
VIS+IR	0.2498 ± 0.0254	0.2213 ± 0.0441	-0.0285	2/5
VIS	0.1964 ± 0.0218	0.1929 ± 0.0407	-0.0035	2/5
IR	0.0277 ± 0.0032	0.0369 ± 0.0009	+0.0092	5/5

The standalone fusion values reproduce the automatic ranking. VIS shows no stable benefit. IR improves in all seeds but remains too low to be operationally useful, and FAM is bypassed in that condition. The difference is an indirect training effect on the learned backbones and neck.

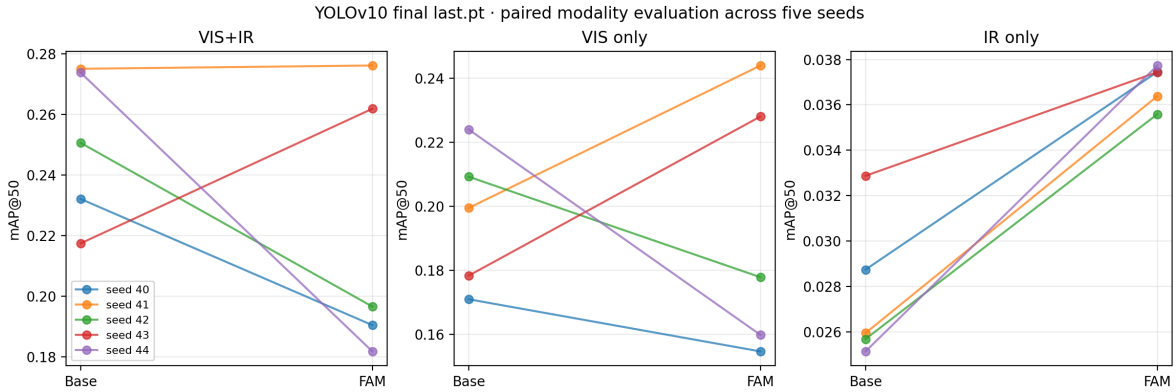


Figure 4.9: Final YOLOv10 paired modality evaluation. The favorable but very small IR-only delta does not compensate for the negative fusion result.

The standalone evaluator produced VIS+IR AP values systematically about 0.0010–0.0021 above the automatic Ultralytics validator. The maximum difference exceeded the predefined 0.002 sanity tolerance by 0.000077. The tolerance was not changed. Since all standalone modality conditions use the same evaluator and the checkpoint ranking is preserved, standalone values are used for the modality comparison while the automatic table remains the primary training result.

4.10.5 Training-horizon interpretation

At seed 42, the historical early Base checkpoint obtained 0.3469 at epoch 16 and the historical FAM checkpoint 0.2965 at epoch 89, compared with 0.2494 and 0.1947 at epoch 200. Base YOLOv10 training losses and learning rate match the old run exactly until its early stop, after which the final run continues. Training losses keep decreasing while the final test score is lower than the historical early score.

This pattern is compatible with late overfitting or specialization to the training sessions, but training loss alone does not establish the cause. The valid conclusion is that the declared 200-epoch protocol yields a negative FAM transfer result. A validation-led replication of YOLOv10 would require the complete-video Stage-A split and a checkpoint rule fixed before any new MtErie evaluation. The YOLO26 comparison below already uses that decision structure, with its own frozen training budget.

4.11 Five-seed Stage-A detector-transfer comparisons

RT-DETRv2 and YOLO26 use the same five-seed Stage-A decision procedure as the alignment ablations in Section 4.2. Each detector has its own Base control, and both branches use the same 3,123 training pairs and 896 whole-video validation pairs. All ten RT-DETRv2 runs complete ten epochs, and all ten YOLO26 runs complete fifty epochs. The primary outcome is the validation-selected best; the last epoch is reported independently.

Table 4.21: Detector-transfer Stage-A validation mAP@50, mean $\pm$ sample standard deviation across five seeds. Comparisons are within each detector: budgets and optimization recipes differ across families.

Configuration	Epochs	Best	Final epoch
RT-DETRv2 Base	10	0.1544 ± 0.0093	0.0908 ± 0.0460
RT-DETRv2 FAM	10	0.1764 ± 0.0255	0.0987 ± 0.0160
YOLO26 Base	50	0.0507 ± 0.0163	0.0013 ± 0.0022
YOLO26 FAM	50	0.0273 ± 0.0149	0.0001 ± 0.0001

Table 4.22: Paired FAM minus Base contrasts on Stage-A validation. Promotion uses only the best-checkpoint row and requires both a mean gain of at least 0.01 and at least four wins.

Detector	Checkpoint	Mean delta $\pm$ SD	Wins	95% CI	Role / decision
RT-DETRv2	best	$+0.0220 \pm 0.0344$	3/5	$[-0.0207, +0.0648]$	do not promote
RT-DETRv2	latest	$+0.0079 \pm 0.0538$	2/5	$[-0.0589, +0.0747]$	diagnostic
YOLO26	best	-0.0234 ± 0.0148	1/5	$[-0.0418, -0.0050]$	do not promote
YOLO26	last	-0.0012 ± 0.0022	0/5	$[-0.0039, +0.0016]$	diagnostic

4.11.1 RT-DETRv2 with FAM

FAM increases mean best-validation mAP@50 from 0.1544 to 0.1764. The mean paired gain of 0.0220 exceeds the required 0.01, but is positive in only three of five pairs. Its median is 0.0396 and its 95% interval spans zero. The candidate therefore fails the sign-consistency condition, not the mean-gain condition. It would be incorrect to describe this result as a negative mean FAM effect or to claim that the gain observed with RT-DETR v1 must disappear in RT-DETRv2.

The final-checkpoint contrast is also positive on average but wins only two pairs and has a wider interval. It does not provide a stable alternative to the primary result. All five pairs use matched recorded runtime and source manifests, transferred detection-head initialization, and paired input traces. These controls reduce avoidable differences; they do not make native CUDA deterministic or force an equal whole-model hash when FAM adds parameters. No Stage B or new MtErie evaluation is performed for this comparison.

4.11.2 YOLO26 dual-backbone Base and FAM

YOLO26 FAM has lower best-validation mAP@50 than Base in four of five pairs. The mean paired difference is -0.0234 , with a 95% interval below zero. Neither the mean-gain nor the four-win condition is satisfied. Both branches have very low final-epoch performance: the unrounded aggregate means from the recorded metrics are 0.001262 for Base and 0.000092 for FAM. Thus the best-checkpoint comparison and the final-epoch diagnostic both need to be retained when interpreting the training behavior.

All runs contain fifty epochs and fifty checkpoint-selection records. Strict loading and finite-parameter checks pass for best and last checkpoints. The five Base–FAM pairs match in initialization and recorded input traces. The ten serialized-best validation replays pass, with maximum absolute live/replay difference 0.0002525 against tolerance 0.003. The live checkpoint-selection values remain the primary metrics in Table 4.21.

This is a measured five-seed FAM effect for the implemented dual-backbone YOLO26 configuration. The low absolute performance of both branches limits its practical interpretation and does not show that stock YOLO26 or every possible RGB–thermal adaptation is unsuitable. The comparison does not enter Stage B.

4.12 Cross-architecture summary

The detector families provide different strengths of evidence. Deformable DETR shows an exploratory increase in median VIS+IR from 0.249 to 0.294 when FAM is added, but its intervals overlap and its protocol predates the final controls. Complete DINO base is weak under the studied configuration, so FAM is not tested on that implementation. YOLOv10 uses a final paired-seed protocol and shows a negative mean FAM effect. RT-DETRv2 and YOLO26 both provide five-seed Stage-A Base–FAM comparisons. RT-DETRv2 has a positive mean gain but insufficient paired wins, whereas YOLO26 has a negative mean gain and very low final performance in both branches.

The thesis therefore does not establish a consistently positive architectural transfer effect. It demonstrates that the FAM concept can be implemented in multiple detector families, that a promising exploratory trend exists in one, and that no controlled transfer comparison demonstrates an improvement meeting its declared stability criterion. The positive RT-DETRv2 mean must be distinguished from the negative YOLO means. These within-detector results do not justify a ranking across different training recipes, horizons, checkpoint policies, and evaluation populations.

Chapter 5

Discussion and Conclusions

5.1 Answers to the research questions

5.1.1 RQ1: experimental reliability

The observed RT-DETR variability is large enough to change single-run model rankings. Nominally identical native-CUDA executions diverge after the first backward pass even when initialization, input, and initial loss agree. The transferred COCO person head reduces amplification of those perturbations but does not remove them, and DCNv2 prevents strict deterministic FAM training in the studied environment.

Reliable comparison consequently requires more than recording a seed. The historical locked protocol combines paired seed values, isolated processes, a fixed training horizon, epoch-10 checkpoints, identical pretrained-head transfer, training-only Modal Dropout, and distributional reporting. It turns the single-run SSJ result into a repeatable selection of standard FAM over the Base RT-DETR configuration.

The validation-led campaign additionally shows that the checkpoint-selection rule materially affects the measured result. Holding out the complete FHL 0401/0402 video and selecting **best** within a fixed ten-epoch budget improves MtErie over **latest** in all five FAM Stage-A seeds. This is checkpoint selection, not early stopping: every run still completes ten epochs. Separating validation-led Stage A from full-data Stage B also prevents an apparent gain from being created by comparing different optimization datasets or code revisions.

The methodological answer to RQ1 is therefore that experimental design is part of the result. Paired replication, a complete-video development split, predeclared promotion rules, and a configuration-matched full-data confirmation are all needed before replacing the baseline. The selected split is still internal, and MtErie remains previously consulted rather than blind.

The acquisition-level confirmation demonstrates the complementary role of freezing data inventories, checkpoint hashes, and contrasts before inference. It provides stronger evidence than another MtErie evaluation, but does not turn internal WiSARD sequences into an external holdout. The mixed-consistency, box-guided, RT-DETRv2, and YOLO26 comparisons apply the same five-seed Stage-A decision structure as the other candidates. No performance decision is taken on an individual seed: the complete budget is run for every seed, the validation-selected **best** is primary, and the

final epoch is a mandatory diagnostic. Promotion requires both a mean paired gain of at least 0.01 and positive gains in at least four of five seeds. The fixed horizon is ten epochs for RT-DETR and RT-DETRv2 and fifty for YOLO26.

5.1.2 RQ2: RT-DETR effectiveness and modality use

In the historical locked comparison, standard FAM improves Base RT-DETR in every paired seed and raises mean mAP@50 by 0.0700. The improvement is supported by COCO mAP, mAP@75, mAR, and the recall analysis. IR feature dropout and SSJ do not provide a reliable increment over FAM; Identity DCNv2 is highly variable, and Grid Sample is close but not consistently superior.

The historical native-sensor modality table evaluates VIS, IR, and fusion on different frame populations and annotations, therefore it cannot support a direct cross-modality ranking. The post-selection audit holds all of these variables fixed for the five current-code FAM checkpoints. On the same 708 paired MtErie frames and VIS ground truth, fusion exceeds VIS in 5/5 seeds by 0.0373 ± 0.0223 , with a 95% interval of $[0.0095, 0.0650]$. This provides the supported evidence that the thermal stream adds information beyond VIS on the selected internal population.

The very low 0.0215 ± 0.0091 masked-IR/VIS-ground-truth score is a different robustness construct. It removes RGB, which supplies the reference for the asymmetric FAM, but still requests boxes in VIS coordinates from IR that has only been resized and padded. A post-hoc native-coordinate diagnostic obtains 0.5618 ± 0.0498 on the same IR counterparts with native IR annotations, ruling out a collapsed thermal branch. Replacing native IR dropout with the paired VIS-coordinate convention fails the frozen seed-40 ablation: it reduces native-IR performance in that screening run without solving paired missing-RGB prediction.

The acquisition-level confirmation strengthens the architectural result while narrowing the modality claim. Historical FAM exceeds Base in all five seeds on both previously unused acquisitions, with gains of 0.0960 on Carnation and 0.1554 on FHL. On the matched Stage-B checkpoints, however, fusion is effectively tied with VIS on Carnation and is worse on FHL. A better fusion operator does not imply that the thermal input helps uniformly on every acquisition.

The answer to RQ2 is therefore conditional. FAM consistently improves the Base RT-DETR reference, and fusion contributes beyond VIS under a same-frame paired evaluation. Modal Dropout supports native-sensor operation under its training contract, but it does not make the asymmetric fusion model robust to every possible missing-reference and coordinate-system intervention.

5.1.3 RQ3: mechanism, levels, and improved alignment

The FAM predicts nonzero transformations at P3 and P4 across all five standard historical checkpoints. Its offset scale and transformed features vary with seed, level, and acquisition session. SSJ changes P3 behavior consistently, increasing offset magnitude and smoothing, even though detection does not improve. Grid Sample is internally more stable but does not establish that a pure warp is equivalent to DCNv2.

The diagnostics do not demonstrate physical RGB-IR registration. DCNv2 mixes geometric sampling, filtering, mask modulation, and channel interaction. The sup-

ported interpretation is a task-oriented geometric-functional transformation. The P5 seed-41 collapse reinforces this distinction: a spatially constant internal branch coexists with the best historical FAM checkpoint because the rest of the detector compensates for it. Smoothly bounding offsets removes the collapse in five new runs but lowers mean accuracy, so a mechanistically effective correction is not automatically a better detector.

The validation-led extension campaign rejects several simple improvement hypotheses. Adding P2 loses in 5/5 seeds, a matched micro-batch control does not explain that deficit, global 800×800 input wins only 1/5 comparisons, and two post-FAM reliability gates fail their promotion rules. These results do not prove that high-resolution features or reliability modeling are generally unsuitable; they reject the implemented recipes under the common ten-epoch protocol.

Zero-offset initialization also fails Stage A: resetting only the initial sampling offsets reduces best-validation mean mAP@50 by 0.0125 and wins only one of five pairs. The final-epoch mean gain of 0.0118 has three positive pairs and does not change the decision based on validation-selected best checkpoints. Both paired confidence intervals include zero. The predictors learn nonzero, distinct weight rows in every inspected level and checkpoint, so their initially identical zero rows do not remain identical. This isolates an initialization intervention from the historical Identity DCNv2 operator, while providing no evidence of improved physical registration or a reliable performance advantage under the tested protocol.

RCRA is the only extension to pass Stage A. Its learned coefficient mainly suppresses the FAM residual at P3 and P4 and reacts to either missing modality in all five seeds; a three-parameter scalar control cannot reproduce the validation gain. In matched full-data Stage B, however, RCRA improves mean mAP@50 by 0.0184 but wins only 3/5 seeds, with an interval spanning $[-0.0550, 0.0918]$. It therefore fails the frozen stability condition. The answer to RQ3 is again asymmetric: reliability-conditioned residual selection is mechanistically active and promising, but its performance gain is not stable enough to replace standard FAM.

Mixed consistency training and Box-Guided P3 are evaluated against the same five standard-FAM Stage-A controls as the other RT-DETR candidates. Mixed consistency reduces best-checkpoint mean mAP@50 from 0.1646 ± 0.0196 to 0.1094 ± 0.0328 , with negative paired differences in all five seeds. Box-Guided P3 obtains 0.1723 ± 0.0233 , but its mean gain of 0.0077 and two positive pairs fail both promotion conditions. Its final-epoch gain is only 0.0005. Neither recipe is promoted to full-data Stage B. These results reject the tested detection recipes, not their underlying mechanistic objectives. Five-seed geometric counterfactuals for box-guided alignment and isolated-modality evaluations for mixed consistency were not performed and their outcomes cannot be inferred from the fusion comparison.

The synthetic stress supplies a second limitation. FAM does not exhibit a universally more favorable signed degradation curve than Base; some FHL translations improve the unmodified Base reference more. The supported mechanism remains a task-oriented geometric-functional transformation, not verified physical registration or invariant tolerance to known shifts.

5.1.4 RQ4: operational relevance

FAM primarily changes the recall side of the precision–recall trade-off. At confidence 0.25, it improves mean recall by 0.0990 in all five paired seeds and reduces the frequency of non-empty frames with at least one missed target. This is relevant to SAR, where a missed person can have high cost. At a common confidence threshold, false positives do not decrease consistently, however, and the 0.01 threshold used for AP collection is not operational because it produces roughly fifty false positives per image.

The post-hoc recall–FPPI comparison in Section 4.5.3 separates this threshold-dependent observation from performance at a matched false-positive budget. At 0.5 FPPI, historical FAM increases mean recall by 0.1102 on MtErie, 0.0981 on Carnation 0025/0026, and 0.1479 on FHL 0407/0408, with positive differences in all five seeds within each population. This supports a better recall–false-positive trade-off at the reported budgets, not universal curve dominance or a validated deployment threshold.

Carnation 0023/0024 exposes a geometric boundary. FAM outperforms Base RT-DETR under the severe scale mismatch, but adding IR remains worse than using VIS alone in every FAM checkpoint. The experiment therefore establishes a relative FAM advantage under mismatch, not correction or mitigation of the underlying sensor misalignment.

On the two larger previously unused internal acquisitions, FAM’s absolute advantage over Base persists in all ten acquisition–seed comparisons. The controlled perturbations show, however, that operational robustness cannot be reduced to a monotone translation or scale curve without calibrated sensor geometry. These findings motivate acquisition-specific validation rather than a universal alignment claim.

The performance gain also has a material computational cost: 77.5% more parameters, 46.2% more proxy GFLOPs, 17.9% more detector latency, and higher CUDA memory on the measured GPU. These values prevent describing FAM as a free or generically real-time improvement.

5.1.5 RQ5: architectural transferability

Consistently positive transfer is not established. Deformable DETR provides a promising exploratory median trend, but its observed ranges overlap and its protocol is not aligned with the final campaign. Complete DINO base is too weak under the studied configuration to justify FAM variants. Under the controlled YOLOv10 protocol, FAM decreases mean VIS+IR performance and wins in only two of five seeds.

The five-seed Stage-A contrasts distinguish RT-DETRv2 from YOLO26 (Tables 4.21 and 4.22). RT-DETRv2 FAM increases best-checkpoint mean mAP@50 from 0.1544 ± 0.0093 to 0.1764 ± 0.0255 . Its paired gain of 0.0220 ± 0.0344 exceeds the 0.01 mean threshold, but only three of five differences are positive; the 95% interval is $[-0.0207, 0.0648]$. This is a positive but unstable mean effect, not evidence of uniformly harmful transfer. The four-of-five consistency condition is not met, so RT-DETRv2 FAM is not promoted to Stage B.

YOLO26 FAM decreases best-checkpoint mean mAP@50 from 0.0507 ± 0.0163 to 0.0273 ± 0.0149 , with a paired difference of -0.0234 ± 0.0148 and only one positive pair. All ten Base and FAM runs complete fifty epochs. Final-epoch means fall to 0.001262 and 0.000092, respectively, showing substantial degradation in both variants.

The selected best checkpoints are predominantly early because they are selected by the declared validation rule over the full completed budget, not because the comparison was restricted to the first epoch. YOLO26 FAM fails Stage A and does not enter Stage B.

The negative YOLOv10 result is scientifically useful. It shows that the RT-DETR gain depends on the surrounding representation, fusion interface, optimization trajectory, or checkpoint regime and cannot be assumed from the local FAM operation alone. RT-DETRv2 indicates that architectural similarity can coexist with a favorable mean effect that remains insufficiently stable across paired seeds. The low YOLO26 scores and final-epoch degradation limit attribution to this dual-backbone port and training recipe, not the detector family in general. None of these results shows that YOLOv10, RT-DETRv2, YOLO26, or DINO is intrinsically unsuitable for RGB–thermal detection.

5.2 Overall interpretation

The most defensible conclusion combines complementary experimental contrasts. The historical locked study establishes that standard FAM improves Base RT-DETR consistently on the internal benchmark and primarily raises recall. The validation-led campaign does not confirm a replacement: RCRA has a positive but unstable Stage-B mean effect, while P2, global upsampling, zero-offset initialization, post-FAM gating, scalar residual calibration, mixed consistency, and box-guided alignment fail their five-seed Stage-A development rules. Standard FAM therefore remains the selected model and the reference for subsequent candidate comparisons.

That choice is strengthened by the previously unused acquisition campaign: historical FAM beats historical Base in 10/10 acquisition–seed contrasts. The detector-transfer comparisons do not establish a general FAM advantage. RT-DETRv2 has a positive mean effect but wins only three of five seeds and fails the same Stage-A consistency requirement. Within YOLO26, FAM loses to Base in four of five paired seeds, while both variants have very low final-epoch scores. Its different optimization budget and evaluation pipeline preclude interpreting the absolute YOLO26 and RT-DETR scores as a controlled cross-family ranking.

The paired modality audit strengthens one part of that conclusion and narrows another. Fusion exceeds VIS on the same 708 frames in all five selected checkpoints, so the thermal stream contributes useful information. Conversely, masked IR in VIS coordinates is not interchangeable with native IR detection. Reporting the coordinate contract is necessary to distinguish sensor complementarity from missing-reference robustness.

At the same time, FAM is internally variable and computationally expensive. One feature level can collapse without visible degradation in the headline metric, a correction that makes the mechanism safer can make the detector worse, and a more adaptive RCRA mechanism can improve the mean without being stable across seeds. Architecture transfer and external scale mismatch further show that an alignment-inspired inductive bias is not sufficient on its own. Controlled synthetic perturbations add that higher absolute detection accuracy and a more favorable signed relative-drop curve are distinct properties.

These results support a broader methodological point: evaluating multimodal fusion should combine output metrics, coordinate-controlled modality tests, paired replication, internal diagnostics, out-of-distribution stress, and cost. Any one of these views alone would have produced an incomplete model selection.

5.3 Threats to validity

5.3.1 Internal validity

Native CUDA execution is nondeterministic, particularly for deformable operators. Pairing seeds controls intended stochastic sources but cannot pair the microscopic CUDA perturbations of two architectures. Five independent checkpoints estimate the resulting distribution but do not exhaust it.

The level ablations are interventions on already trained networks. Bypassed representations are outside the training path and cannot be interpreted as the performance of a newly trained P3-only or P3+P5 architecture. The P5 audit and the choice of a bounded correction followed observation of the MtErie checkpoint. Freezing one hypothesis before five new trainings reduces further adaptation, but the correction remains post-hoc with respect to the internal benchmark.

The standalone YOLOv10 evaluator differs slightly from the automatic Ultralytics validator. The predefined tolerance is recorded as missed rather than changed after inference. Comparisons among modalities use the same standalone evaluator, while automatic VIS+IR results remain primary for the training protocol.

The whole-sequence split, promotion thresholds, and Stage-B confirmation rule were frozen before their respective comparisons. Nevertheless, the candidate sequence was designed within the same study and after extensive prior experimentation. Stage A is therefore an internal model-development procedure, not an independent replication. The Modal Dropout coordinate-system ablation uses only one predeclared screening seed and supports rejection of that recipe, not a five-seed population estimate.

The zero-offset initialization, mixed-consistency, and Box-Guided comparisons reuse the five completed standard-FAM Stage-A controls. Shared model, training, data, and split settings were checked, and reconstructed initialization hashes match for all five seeds. For zero-offset initialization, the intervention additionally preserves all unaffected parameters and random-number-generator state. Reconstructing these initial states does not certify identical historical source snapshots or environments. Mixed consistency adds student-path random draws that can alter later Modal Dropout realizations. Seed pairing therefore evaluates the complete training intervention rather than isolating the consistency loss under identical intermediate tensors. Strict checkpoint loading and recorded replay checks constrain execution and evaluation errors, but cannot remove these attribution limits or native-CUDA divergence.

5.3.2 Construct validity

mAP@50 emphasizes detection at IoU 0.50 and is not a complete measure of localization quality or SAR utility. Secondary COCO mAP, mAP@75, mAR, and error metrics reduce dependence on one score but do not measure response time, operator workload,

alarm calibration, energy consumption, or consequences of false alarms.

Recall–FPPI curves make the cost in false detections explicit but still count boxes per frame, not distinct alarm events. Consecutive frames are correlated, and tracking or temporal aggregation would change the operator-facing alarm rate. These curves describe performance on evaluation sets that had already been examined in earlier analyses. The shaded bands show variability across five training seeds on these fixed sets, not uncertainty about performance on new acquisitions. The confidence sweep does not establish a confidence threshold for operational use, but it characterizes the recall–false-positive trade-off.

Offset magnitude, feature cosine similarity, RMSE, smoothing, and PCA are internal proxies. Without camera calibration or ground-truth correspondences, they do not validate physical registration. PCA figures also depend on a low-dimensional projection and are used illustratively.

Native-sensor and paired masked-modality AP operationalize different constructs. The former changes the image canvas and ground truth with the sensor, while the latter fixes a VIS-coordinate population and zeros channels. The native-IR diagnostic is therefore evidence against branch collapse, not a direct ranking against fusion. Likewise, `adapt_ir2rgb` is a resize-and-pad transformation and should not be interpreted as calibrated registration.

The box-centre displacement used by Box-Guided P3 is a weak geometric proxy, not dense pixel correspondence. Detection results alone do not establish whether that target was learned consistently across seeds, let alone physical registration or geometric generalization. Similarly, the signed relative drop under synthetic translation and scaling confuses sensitivity with the possibility of accidentally compensating a pre-existing mismatch.

5.3.3 Statistical conclusion validity

Five paired runs permit estimation of effect direction and variability but produce wide intervals. Exact Wilcoxon tests have coarse resolution, and the smallest attainable two-sided value for five nonzero signed pairs is 0.0625. The reported tests are exploratory and are not corrected for the number of contrasts. Checkpoints, rather than frames, boxes, diagnostic samples, or feature cells, are the experimental units.

The Stage-A comparisons of zero-offset initialization, mixed consistency, Box-Guided P3, RT-DETRv2, and YOLO26 likewise retain five paired checkpoints per contrast. Their confidence intervals describe between-seed variability on the fixed validation split, not uncertainty across new acquisitions. Selection is governed by the predeclared mean-gain and sign-consistency thresholds, not a post-hoc significance filter.

5.3.4 External validity

MtErie was repeatedly consulted during study development and is not a blind holdout. Its locked and Stage-B comparisons are internally useful but may overstate performance on a new acquisition. The official 273-frame FHL validation remains too small and shifted for stable selection. The replacement 896-frame whole-video split is much more informative and removes same-video adjacency, but FHL 0401/0402 and the remaining FHL training video still belong to the same acquisition campaign.

Carnation 0025/0026 and FHL 0407/0408 were unused before the frozen confirmation and contain substantially more paired frames, but they remain internal WiSARD acquisitions. They strengthen acquisition-level replication without demonstrating generalization outside WiSARD, to a different sensor platform, or to the SAR population at large.

Carnation is one deliberately severe scale-mismatch case. It reveals a specific boundary but cannot estimate performance across SAR domains. The study uses one dataset, one main RT-DETR size, one YOLOv10 size, one RT-DETRv2 configuration, one YOLO26 size, and one GPU for latency. Different cameras, calibration, target scales, thermal conditions, hardware, or training budgets may change the conclusions.

Deformable DETR and DINO results use an exploratory protocol. RT-DETRv2 and YOLO26 follow the five-seed Stage-A decision procedure within the two-stage framework, but neither satisfies the promotion rule and therefore neither enters Stage B. Detector-specific recipes, budgets, and evaluators remain different. Their Stage-A validation scores also have a different population and checkpoint role from the full-data MtErie results. These tables cannot form a quantitatively homogeneous architecture leaderboard with final RT-DETR and YOLOv10.

5.4 Future work

5.4.1 Independent confirmation of the model-selection protocol

This thesis implements a train-derived, whole-video validation protocol and shows that the five validation-selected FAM checkpoints outperform their corresponding epoch-10 checkpoints on MtErie in all five Stage-A seeds. The next priority is therefore not to create that split again, but to test whether the rule transfers beyond the FHL acquisition campaign. A genuinely independent paired sequence should be reserved before architecture development, and the checkpoint metric, horizon, promotion thresholds, and allowed search space should be fixed before observing it.

The RT-DETR ten-epoch design deliberately avoids early stopping. Future work may compare fixed-budget checkpoint selection with early stopping, but only under the same independent validation population and equal maximum compute. This would help determine whether the late YOLOv10 degradation reflects overfitting, schedule mismatch, or a detector-specific checkpoint problem.

5.4.2 Improving the alignment architecture

The negative and unstable results narrow the next useful hypotheses. Repeating P2, global 800×800 input, or another learning-rate search on the same post-FAM gate would add little evidence. More targeted directions are:

1. use camera calibration, dense paired cross-modal annotations, or a controlled acquisition rig with known physical geometry to supervise and evaluate alignment explicitly;

2. train auxiliary modality-specific heads or coordinate-consistency losses so that native IR and paired VIS-coordinate robustness are not forced into one ambiguous objective;
3. evaluate RCRA on an independent sequence and with more seeds before altering its descriptors or optimization, to distinguish a real conditional-alignment effect from training variance;
4. revisit mixed native/paired consistency only with a new mechanism and an independent development population; the tested teacher–student formulation should not be retuned on the same FHL validation;
5. reduce FAM or RCRA cost through channel compression or selective application to the most useful levels, followed by a complete inference benchmark rather than training-time measurements alone.

These candidates should not be searched retrospectively on MtErie. A small number of mechanistically justified variants should be frozen on an independent development protocol and followed by paired replication and the same internal audit used in this thesis.

5.4.3 Broader evaluation and deployment

Further work should include multiple external RGB–thermal acquisitions, calibrated scale and translation experiments, additional missing-sensor patterns, confidence calibration, and end-to-end latency including image loading, preprocessing, transfer, post-processing, and visualization. Deployment experiments should report power and memory limits of the target UAV hardware rather than extrapolating from a laptop GPU.

5.5 Conclusions

This thesis evaluated RGB–thermal fusion for UAV Search and Rescue from a reliability-first perspective. Its study of the inherited RT-DETR Feature Alignment Module combines paired multi-seed comparison, direct inspection of learned transformations, temporal and whole-video validation audits, acquisition-level stress testing, and configuration-matched candidate confirmation.

The historical locked study shows that standard FAM improves Base RT-DETR in all five paired seeds and consistently increases recall. The validation-led campaign rejects P2, global upsampling, zero-offset initialization, post-FAM gates, scalar residual control, mixed consistency, and box-guided alignment under the common five-seed protocol. RCRA is the only candidate to pass Stage A and learns the intended conditional mechanism, but its full-data gain appears in only 3/5 seeds. Under the rule fixed before Stage-B evaluation, standard FAM therefore remains the selected model.

The corrected paired modality audit shows that this FAM model benefits from thermal information beyond VIS on the same 708 frames. It also exposes a sharp coordinate-contract boundary: paired masked IR with VIS ground truth is very weak,

while native IR on native annotations remains functional. A one-seed attempt to replace native dropout with the paired contract fails to resolve the first condition and sharply reduces native-IR performance in that run.

The frozen confirmation on Carnation 0025/0026 and FHL 0407/0408 strengthens the central architectural result: historical FAM exceeds Base in all ten acquisition-seed comparisons, with a macro-average gain of 0.1257 mAP@50. At the same time, fusion does not uniformly exceed VIS on the matched Stage-B checkpoints. The synthetic intervention also rejects the stronger claim that FAM always degrades more gracefully under IR translation or rescaling.

The within-family transfer comparisons delimit the architectural result. RT-DETRv2 FAM improves the best-checkpoint mean by 0.0220 but wins only three of five seeds, so the Stage-A stability condition prevents promotion. YOLO26 FAM has a lower best-checkpoint mean than its paired Base control and wins only one of five seeds; both variants degrade strongly over the fifty-epoch horizon. Full-budget replication and validation-selected checkpoints are essential to interpret this result as a property of the tested recipe, without extending it to all uses of FAM or all YOLO26 architectures.

The principal outcome is therefore both architectural and methodological. FAM is the strongest confirmed RT-DETR configuration in this study, but it is not a universal alignment solution and no new architecture is shown to improve it reliably. Repeated runs, predeclared development and confirmation stages, coordinate-controlled modality evaluation, internal diagnostics, frozen acquisition confirmation, controlled stress cases, and computational cost are necessary to determine whether an apparent fusion gain is reliable, interpretable, and transferable.

Appendix A

Supplementary Results and Artifact Provenance

A.1 Secondary RT-DETR metrics

Table A.1: Final VIS+IR secondary metrics, mean $\pm$ sample standard deviation over five training checkpoints. COCO mAP averages IoU thresholds from 0.50 to 0.95.

Configuration	COCO mAP	mAP@50	mAP@75	mAR@100
Base	0.1059 $\pm$ 0.0243	0.3080 $\pm$ 0.0677	0.0474 $\pm$ 0.0128	0.2474 $\pm$ 0.0302
FAM	0.1351 $\pm$ 0.0096	0.3780 $\pm$ 0.0440	0.0692 $\pm$ 0.0043	0.2798 $\pm$ 0.0141
FAM + IR Dropout	0.1360 $\pm$ 0.0165	0.3871 $\pm$ 0.0469	0.0698 $\pm$ 0.0079	0.2764 $\pm$ 0.0222
FAM + SSJ	0.1297 $\pm$ 0.0095	0.3749 $\pm$ 0.0185	0.0625 $\pm$ 0.0086	0.2719 $\pm$ 0.0331
Identity DCNv2	0.1134 $\pm$ 0.0305	0.3280 $\pm$ 0.0943	0.0560 $\pm$ 0.0141	0.2556 $\pm$ 0.0357
Grid Sample	0.1331 $\pm$ 0.0077	0.3701 $\pm$ 0.0345	0.0631 $\pm$ 0.0044	0.2849 $\pm$ 0.0115

A.2 Low-threshold qualitative montage

A.3 Versioned compact artifacts

Frame-level prediction outputs and diagnostic JSON files are retained in the local `out/` directories. The following compact artifacts are stored with the thesis source:

Artifact	Content
<code>results/rtdetr_error_analysis_checkpoints.csv</code>	Fifty checkpoint–threshold summaries for Base and FAM on MtErie.
<code>results/rtdetr_recall_fppi_summary.csv</code>	Nine population–budget summaries of historical Base and FAM recall, paired differences, and wins.

Artifact	Content
results/rtdetr_recall_fppi_budgets.csv	Ninety checkpoint–budget points, including attained FPPI, descriptive confidence thresholds, TP, and FP.
results/rtdetr_recall_fppi_paired.csv	Forty-five within-seed recall differences; repeated budgets and populations are not independent replications.
results/rtdetr_recall_fppi_grid.csv	Mean and sample SD on the common 301-point FPPI grid for each configuration and population.
results/rtdetr_recall_fppi.json	Complete 30-unit protocol, checkpoint and inventory hashes, software provenance, and cache references.
results/rtdetr_carnation_stress_test.csv	Thirty checkpoint–modality units from the frozen Carnation stress test.
results/rtdetr_compute_benchmark.csv	Parameter, operation-proxy, latency, throughput, and CUDA-memory summary.
results/rtdetr_fam_sequence_checkpoint_evaluation.csv	Five-seed Stage-A FAM comparison of validation-selected best and epoch-10 latest .
results/rtdetr_fam_residual_alignment_stage_a_validation.csv	Paired Stage-A FAM–RCRA validation results.
results/rtdetr_fam_scalar_alignment_control_stage_a_validation.csv	Three-parameter residual-control attribution results.
results/rtdetr_fam_rcra_full_data_stage_b_evaluation.csv	Matched full-data FAM–RCRA Stage-B comparison.
results/rtdetr_fam_full_data_paired_modality_evaluation.csv	Selected-FAM modality interventions on the 708 common paired frames.
results/rtdetr_fam_full_data_native_ir_coordinate_diagnostic.csv	Post-hoc native-IR coordinate diagnostic on the corresponding 708 IR frames.
results/rtdetr_fam_paired_vis_modal_dropout_probe_evaluation.csv	Seed-40 Modal Dropout coordinate-system ablation.
results/yolo_final_modality_evaluation.csv	Thirty final YOLOv10 checkpoint–modality evaluations.
results/rtdetr_unused_acquisition_confirmation.csv	Fifty frozen evaluations on previously unused internal WiSARD acquisitions.
results/rtdetr_synthetic_geometric_stress.csv	Six hundred identity and IR-perturbation evaluation points.

Artifact	Content
results/stage_a_five_seed_comparison.csv	Thirty-five run records: five seeds each for the FAM reference, mixed consistency, box-guided alignment, and RT-DETRv2/YOLO26 Base and FAM; best and final validation scores.
results/stage_a_five_seed_comparison.json	Configuration summaries, paired differences, sample deviations, medians, intervals, promotion decisions, and provenance for these Stage-A comparisons.
results/rtdetr_fam_zero_offset_stage_a.csv	Five paired standard-FAM and zero-offset-initialization records, with best/final validation scores, selection epochs, and replay values.
results/rtdetr_fam_zero_offset_stage_a.json	Zero-offset initialization protocol, paired summaries, initialization/replay audits, provenance, and Stage-A decision.
images/rtdetr_additive_fam_error_summary.png	Paired error summary at confidence 0.01.
images/rtdetr_additive_fam_threshold_sensitivity.png	Multi-threshold error sensitivity analysis.
images/rtdetr_recall_fppi.pdf	Recall-FPPI curves for historical Base and FAM on MtErie, Carnation 0025/0026, and FHL 0407/0408.
images/rtdetr_carnation_paired_map50.png	Carnation paired-seed plot.
images/yolov10_final_modality_paired_map50.png	Final YOLOv10 paired modality plot.
images/rtdetr_synthetic_geometric_stress_curves.png	Signed relative-drop curves for controlled IR translation and rescaling.
images/experimental_design.pdf	Vector diagram of the five-seed experimental design, Stage-A selection, conditional Stage-B confirmation, and separate audits.
images/rtdetr_fam_architecture.pdf	Vector diagram of the implemented dual-backbone RT-DETR and per-level FAM.
images/rtdetr_final_paired_map50.pdf	Final paired-seed RT-DETR comparison generated from the primary training outputs.
images/fam_p5_seed41_collapse.png	Thesis-readable P5 collapse diagnostic assembled from the saved feature diagnostic and JSON statistics.
images/yolov10_dual_backbone_fam.pdf	Vector diagram of the implemented YOLOv10-s transfer architecture and modality contract.

The artifact README records protocol identifiers, completeness markers, and checksums of the larger local aggregates. The Stage-A CSV retains per-seed values and run provenance to make the aggregate tables auditable; these records are not separate single-run claims. Archived exploratory outputs are not pooled with the five-seed esti-

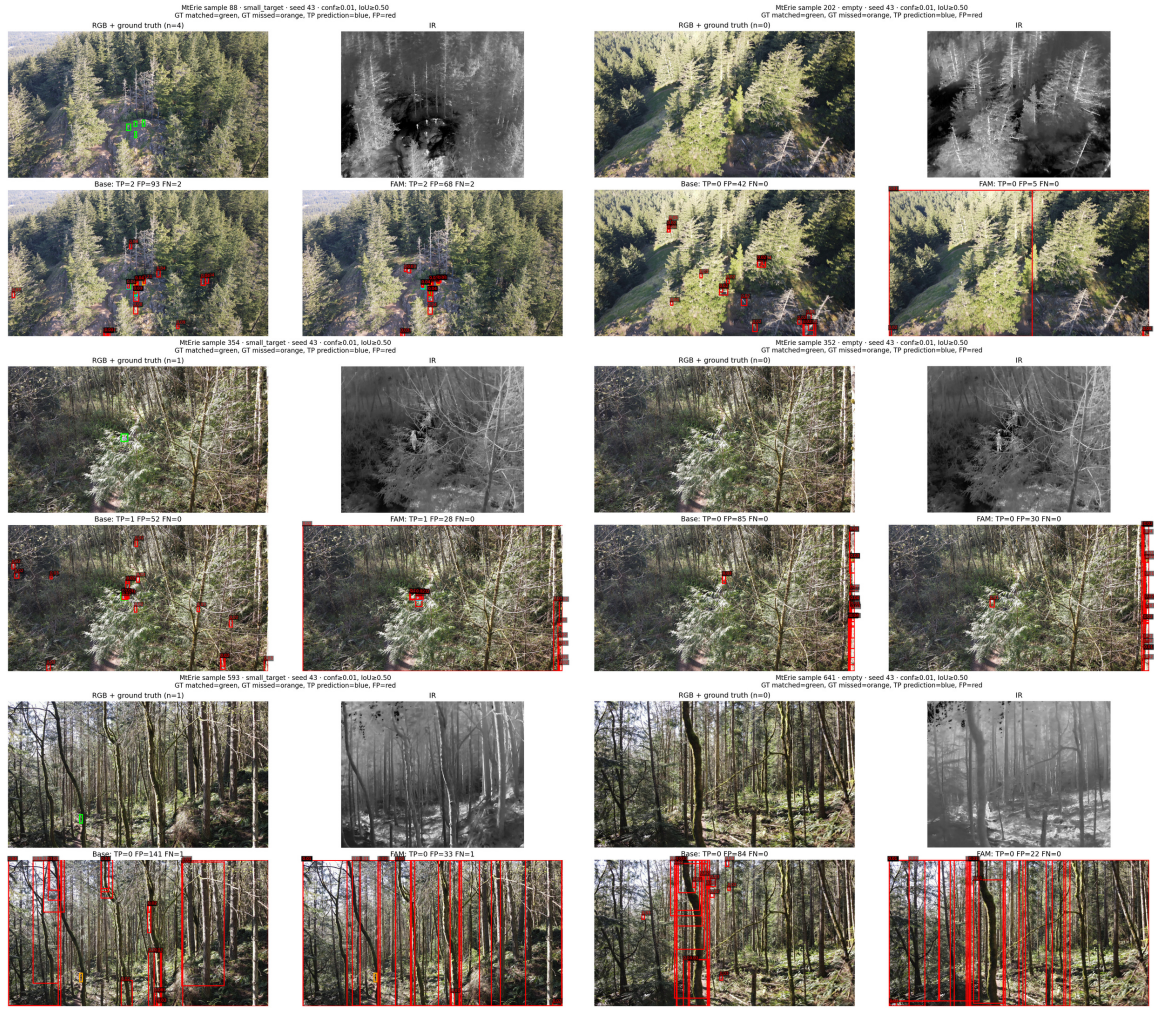


Figure A.1: The six predefined qualitative frames at confidence threshold 0.01. Titles count all detections, while at most twenty high-score predictions are drawn for readability. The density of low-confidence false positives illustrates why 0.01 is a metric-collection threshold rather than an operational setting.

mates. The configuration summaries distinguish live best-selection scores from replay checks and from final-epoch diagnostics.

A.4 Code availability

Model implementations, preprocessing code, experimental configurations, and analysis scripts are maintained in the project repository. The generated vector diagrams and comparison figures can be regenerated with `scripts/generate_thesis_figures.py`. The recall-FPPI figure and compact tables are instead regenerated from the complete local prediction caches with `scripts/run_rtdetr_recall_fppi.py` using `-summarize-only`; this does not require new inference. Historical public project material is available at https://github.com/DonatoFe11/SAR_Fusion. Checkpoints required for the reported final diagnostics are preserved locally because model-weight files were excluded from the experiment tracker uploads.

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